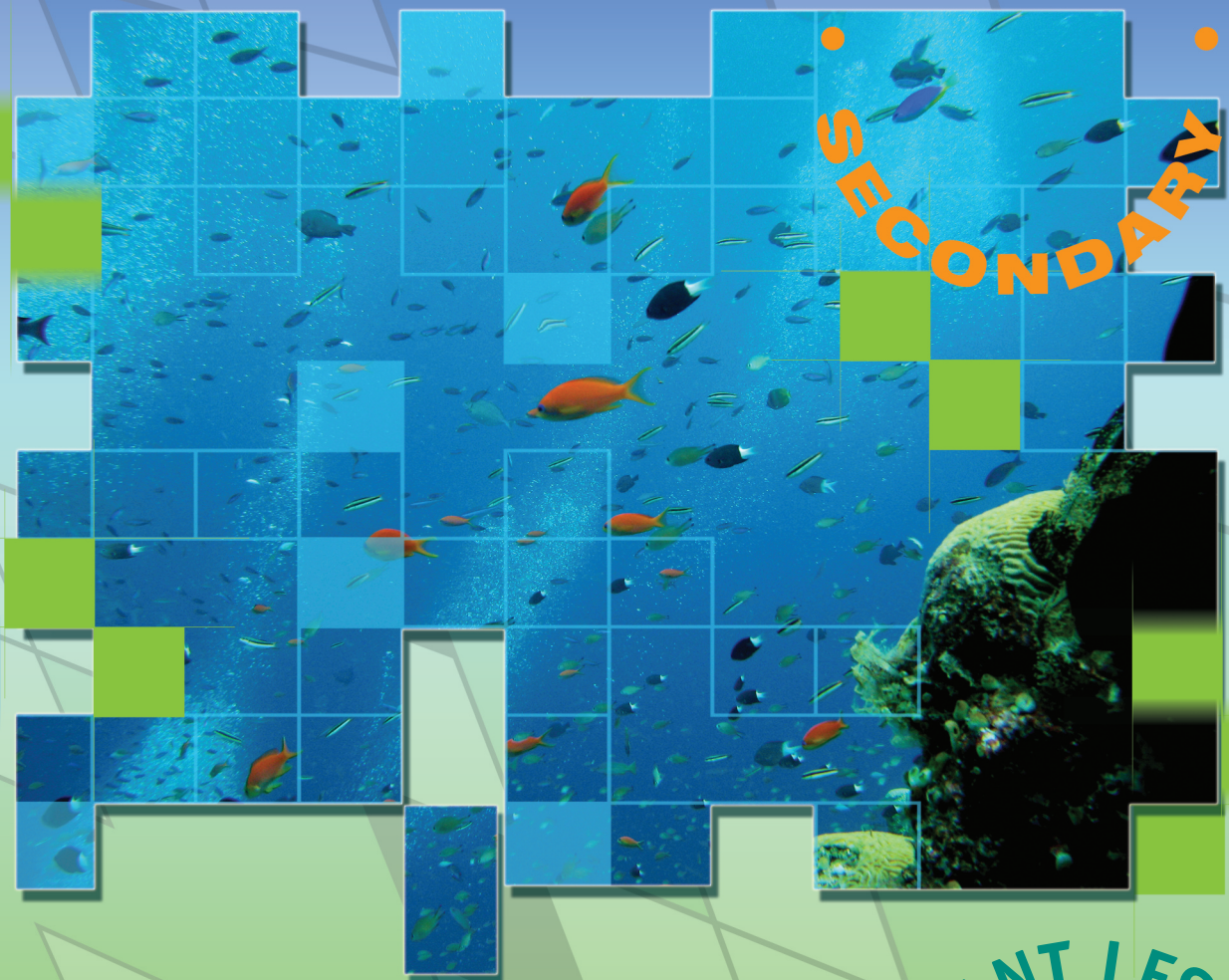


CHALLENGING SCIENCE

CROSSWORDS

JUNIOR

SECONDARY



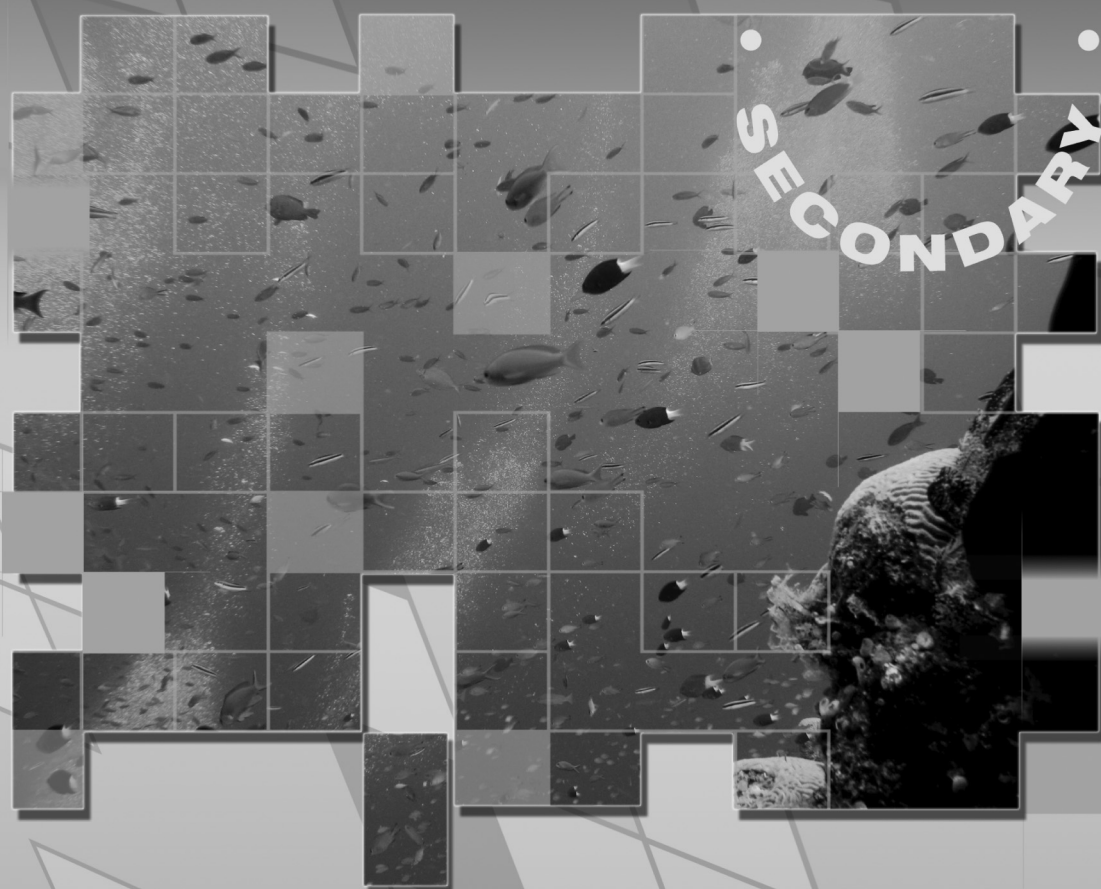
Dr Raimund Pohl



 Quality ready-to-use
resources

CHALLENGING SCIENCE CROSSWORDS

JUNIOR



SECONDARY

Dr Raimund Pohl



Acknowledgements

I would like to thank the students of Cheltenham Girls' High School for trialling these crosswords, the teachers who also assisted by their positive criticisms and last but not least my wife, who gave up her precious time in proofing these crosswords.

Dr Raimund Pohl

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INTRODUCTION



This series of *Challenging Science Crosswords* is a unique, student-friendly approach to increasing science literacy. They have been thoroughly trialled in classrooms, with positive feedback from students who find the crosswords great for revision and fun to do. Teachers have found the crosswords to be excellent springboards for promoting discussion of science concepts and issues, and for introducing or revising core curriculum. They also love the opportunity to target every student's ability and confidence level – the crosswords contain a wordlist at the bottom of the page. This list can easily be covered when photocopying to create a more challenging activity or left on to provide helpful hints for students.

Challenging Science Crosswords need no further preparation, and are stimulating, educational activities, perfect for last-minute teaching or integration into a science unit.

What the students said: *Interesting and not too difficult ... Great for revision ... Learnt something new ... Challenging – I love science crosswords ... They are fun ... Good material to study with ... Educational! ... The word bank helped a lot ... I tried these and I remember.*

What the teachers said: *A real help for revision and reinforcing of ideas ... Good to see something that makes revision enjoyable ... Why didn't I think of that? ... The word bank makes it great for the junior years.*

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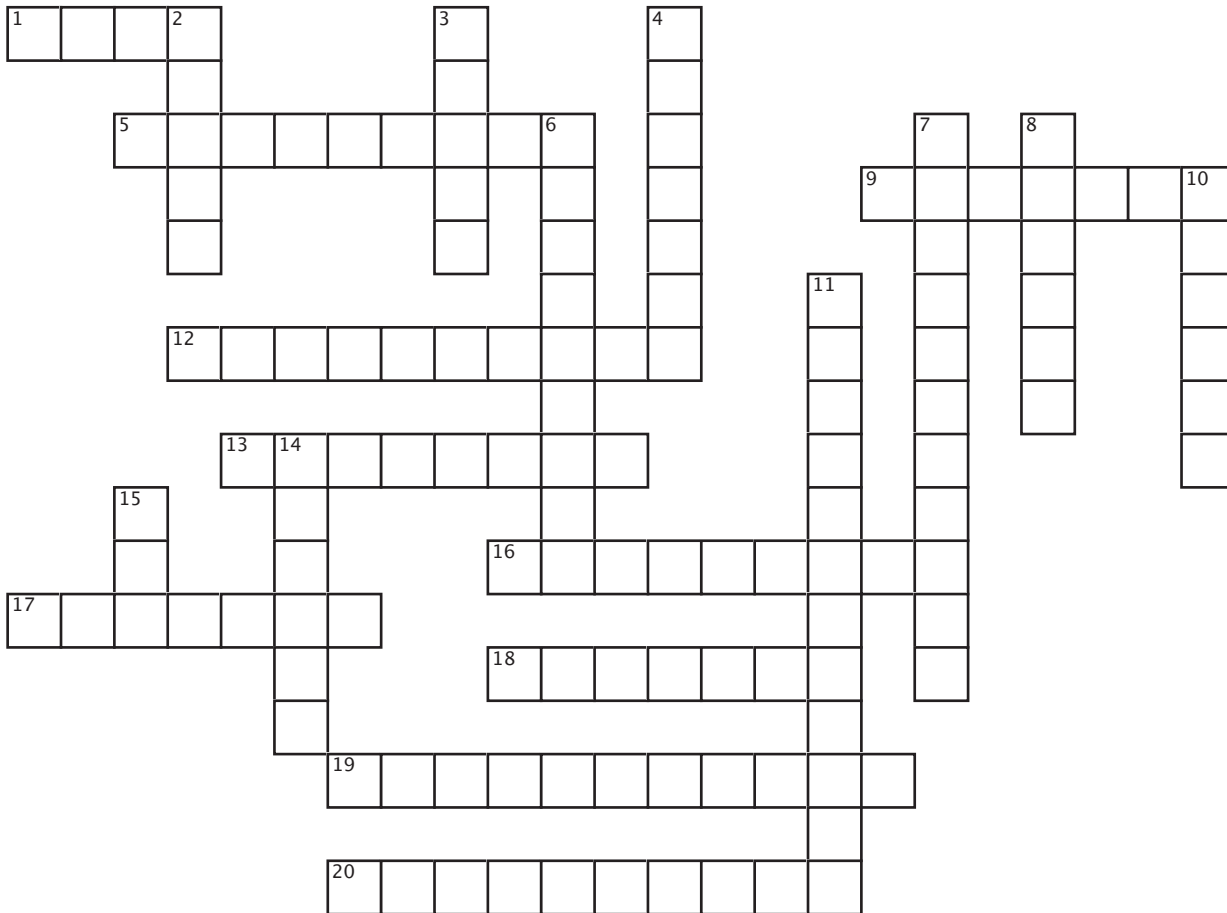
SKILLS, CAREERS AND CONCEPTS

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EARTH'S FORMATION



ACROSS

1. Supernova energy and gas clouds collide to form a _____
5. An exploding star emitting vast amounts of energy
9. Earth's primordial atmosphere was made from this vapour
12. Least dense layer of the Earth
13. About 3.5 billion years ago the atmosphere cooled sufficiently to form what type of precipitation?
16. Nickel and iron form this part of the Earth's interior
17. Force that causes accretion
18. Term used to describe the amount of matter in an object
19. Small bodies that orbited the Sun during the formation of the planets
20. Objects that come from space and reach the Earth's surface

DOWN

2. Name of Earth's outer solid layer
3. Atmospheric layer responsible for preventing harmful radiation from reaching the Earth
4. Earth's second atmosphere is composed mainly of this gas
6. An increase in a planet's size by gradual external addition or fusion
7. The outer part of the Earth consisting of the crust and upper mantle; means 'rock sphere'
8. The inner core of the Earth is in this state of matter
10. An immense cloud of mainly hydrogen gas and dust in interstellar space
11. The age of the solar system is approximately 4.5 _____
14. Name of a meteor impact
15. Abbreviation for 'millions of years ago'

WORD LIST

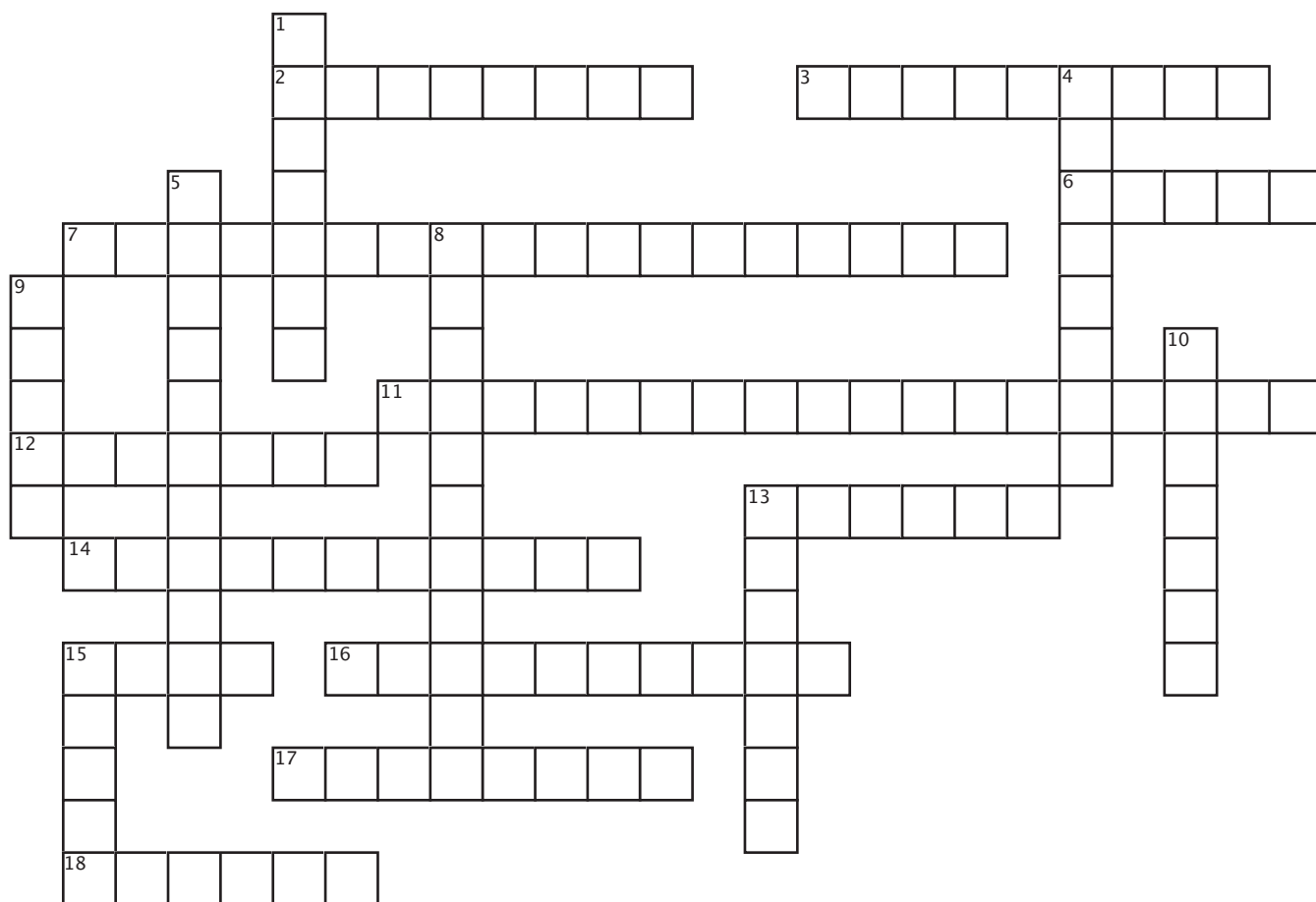
☛ supernova
☛ disc
☛ nebula
☛ planetismal

☛ accretion
☛ gravity
☛ meteorites
☛ inner core

☛ crust
☛ methane
☛ crater
☛ atmosphere

☛ lithosphere
☛ silicon
☛ density
☛ liquid

☛ acid rain
☛ ozone
☛ mya
☛ billion years



ACROSS

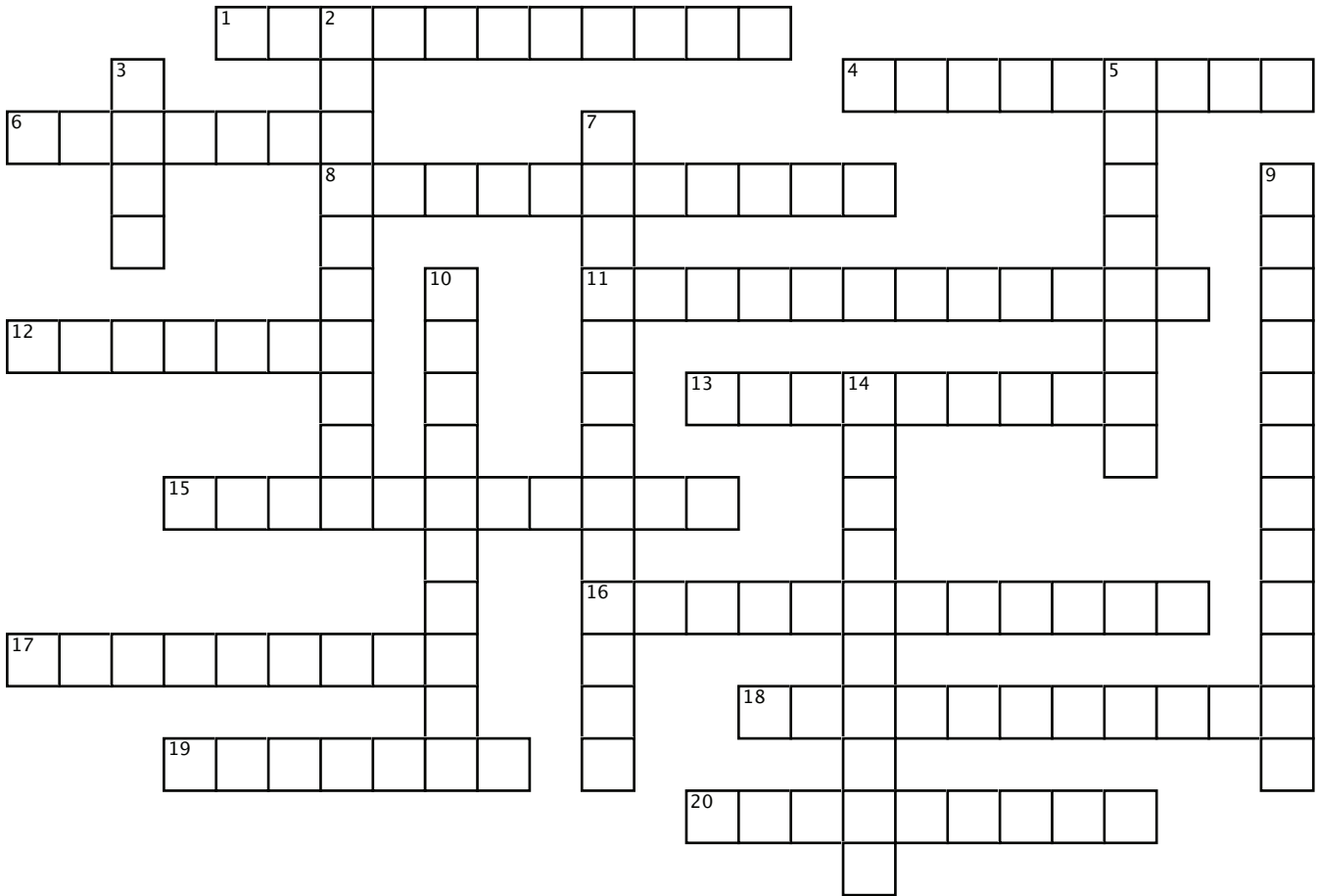
2. Another word for 'spinning about an axis'
3. Atmospheric gases, except oxygen, come from ____
6. The name of the interaction of the Moon and the Sun that causes oceanic levels to rise and fall
7. The breaking down of rocks by acids, and other substances, that alter their composition is called ____
11. The breaking down of rocks into sediments by ice, abrasion, etc. is called ____
12. Rocks that form from lava or magma
13. The colour of a scratch made by a rock or mineral
14. Rocks formed from intense heat and pressure
15. Innermost layer of the Earth
16. Name of gases that trap heat
17. Rocks are made from these
18. The softest rock on Earth; used on babies

DOWN

1. The process of rocks, sediments and soil swept away by water or wind
4. Most common atmospheric gas
5. Compacted, layered rocks
8. Crust is part of this layer
9. Pathway of a body around a planet or star
10. The hardest naturally occurring substance on Earth
13. The tilt and revolution of the Earth around the Sun causes four natural divisions termed ____
15. Thinnest layer of the Earth

WORD LIST

- | | | | | |
|-------------|------------|---------------------|---------------------|---------|
| core | orbit | sedimentary | chemical weathering | diamond |
| crust | greenhouse | metamorphic | erosion | talcum |
| lithosphere | nitrogen | igneous | minerals | seasons |
| rotation | volcanoes | physical weathering | streak | |
| | tides | | | |



ACROSS

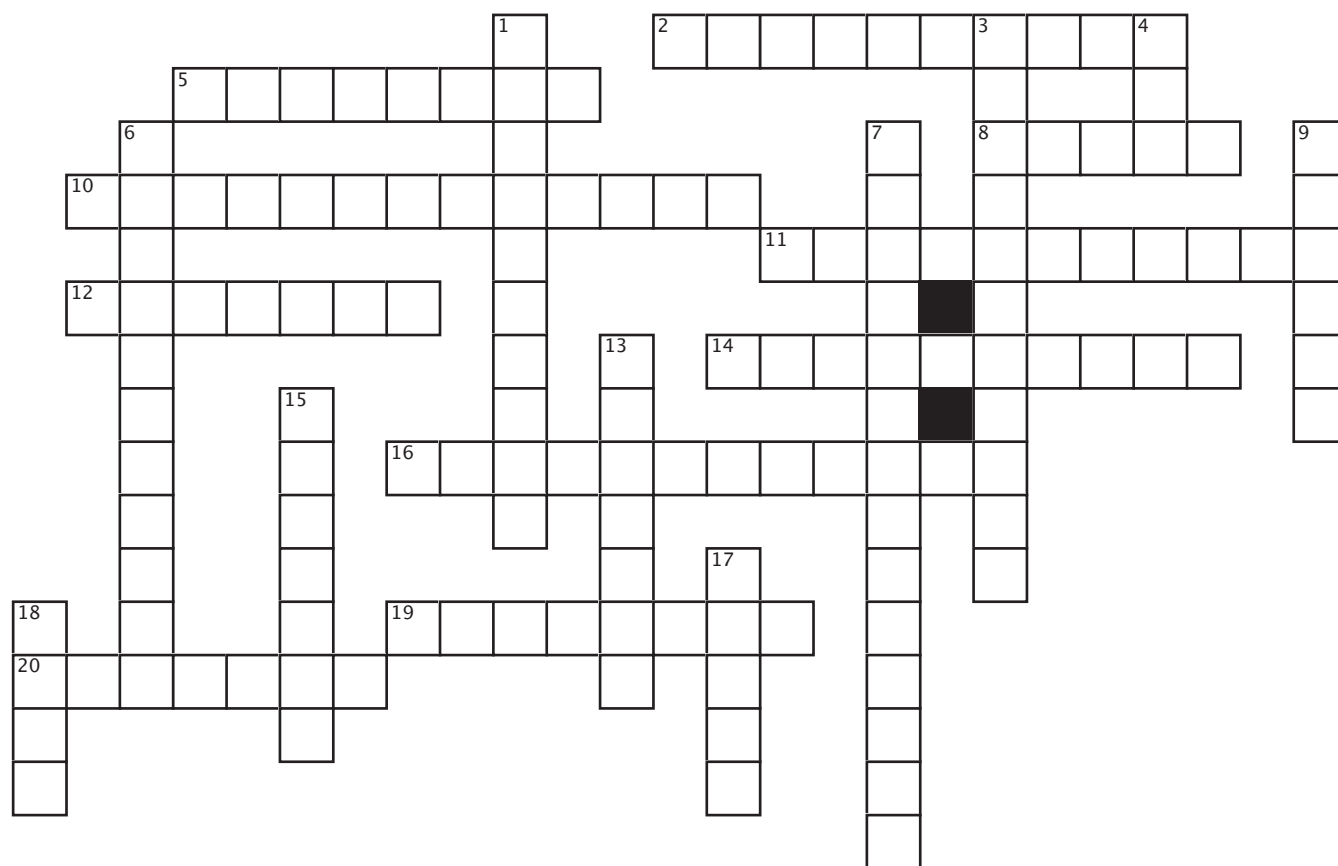
1. The process of changing a liquid to a gas
4. Caused by large electrical field differences within clouds or between the air and the ground
6. Word used to describe the typical weather of a region
8. The study of weather
11. Process that causes clouds to form
12. A rotating air mass caused by temperature differences between cold and warm fronts
13. Rotation of low pressure areas in the southern hemisphere
15. Instrument used for measuring heat
16. The result of an unstable air mass convecting violently in a humid layer
17. Weather map line with solid black triangles
18. Most important greenhouse gas in the atmosphere
19. A line on a weather map connecting points of equal atmospheric pressure
20. Used to measure atmosphere pressure

DOWN

2. Used to measure wind speed and direction
3. Flow of air from a hot region to a cold region causes this
5. Most abundant gas in Earth's atmosphere
7. Any form of water that falls onto the Earth's surface
9. Forms when an air mass cools over a cool land or ocean surface
10. Used to measure humidity
14. Process where air is recirculated vertically from hot to cold

WORD LIST

- | | | | | |
|---------------|--------------|---------------|------------|--------------|
| isobars | condensation | high pressure | nitrogen | water vapour |
| precipitation | wind | convection | barometer | climate |
| meteorology | thunderstorm | evaporation | hygrometer | anemometer |
| cold front | cyclone | clockwise | lightning | thermometer |



ACROSS

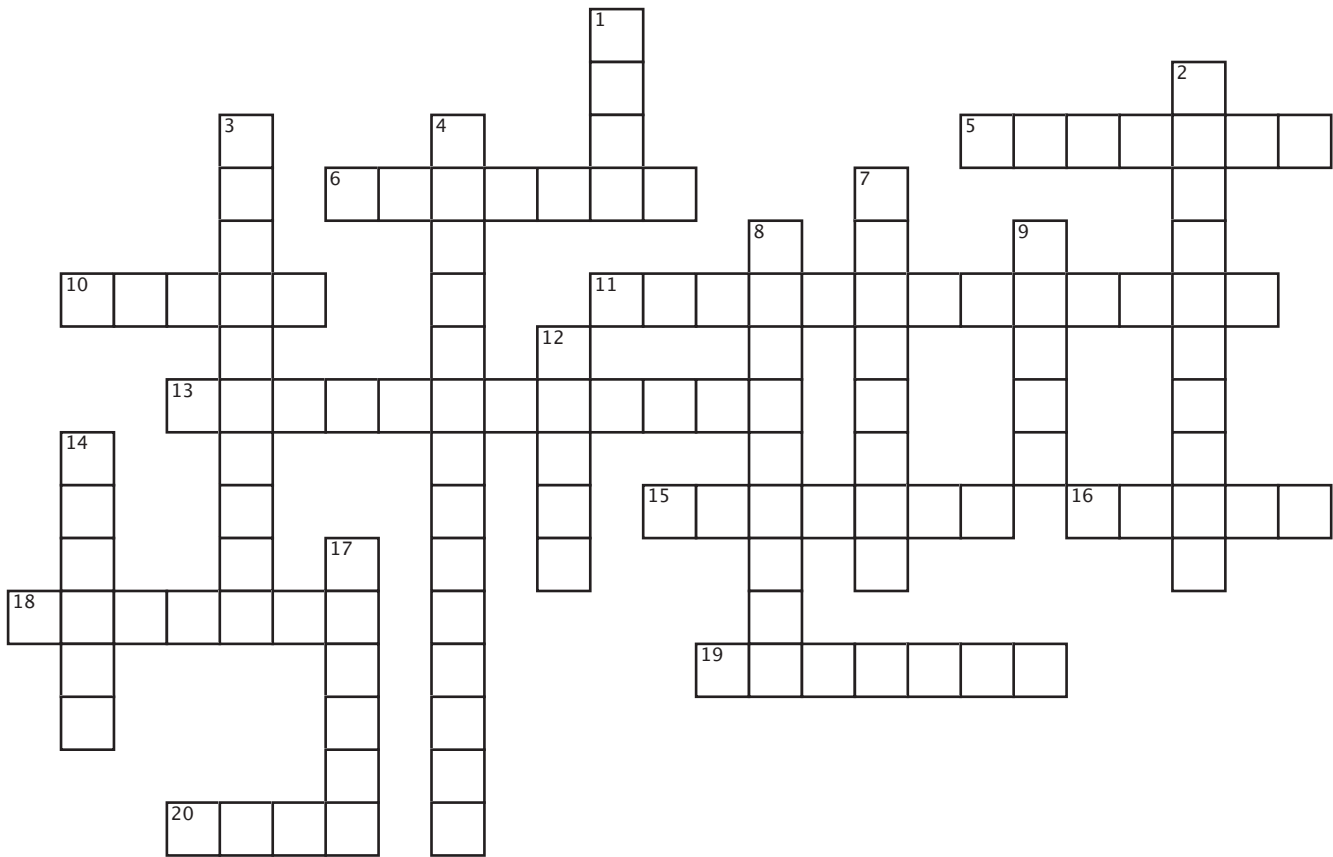
2. The units used to measure the height of the atmosphere
5. Most abundant gas in the atmosphere
8. Atmospheric layer that prevents harmful radiation from penetrating Earth's surface
10. Gas that traps heat
11. Name given to the study of the atmosphere
12. Pollutant gas derived from humans, animals, burning coal, etc.
14. Layer of atmosphere that reflects radiation
16. Layer between troposphere and mesosphere
19. Made by gases dissolved in moisture
20. A gaseous suspension of fine solid or liquid particles

DOWN

1. The middle layer of the atmosphere
3. The lowest layer of the atmosphere
4. Energy source that moves air masses
6. A naturally occurring greenhouse gas
7. Noxious gases made from nitrogen that come from vehicle exhausts
9. Gas produced by green plants
13. The term used for burning space debris and asteroid material
15. Very rare atmospheric gas; home of Superman
17. Aerosols, air molecules make this energy source scatter
18. Causes decreased visibility as the result of car exhausts

WORD LIST

- | | | | | |
|----------------|--------------|--------------|-----------------|------------|
| nitrogen | ionosphere | Krypton | aerosol | methane |
| oxygen | mesosphere | meteorology | nitrogen oxides | meteors |
| troposphere | stratosphere | water vapour | acid rain | light |
| carbon dioxide | ozone | haze | Sun | kilometres |



ACROSS

5. A planet with a ring of ice particles, named after the Roman god of water
6. Force responsible for causing tides
10. A planet that has liquid water and a breathable atmosphere
11. Name of the person who calculated the path of planets around the Sun
13. Name of the shadow when the Moon is between the Earth and the Sun
15. Name of a comet that reappears every 76 years
16. Name of the system that includes the Sun and all objects that orbit it
18. The largest planet in our solar system
19. The planet that is closest to the Sun
20. The planet named after the Roman god of war

DOWN

1. As the Earth orbits the Sun, seasons occur; they occur because of Earth's ____
2. The most likely source of many short-period comets
3. Direction in which the Earth rotates about its axis
4. Scientist who was punished by society for stating that the Sun was the centre of the universe
7. Radiation belt surrounding Earth
8. A body that orbits a planet
9. This planet has an atmosphere of sulfuric acid and carbon dioxide
12. The planet named after the Roman god of the dead
14. The sixth planet from the Sun
17. The planet named after the earliest supreme god in Greek mythology

WORD LIST

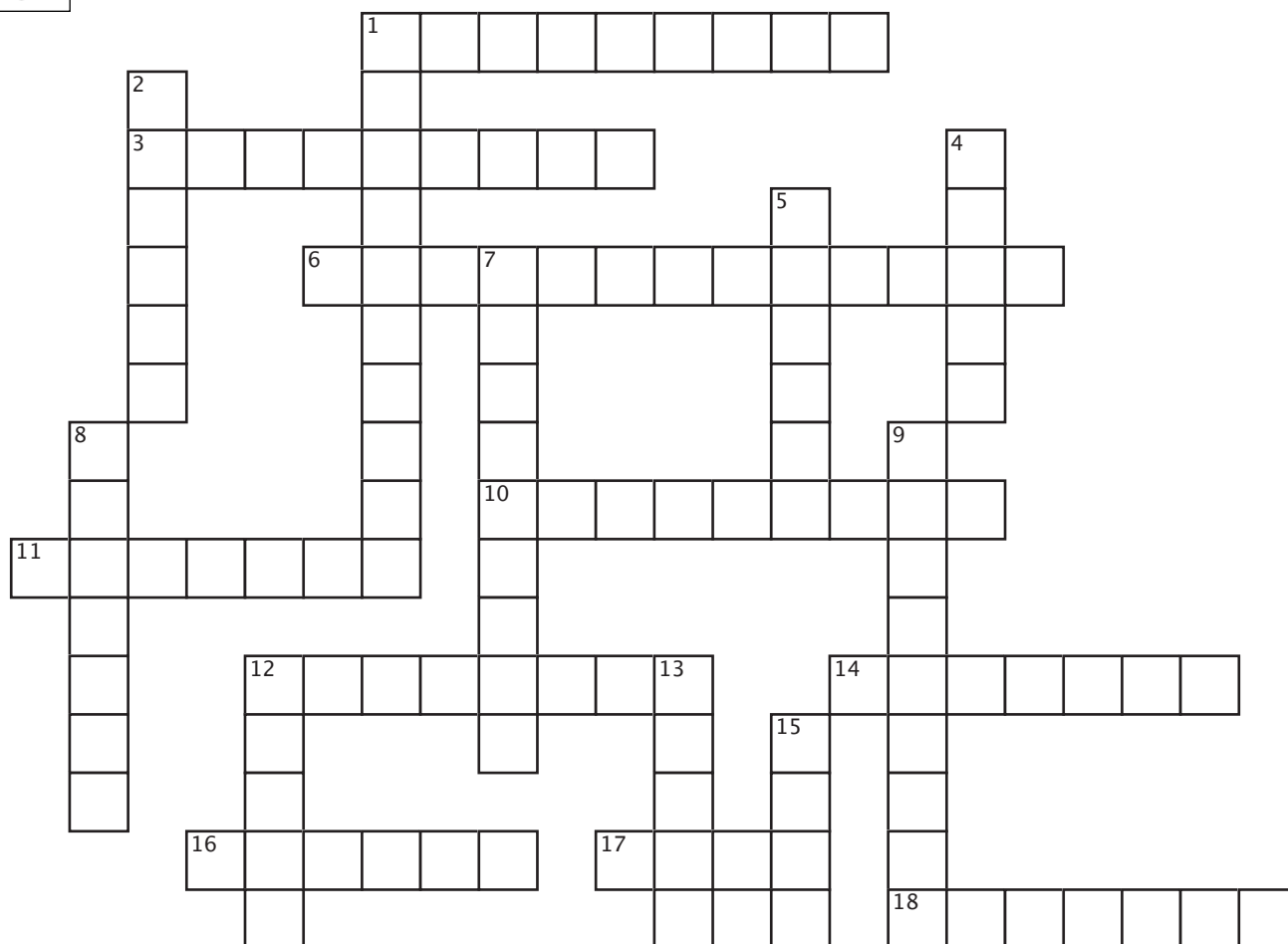
solar
 Mercury
 Earth
 Jupiter

Kuiper Belt
 Van Allen
 Johanne Kepler
 Galileo Galilei

west to east
 solar eclipse
 gravity
 tilt

satellite
 Halleys
 Uranus
 Saturn

Neptune
 Venus
 Mars
 Pluto



ACROSS

1. Not a science; used to create horoscopes
3. Earliest known calendar, made by people living in north Africa near the Mediterranean Sea
6. The Southern Cross is one of these
10. A unit of distance in space
11. Largest planet in our solar system
12. When the Earth is at its furthest or shortest distance from the Sun
14. Theory of how our universe began
16. A unit of measurement equivalent to 3.25 light years
17. A luminous mass of gas that is generated by nuclear reactions
18. Equal length of day and night

DOWN

1. A scientist who studies stars and galaxies
2. A diffuse mass of interstellar dust or gas or both, visible as luminous patches or areas of darkness
4. Natural satellites
5. A large-scale aggregate of stars, gas and dust
7. A natural or artificial body that orbits a planet
8. Name of the first man-made satellite
9. Another word for size and brightness
12. System of planets around a star
13. Third rock from the Sun
15. Planet named after the Roman god of war

WORD LIST

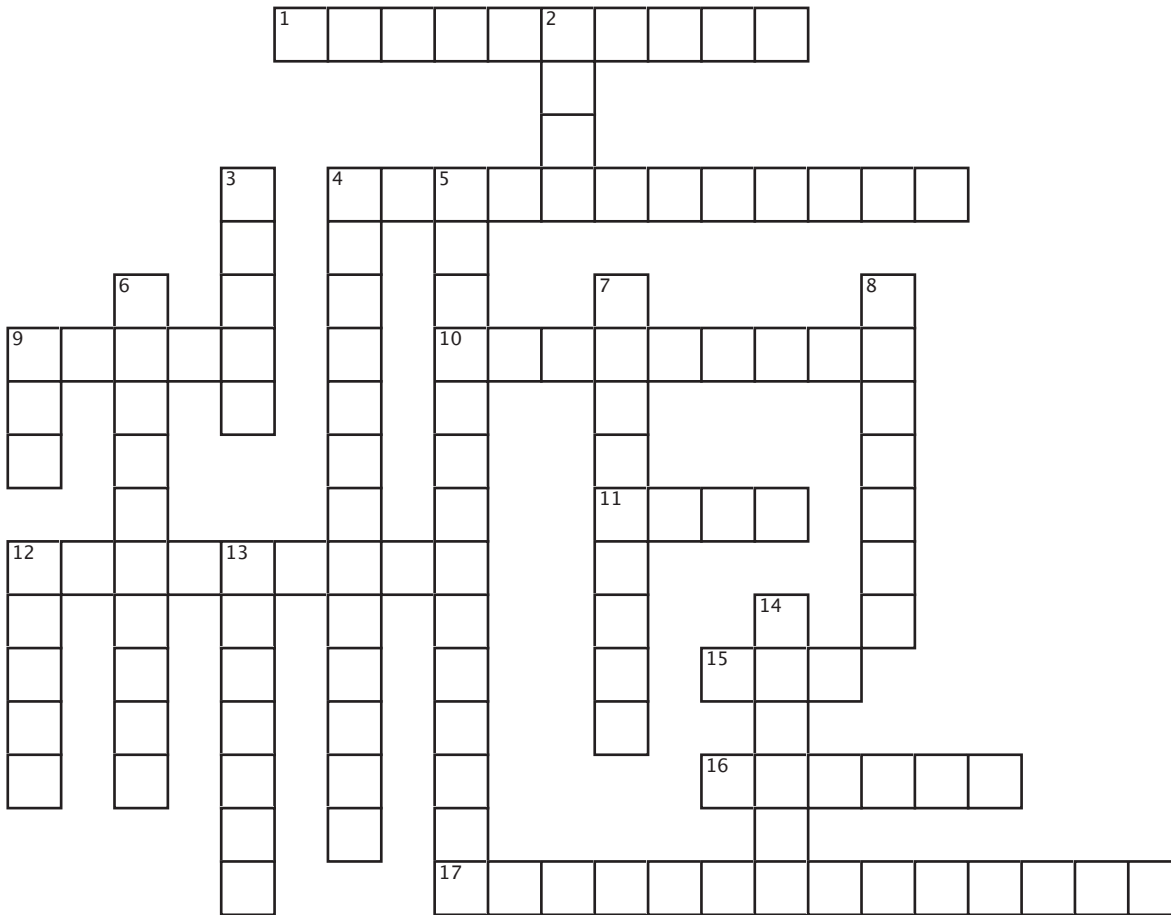
big bang
 solar
 galaxy
 nebula

constellation
 light year
 star
 parsec

satellite
 equinox
 astrology
 Earth

Jupiter
 astronomer
 solstice
 Egyptians

Sputnik
 magnitude
 moons
 Mars



ACROSS

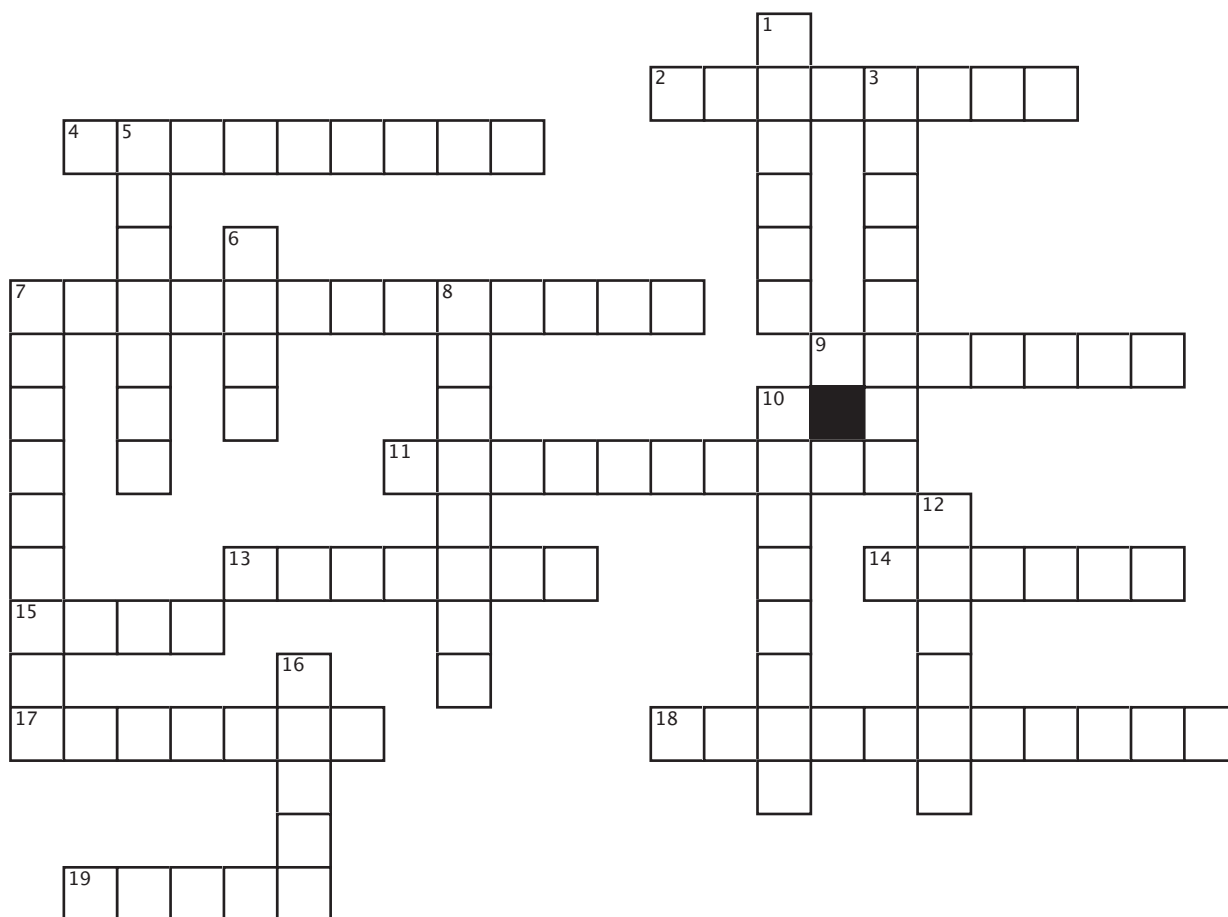
1. Less than 1% of this resource is available for human use
4. The proper preservation and management of natural resources
9. Destruction of this gas layer permits increased electromagnetic radiation to reach Earth's surface
10. Form of energy that comes from any natural resource and can be replenished naturally over time
11. A solid fossil fuel
12. A contaminant of the environment
15. The source of photovoltaic energy
16. Process essential for the production of goods
17. To keep in existence

DOWN

2. A renewable energy form derived from the oceans
3. The reaction of a fossil fuel produces carbon dioxide and what other greenhouse gas?
4. Product of fossil fuel combustion
5. Gases emitted from motor vehicles that increase global warming; not carbon dioxide or water
6. Coal is an example of this type of energy resource
7. An action all humans should carry out
8. A gas made from waste materials
9. Substance obtained from the ground to make synthetic materials
12. A recyclable resource made from cellulose
13. Fuel used in nuclear fission
14. Type of nuclear energy process that would be almost non-polluting

WORD LIST

- | | | | | |
|----------------|-------------|-----------------|-------------|----------------|
| carbon dioxide | mining | fusion | ozone | coal |
| water | fossil fuel | renewable | fresh water | conservation |
| pollution | oil | uranium | methane | paper |
| Sun | wave | nitrogen oxides | recycling | sustainability |



ACROSS

2. A soft mineral containing fluorine that fluoresces under ultraviolet light
4. Type of rock that produces sand grains when rubbed
7. The process of consolidating sediments and converting them into rock
9. An extremely hard, highly refractive crystalline form of carbon
11. Concentrated form of uranium oxide
13. A common, coarse-grained, light-coloured, hard igneous rock used in monuments and for building
14. A hard, dense, dark volcanic rock
15. A mineral made from hydrated silicon dioxide and used in jewellery making
17. A highly prized, glassy, green mineral made from beryllium, aluminium and silicon oxides and used in jewellery making
18. Another name for ruby
19. A rock made from layers of fine-grained sediments deposited in still water

DOWN

1. A mineral with a hardness of 7 which is colourless to white
3. A fine-grained extrusive volcanic rock, usually has flow lines
5. A common mineral consisting of calcium fluoride phosphate or calcium chloride phosphate
6. A mineral that has cleaves into very thin leaves, and is electrically resistant
7. This rock is made from ancient coral reefs
8. A grey, fine-grained volcanic rock
10. A gemstone that is usually blue but may be any colour except red
12. A coarse-grained, intrusive igneous rock
16. A fine-grained, grey metamorphic rock that splits into thin, smooth layers

WORD LIST

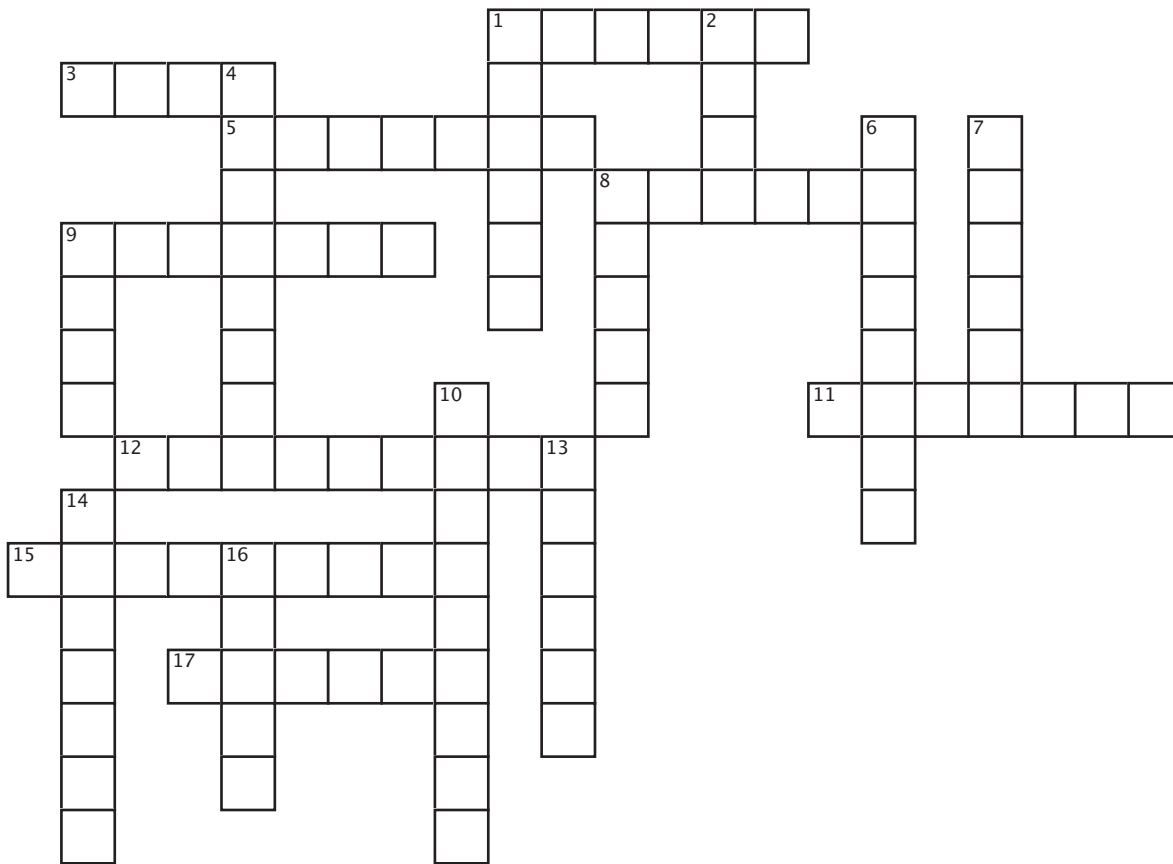
sandstone
 limestone
 andesite
 basalt

rhyolite
 slate
 granite
 gabbro

diamond
 lithification
 opal
 emerald

sapphire
 apatite
 fluorite
 quartz

mica
 yellowcake
 shale
 carborundum



ACROSS

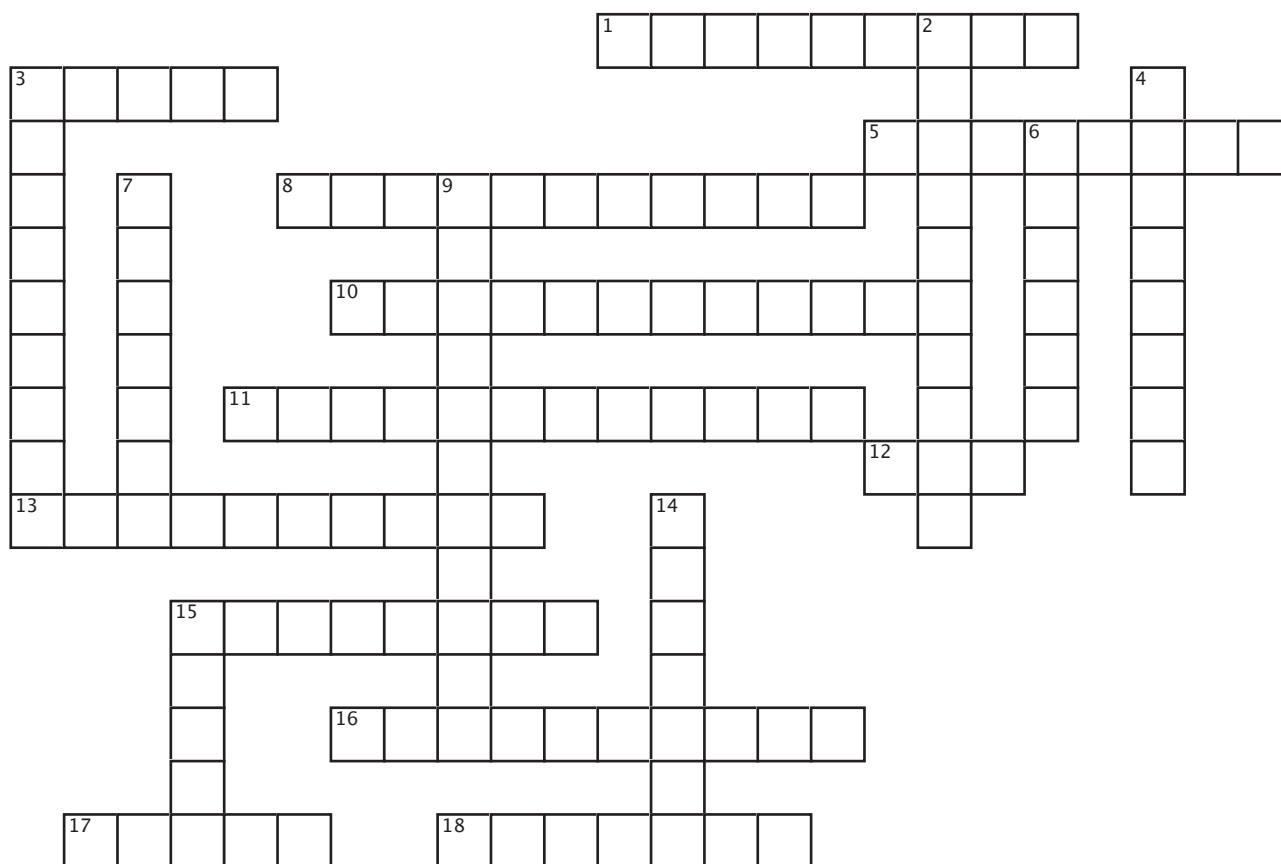
1. Igneous rock that floats in water and is used in abrasives and soaps
3. Metamorphic rock used in cosmetics, paper making and electrical insulation
5. This rock ore and its associated minerals are used in the manufacture of vehicles, inks and magnets
8. This yellow, non-metal element is derived from sulfide rocks and is used in fungicides, explosives, rubber manufacture and antibiotics
9. Attractive igneous rock used for table tops and ornamenting buildings
11. Hardest naturally occurring substance on Earth; made from carbon
12. Green igneous mineral used in X-ray machines, communication devices and jewellery; its main atom is the fourth atom in the Periodic Table
15. This rock is used in the construction and decoration of buildings
17. Rock used for road making

DOWN

1. Known as fool's gold, it is used in the manufacture of sulfuric acid and costume jewellery
2. A brown to black, shiny, sedimentary rock, used as fossil fuel
4. Reddish mineral that contains mercury
6. This mineral is used in pencils and greases. It is made from carbon atoms
7. This mineral is used in the making of plaster of Paris, fertilisers, soil conditioners and Gyprock
8. A sedimentary rock that can be used in brick manufacture and oil
9. This soft metal occurs in metamorphic and igneous rocks and is used in jewellery, medicine and the making of medals
10. This sedimentary rock is important in the making of cement
13. Metamorphosed limestone used in sculpting
14. This mineral is a form of calcium carbonate and is used in the manufacture of antacids, glass, paper, photography and waste treatment
16. Metamorphic rock used for floor tiles and roofing

WORD LIST

basalt	marble	calcite	coal	pumice
granite	talc	gypsum	diamond	sandstone
slate	gold	pyrite	sulfur	shale
limestone	beryllium	iron ore	graphite	cinnabar



ACROSS

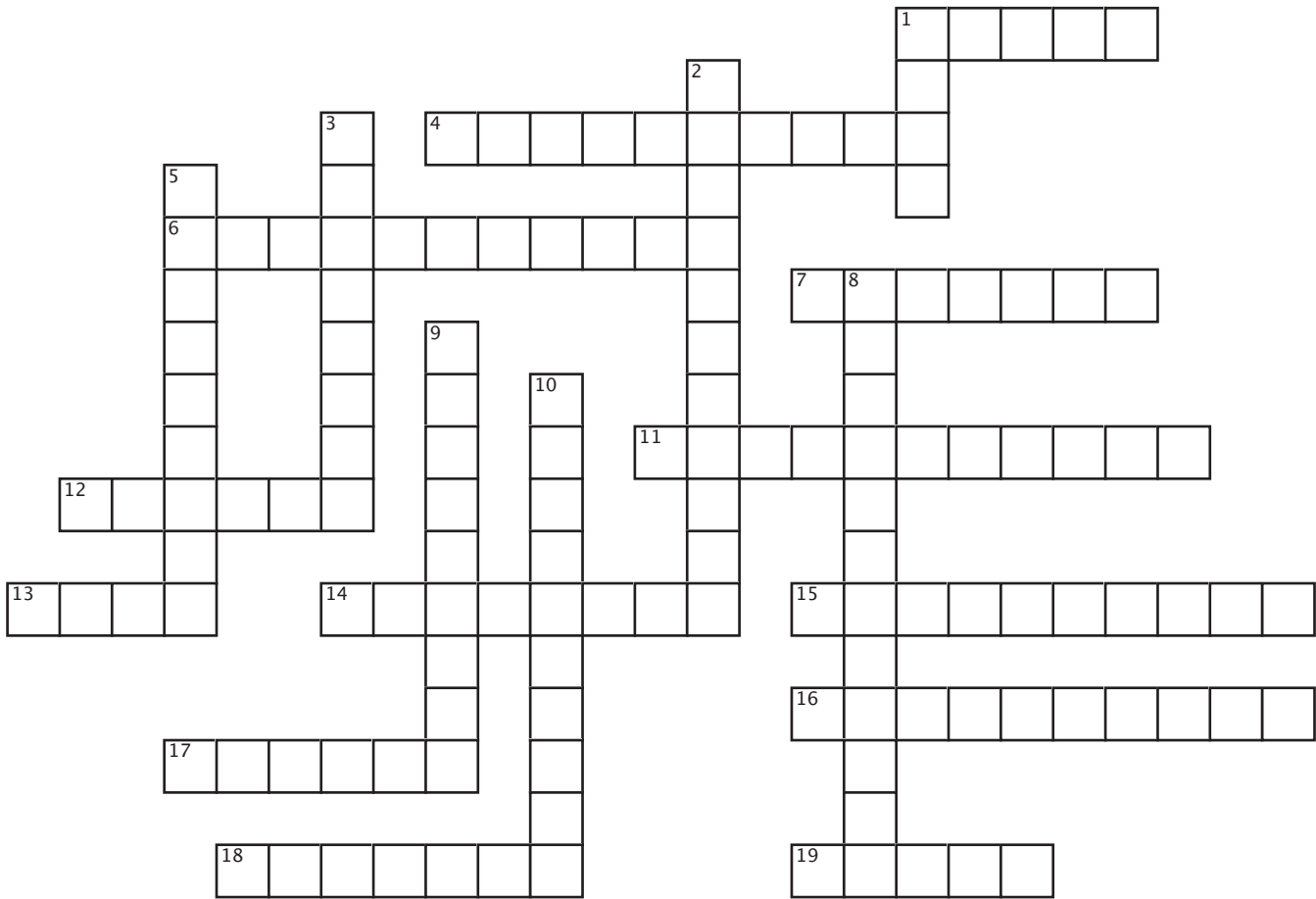
1. A lens for viewing distant objects with one eye
3. All living things are made from these
5. A thin, flexible layer of tissue covering surfaces, structures or organs of an animal or a plant
8. An organism made up of one cell
10. Method by which unicellular organisms reproduce
11. Organisms that consume their food
12. Common method of identifying organisms
13. Used to see microscopic things
15. Animal cells do not have this
16. Organisms that make their own food
17. Place where microscope slide is placed
18. Large structure in cells that contains hereditary material

DOWN

2. Inventor of the microscope
3. The watery component of a cell
4. Single celled organism
6. Another word for living
7. Another word for non-living
9. A chlorophyll pigmented structure in algae and green plants
14. A storage chamber within a cell
15. Tiny hair-like protrusion for propelling some micro-organisms

WORD LIST

* biotic	* nucleus	* unicellular	* cell wall	* cell division
* cells	* membrane	* stage	* Leewenhoek	* key
* monocular	* cytoplasm	* abiotic	* autotrophs	* bacteria
* microscope	* chloroplast	* vacuole	* heterotrophs	* cilia



ACROSS

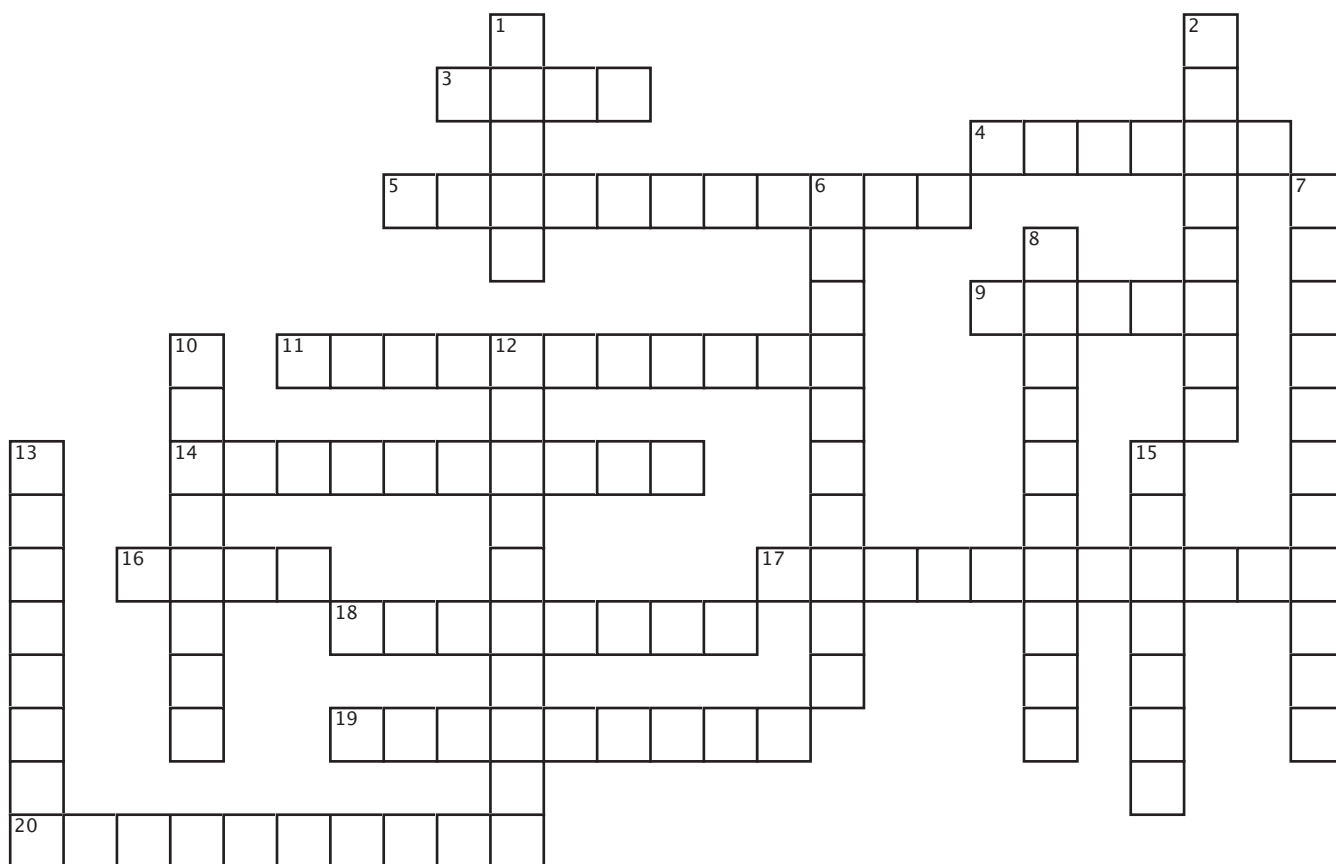
- 1. A group of immobile organisms that feed on decaying wood or other tissue
- 4. Platypus and echidna are well known examples
- 6. These are characterised by a thick, often spiny body with five arms extending from a central disk. They are found in rock pools.
- 7. Highest form of classification
- 11. All backboned animals are in this classification
- 12. The name of the kingdom that has chloroplasts and cell walls, and are not mobile
- 13. All birds are in this classification
- 14. Lizards, snakes and turtles belong to this class
- 15. Octopus and squid belong to this classification meaning 'head-foot' in Latin
- 16. Animals that have an exoskeleton, a segmented body and limbs in pairs
- 17. The largest marine animal belongs to this class
- 18. Another name for rats and mice
- 19. A group of species exhibiting similar characteristics as in Homo

DOWN

- 1. Cold-blooded vertebrates that swim
- 2. Collective word for prawns, lobsters, etc.
- 3. Humans, apes, monkeys and lemurs belong to this order
- 5. Of dolphins, porpoises and whales
- 8. Not a vertebrate
- 9. Tasmanian tiger belonged to this group
- 10. Frogs, toads and newts are in this classification

WORD LIST

- | | | | | |
|--------------|----------------|------------|---------------|--------------|
| * marsupial | * vertebrates | * kingdom | * aves | * cephalopod |
| * mammal | * invertebrate | * plants | * reptiles | * cetaceans |
| * monotremes | * crustaceans | * primates | * amphibians | * fish |
| * rodents | * arthropods | * fungi | * echinoderms | * genus |



ACROSS

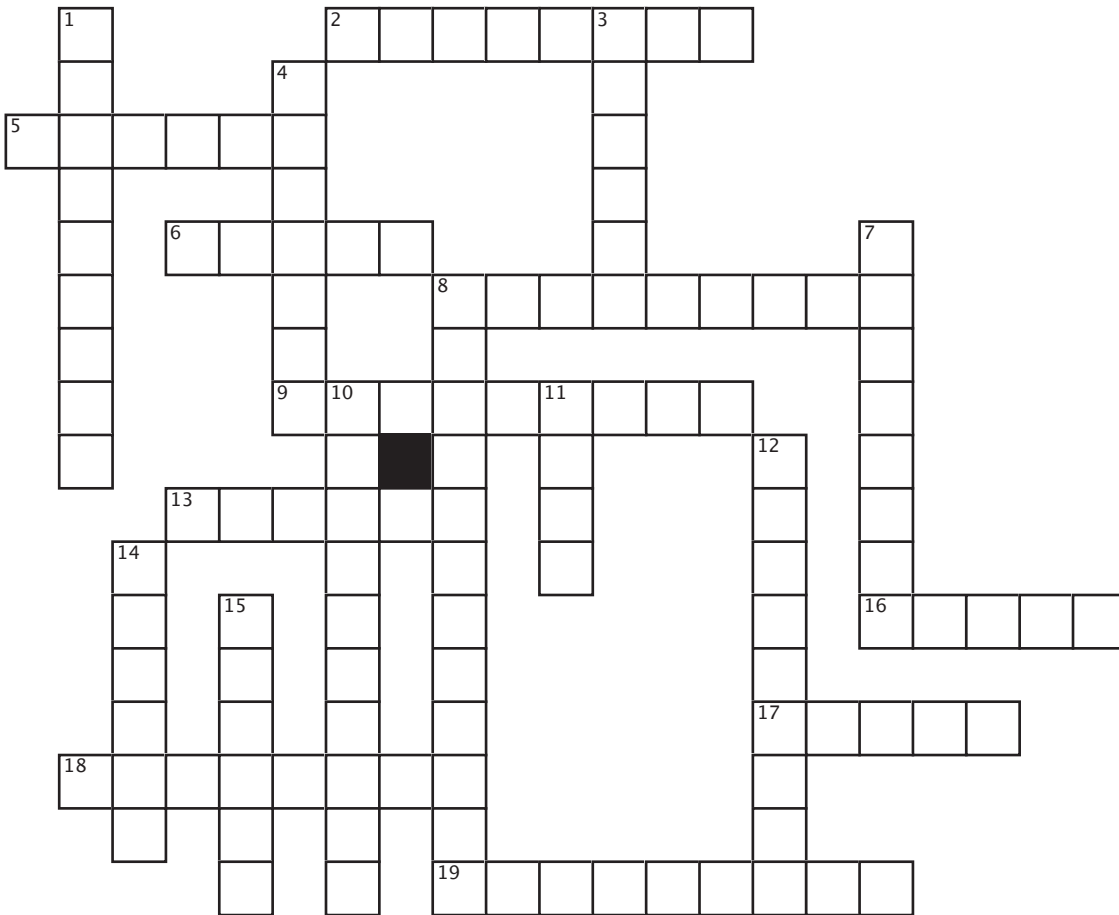
3. These animals have scales or tough skin, fins, live in water and breathe through gills
4. Name of monotreme young
5. Means warm-blooded
9. These organisms have no jointed covering, no shell and a long soft body
11. This class of molluscs is characterised by a prominent head and tentacles
14. These organisms have a vascular organ that develops in pregnancy, give birth to live young, produce milk and are homeothermic
16. The colour of an echidna's milk
17. Person who preserves animals for museum exhibits
18. Study of animal classification
19. Opposite of endotherm
20. These are ectothermic, have moist skin, no scales and lay eggs

DOWN

1. Living descendants of dinosaurs
2. These usually aquatic organisms have no backbone but a shell and a soft body
6. These animals have a pouch with teats where the young are fed and carried. They are homeothermic and have hair or fur
7. This process occurs in all animals: the conversion of sugars and oxygen to carbon dioxide and water
8. These animals are warm-blooded with a high metabolism, have hair, produce milk to feed their young and lay eggs
10. These ectotherms have scaly skin and lay eggs
12. These animals have many segmented limbs, and a segmented, covered body
13. These organisms have a tough, soft outer covering, have tentacles and are usually marine
15. All these endothermic animals have fur or hair and suckle their young

WORD LIST

- | | | | |
|--------------|--------------|---------------|---------------|
| * pink | * birds | * cnidaria | * endothermic |
| * monotremes | * puggle | * fish | * ectotherm |
| * placentals | * amphibians | * cephalopods | * taxidermist |
| * marsupials | * reptiles | * worms | * respiration |
| * mammals | | | |



ACROSS

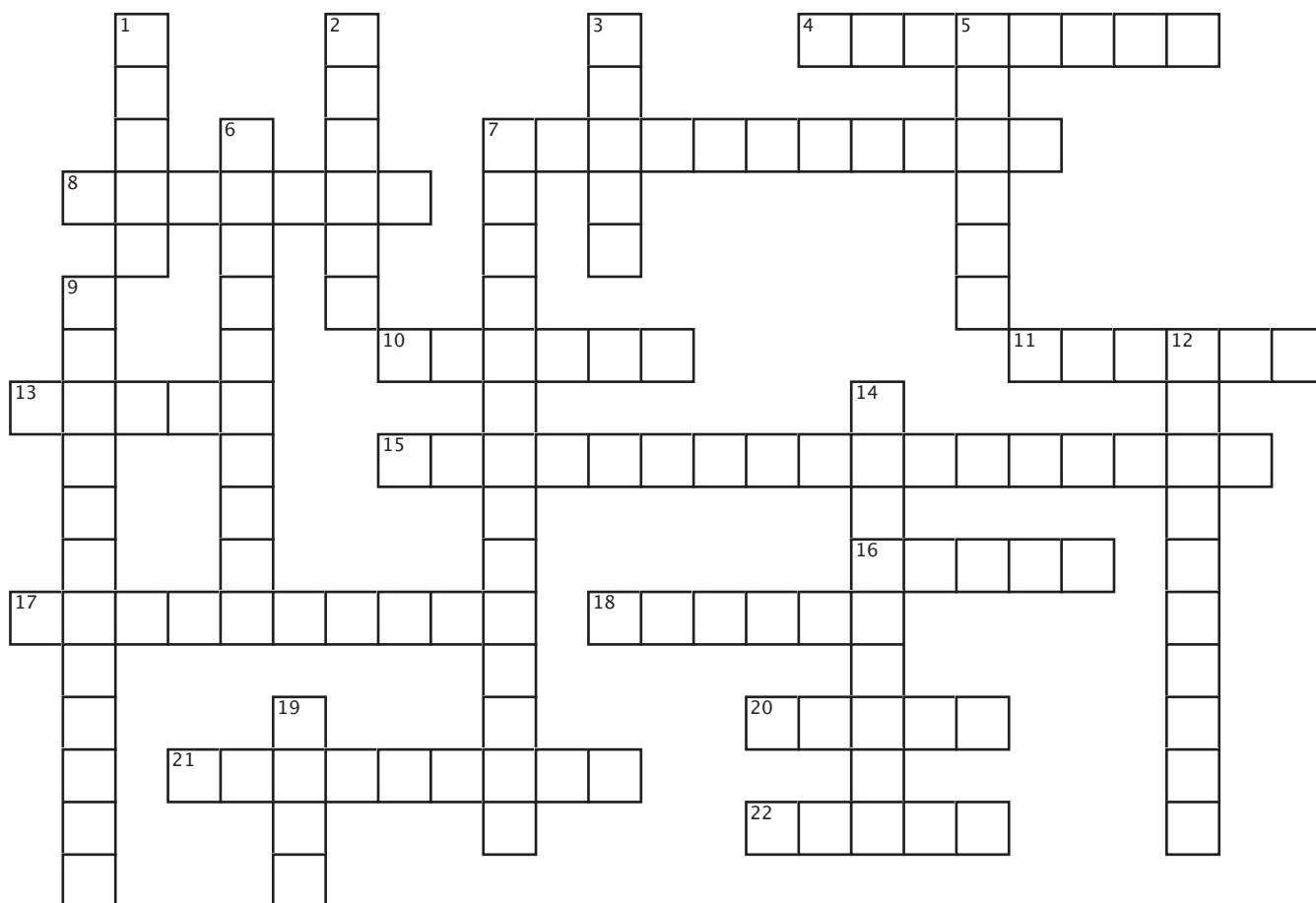
2. Kingdom of unicellular, eukaryotic micro-organisms
5. Another word for maggots
6. Name of dragonfly young
8. Microscopic organism with antennae that swims
9. Long tubular organism found in gardens
13. An edible bivalve that attaches itself to rocks
16. Small, freshwater, lobster-like organism
17. Worm-like, blood-sucking organism
18. Worm-like, flat organism with a triangular head.
Usually glides along the bottom of creeks
19. Makes honey but does not sting

DOWN

1. Found hovering over creeks and aquatic plants
3. An aquatic, colonial organism that has a porous skeleton made from silicon or carbonate material
4. Small, white or yellow ant that eats cellulose
7. These are commonly seen around flowers during spring and early summer
8. Beetle that rows across water
10. Insects belong to this phylum
11. An organism with a slender body, usually orange and black, with a constricted abdomen, two pairs of wings, and mouths adapted for biting or sucking. The female has a sting
12. Deadly Australian spider
14. Limestone is made from these organisms when they die
15. A small, terrestrial, grey crustacean that has a convex, segmented, flexible body

WORD LIST

- | | | | | |
|--------------|------------|--------------|-----------------|-------------|
| * native bee | * termite | * arthropods | * water boatman | * nymph |
| * earthworm | * sponge | * funnel web | * yabby | * larvae |
| * dragonfly | * corals | * oyster | * leech | * flat worm |
| * slater | * marchfly | * wasp | * water flea | * protista |



ACROSS

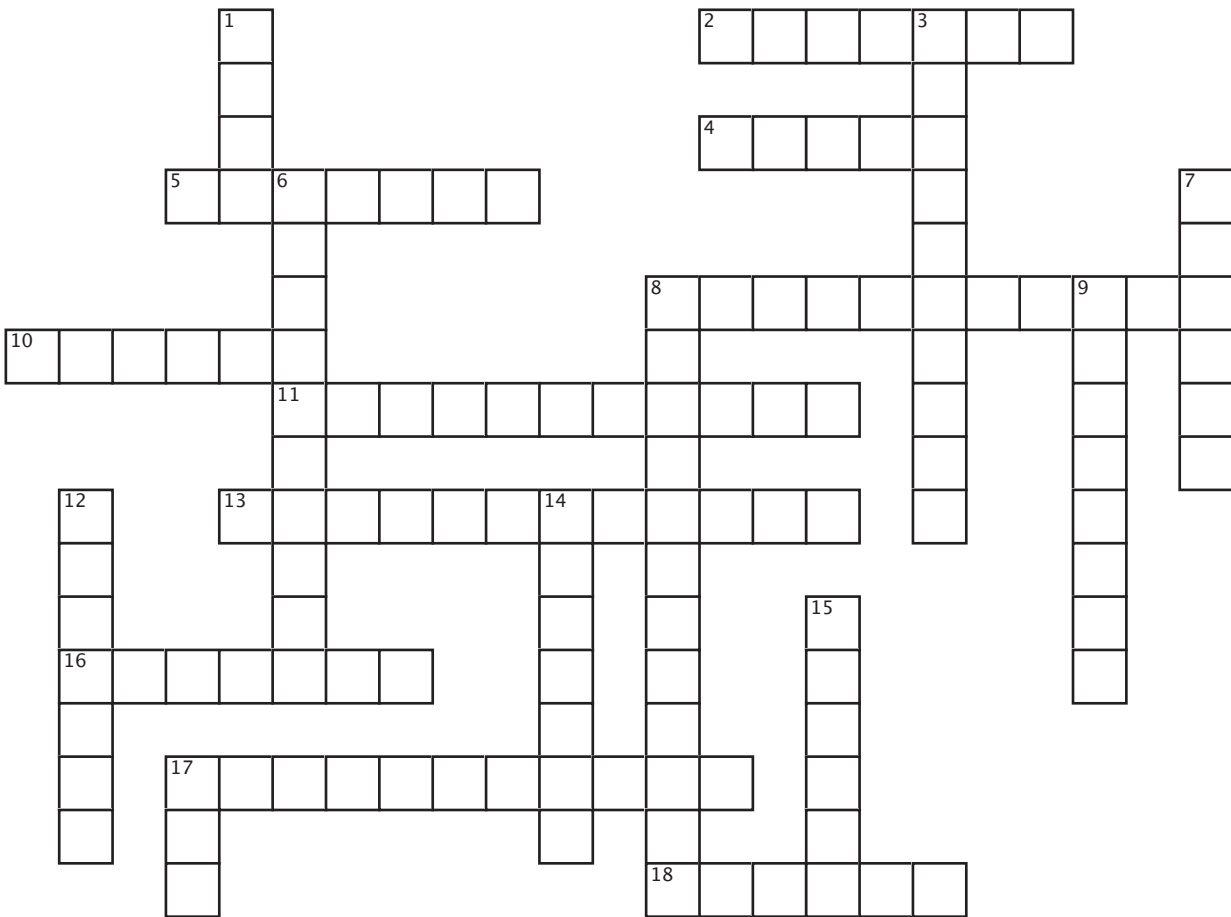
4. Originally thought to be a swimming hoax
7. Spiky lizard from the Simpson desert
8. Small hopping kangaroo
10. A reptile that climbs trees, swims, runs quickly and can stand upright
11. Large, slow possum
13. Native dog
15. Reptile with a ruffle that lives in northern Australia
16. Delicate lizard that can walk on glass
17. Snake that is black on top and red underneath
18. Ocean-going reptile that lays eggs in the beaches of north Queensland
20. Sleeps in a tree almost 20 hours a day
21. Ancient-living reptile found in northern Australia
22. Large-eared, small, desert-living bandicoot

DOWN

1. Grey and pink parrot
2. Close relative to the koala
3. Native cat
5. The most dangerous snake anywhere!
6. Lizard with dark coloured tongue
7. Noisy fighters from Tasmania
9. Small upright bird that swims
12. Small, gold and black toadlet that lives in the Snowy Mountains
14. Australia's largest eagle
19. Name of baby kangaroo

WORD LIST

- | | | | |
|-----------------------|--------------|-------------------|-----------------|
| * blue tongue | * bilby | * Tasmanian devil | * fairy penguin |
| * thorny devil | * koala | * joey | * wedge tail |
| * turtle | * wombat | * galah | * dingo |
| * taipan | * corroboree | * platypus | * cuscus |
| * frill necked lizard | * quoll | * crocodile | |
| * red bellied | | | |
| * goanna | | | |
| * gecko | | | |
| * wallaby | | | |



ACROSS

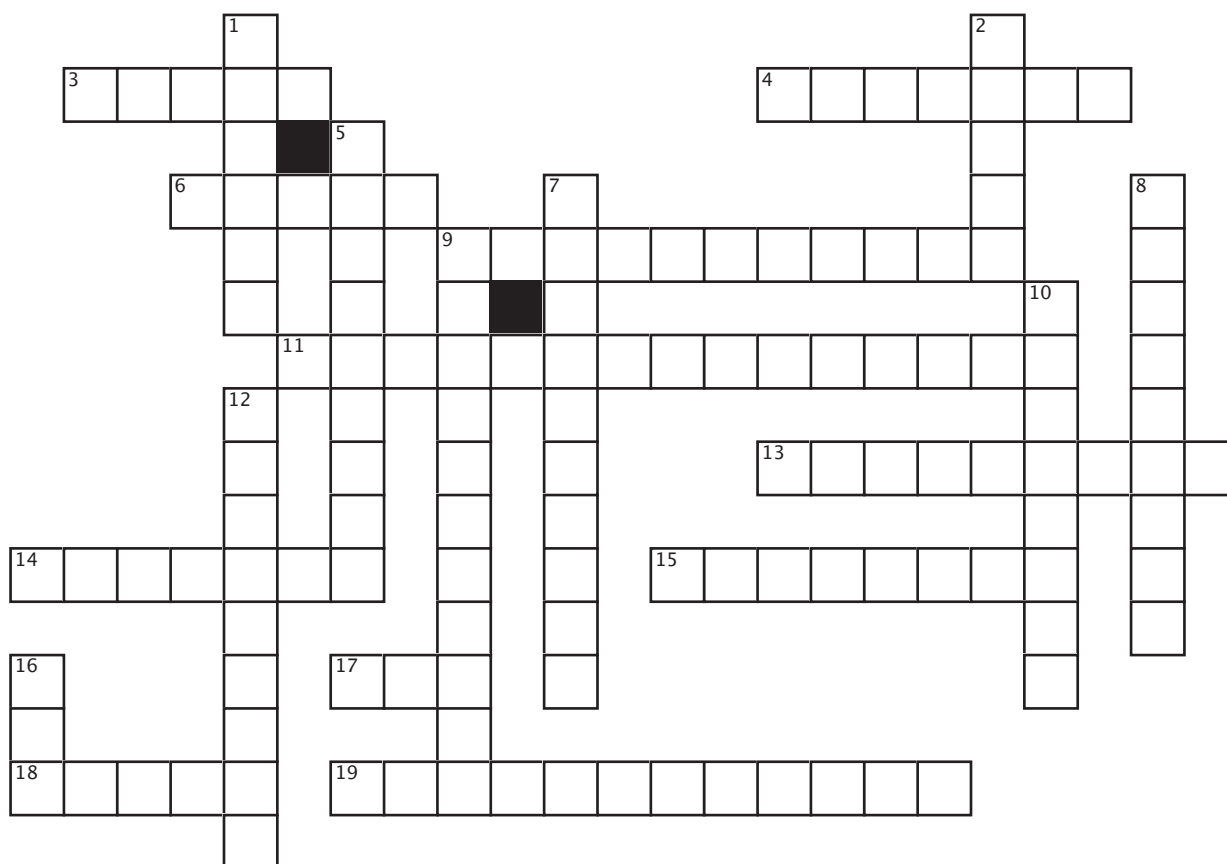
- 2. Tissue that makes bones move
- 4. Male reproductive cell
- 5. Cell reproduction process that transfers genetic information
- 8. Human cells have 46 of these
- 10. Fertilised ovum
- 11. Energy powerhouse within a plant cell
- 13. Energy powerhouse within an animal cell
- 16. Duplication of a cell in animals
- 17. Shape of DNA
- 18. Cell used in conducting electrical impulses

DOWN

- 1. A hereditary unit that occupies a specific location on a chromosome
- 3. Third line of defence
- 6. Opposite of non-infectious
- 7. Male gametes are made in these
- 8. Growth in multicellular organisms is a function of —
- 9. A sudden change in the DNA sequence
- 12. A specific substance that activates target cells
- 14. Female equivalent to testes
- 15. Abnormal cell function may cause this disease
- 17. Hereditary polymer in nucleus

WORD LIST

- | | | | | |
|----------------|---------------|-----------------|----------------|-----------|
| * meiosis | * DNA | * ovaries | * lymphocyte | * neuron |
| * mitosis | * gene | * cell division | * infectious | * hormone |
| * mitochondria | * chromosomes | * mutation | * double helix | * sperm |
| * chloroplast | * testes | * cancer | * muscles | * zygote |



ACROSS

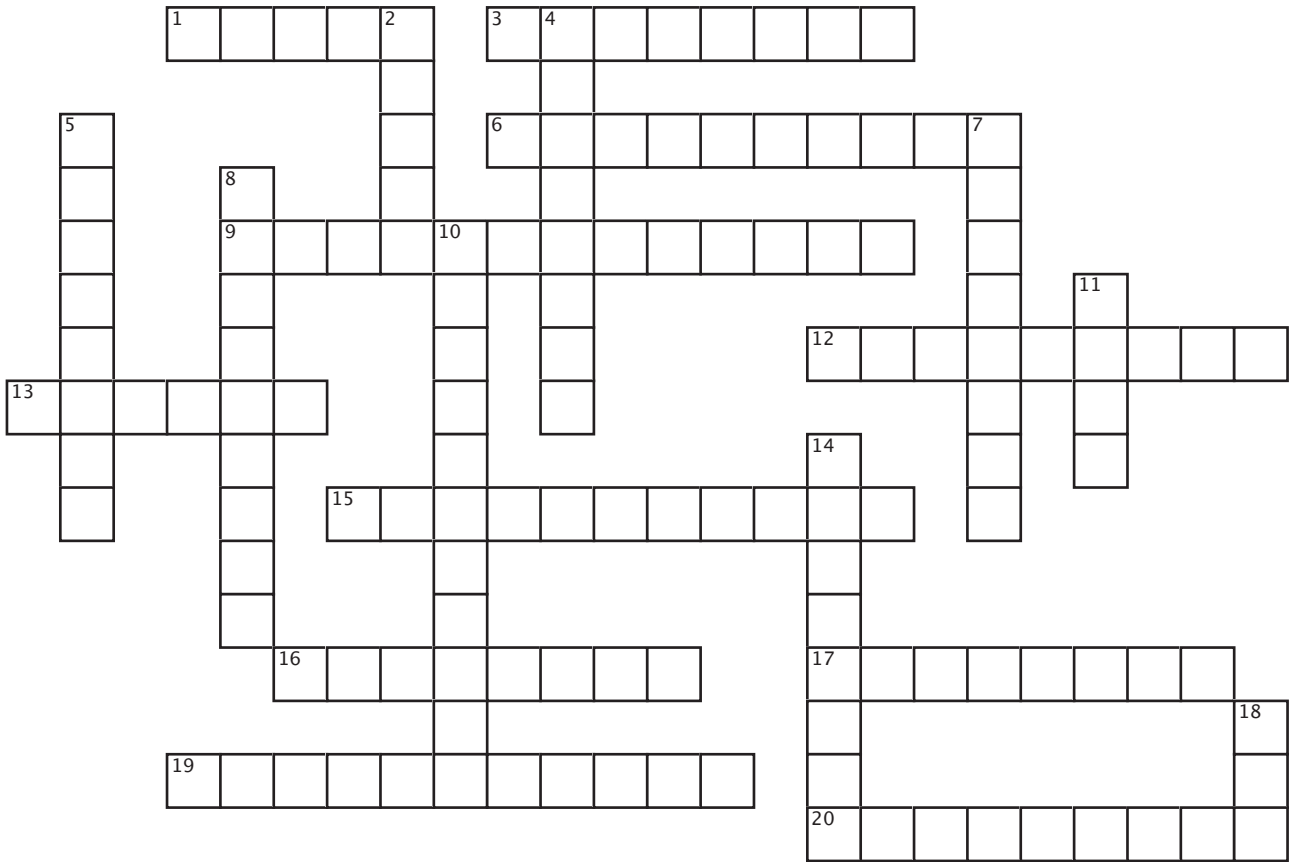
3. The number of dimensions a cell has
4. Part of a cell that contains the hereditary material and controls the cell's functions
6. The hereditary units found in DNA
9. Threadlike structures that carry genetic information and are found in the nucleus
11. Name of the double layer covering the nucleus
13. Long, hair-like structure used for cell propulsion
14. Place for storing substances in a cell
15. Very small cells with no nucleus which may cause illness
17. Found as a double helix
18. Minute holes in the cell membrane
19. The cell's powerhouses

DOWN

1. A very large cell
2. All organisms are made from these
5. Cell walls are made from this substance
7. Components of a cell
8. Basic shape of a cheek cell
9. A structure only found in green plant cells
10. Animal cells don't have this structure
12. Jelly-like liquid found in the body of the cell
16. Plant vacuole is filled with cell ____

WORD LIST

- | | | | | |
|---------------|---------------|-------------|----------------|--------------------|
| * nucleus | * pores | * three | * cytoplasm | * cells |
| * chromosomes | * cell wall | * irregular | * mitochondria | * sap |
| * DNA | * genes | * bacteria | * hen egg | * nuclear membrane |
| * vacuole | * cellulose | * flagellum | * organelles | |
| | * chloroplast | | | |



ACROSS

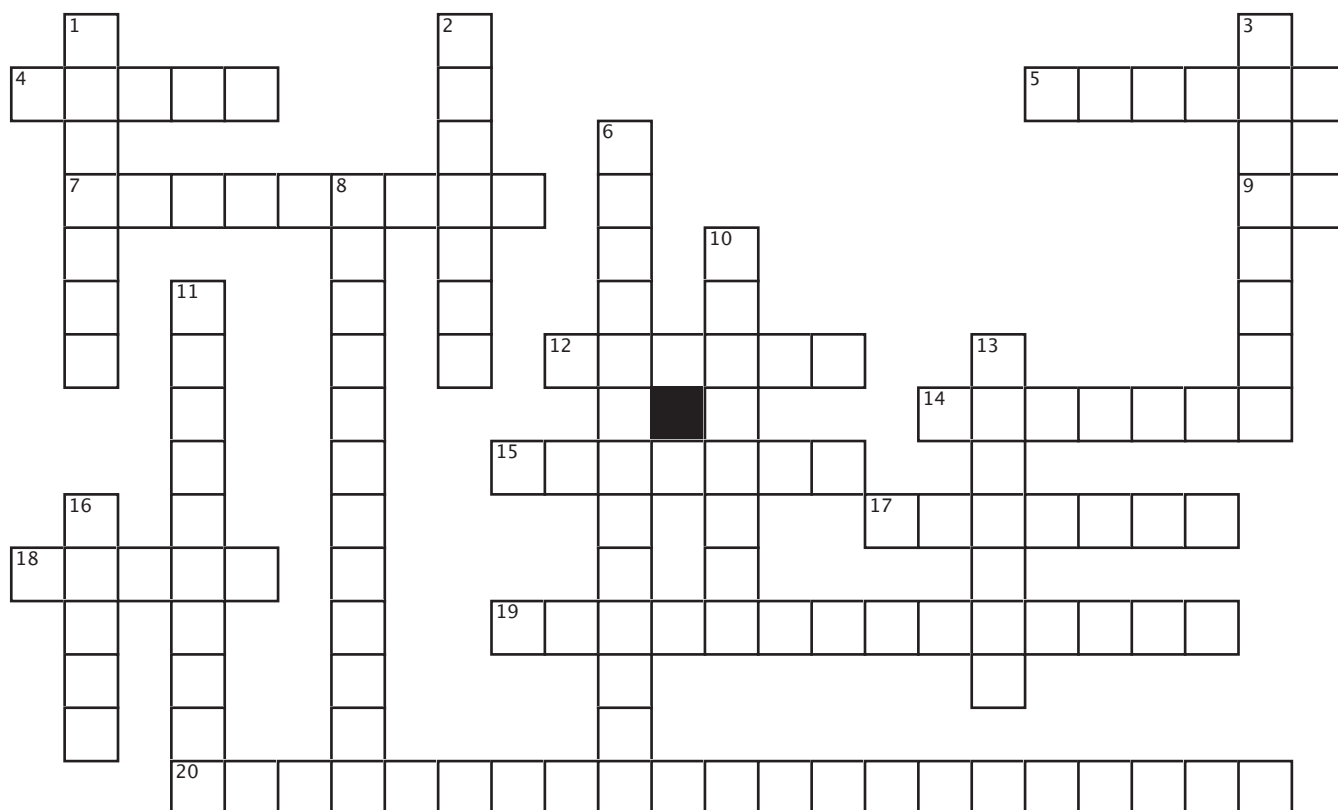
1. Arms that hold slide in place
3. Simple microscope magnification is usually ____
6. Type of stage that can move the slide around using two turning knobs
9. Lens closest to object being looked at
12. Microscope with two eye pieces
13. Rotating part that holds several lenses
15. Highest magnification with a simple compound microscope
16. Microscope used for seeing specimen components greater than 10 000 times
17. Adjustment that prevents lens from hitting the slide
19. Component that illuminates the object
20. Rotating disc under stage with different size holes

DOWN

2. The flat platform where slides are put
4. Lens you look through
5. Microscope system that has two lens systems for magnification
7. Always used at the beginning to view a specimen
8. Lens used to focus light onto specimen
10. Large knob on side of microscope
11. Part that connects eyepiece to objective lens
14. When specimen is viewed, it is reversed and ____
18. Part that supports the tube and connects it to the base

WORD LIST

- | | | | | |
|------------------|----------------|-------------|-------------|----------------|
| * eyepiece | * turret | * condenser | * low power | * mechanical |
| * objective lens | * coarse focus | * diaphragm | * compound | * ten times |
| * light source | * stage | * rack stop | * electron | * binocular |
| * tube | * clips | * inverted | * arm | * one thousand |



ACROSS

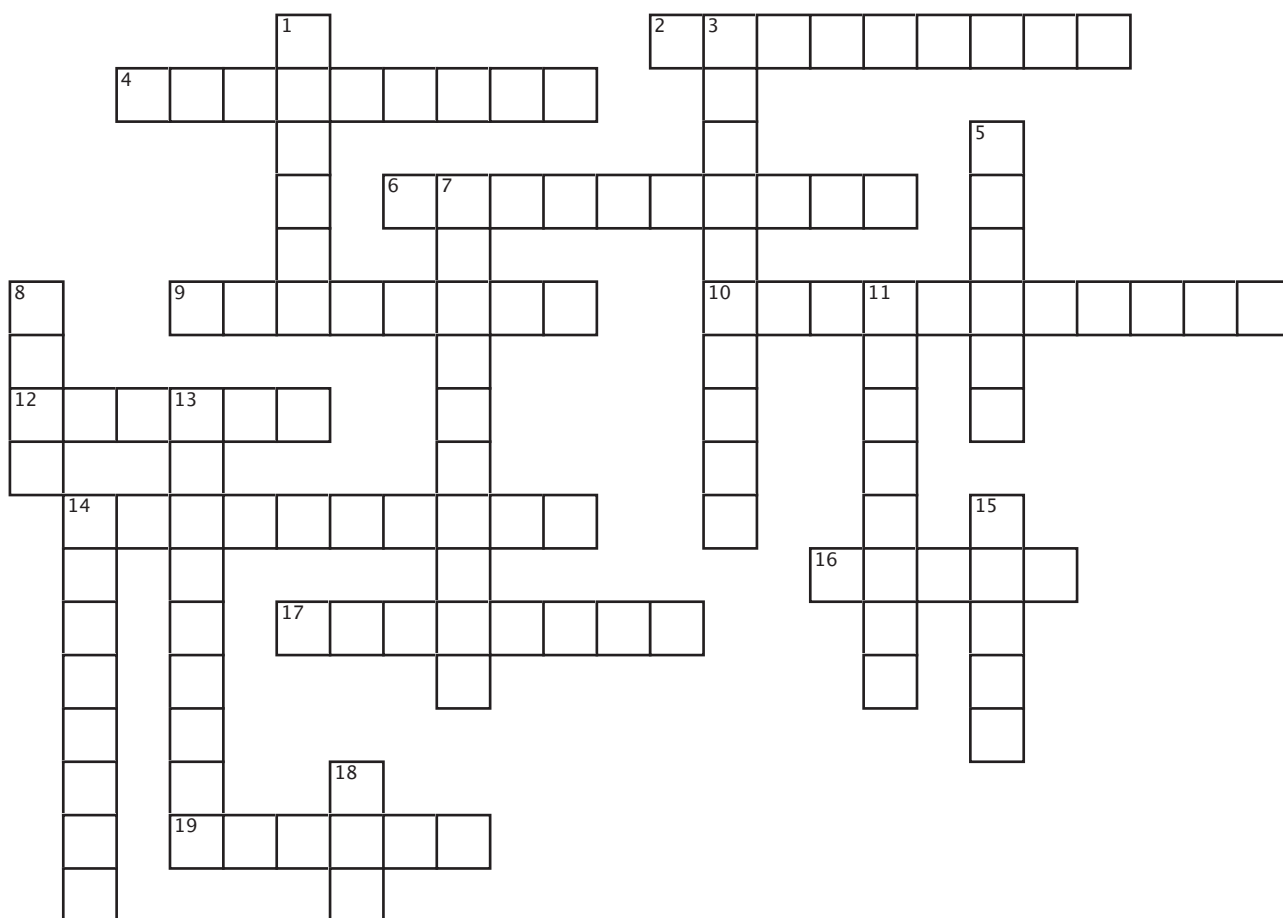
4. Micro-organism used in making bread and champagne
5. Kingdom in which bacteria are placed
7. Bacteria reproduce this way
9. Abbreviation of tubercle bacillus: an infectious disease of vertebrates
12. Latin for spherical
14. This organism transmits bubonic plague (Black Death) bacteria
15. Largest form of a protista
17. This protista is found in waterways and causes abdominal cramps, weight loss and fatigue
18. Correct term for hair-like structure found on the surface of a cell or unicellular organism
19. A bacterium that inhabits the intestines of most vertebrates
20. This bacteria causes meningitis

DOWN

1. Clostridium tetani bacterium causes this disease
2. A microscopic, single-celled, green, freshwater organism that has a single flagella
3. Kingdom in which single-celled organisms are placed
6. Toxic blue-green bacteria
8. Means 'acid-loving bacteria'
10. These micro-organisms have only one strand of DNA
11. A bluish-green fungus that grows as mould on decaying fruits and ripening cheese and is used in the production of an antibiotic
13. A protozoan that causes chills, fever and sweating and is transmitted by the bite of an infected female anopheles mosquito
16. This microscopic, disease-causing particle is not a micro-organism

WORD LIST

- | | | | | |
|------------|----------------|-------------------|----------------|-------------|
| * monera | * malaria | * faecal coliform | * rat flea | * virus |
| * protista | * penicillin | * euglena | * neisseria | * asexually |
| * tetanus | * seaweed | * cyanobacteria | * meningitidis | * bacteria |
| * coccus | * acidophilous | * TB | * giardia | * cilia |
| | * yeast | | | |



ACROSS

2. An important complex carbohydrate that cannot be digested by the human body
4. Unit of energy
6. Proteins cannot be made without these
9. The combustion of glucose provides this kind of energy
10. Name of the process by which stored energy and oxygen are released
12. Any of a group of organic compounds of nitrogen
14. General term for all processes occurring in cells
16. Correct term for fat
17. Essential, soluble, organic substances needed in minute quantities
19. Glucose, fructose and lactose belong to this group

DOWN

1. Solution used to test for starch
3. Sodium and potassium are needed for this energy to work properly
5. Solution used to test for proteins
7. Type of energy used for actions
8. Energy given off as a result of cell respiration
11. Substance used for building new cells and tissues
13. Any metabolised substances used to give energy and build tissue
14. These come mainly from fruit and vegetables
15. Ethanol plus food sample, when filtered, indicates the presence of this substance
18. Energy is stored as this in the body

WORD LIST

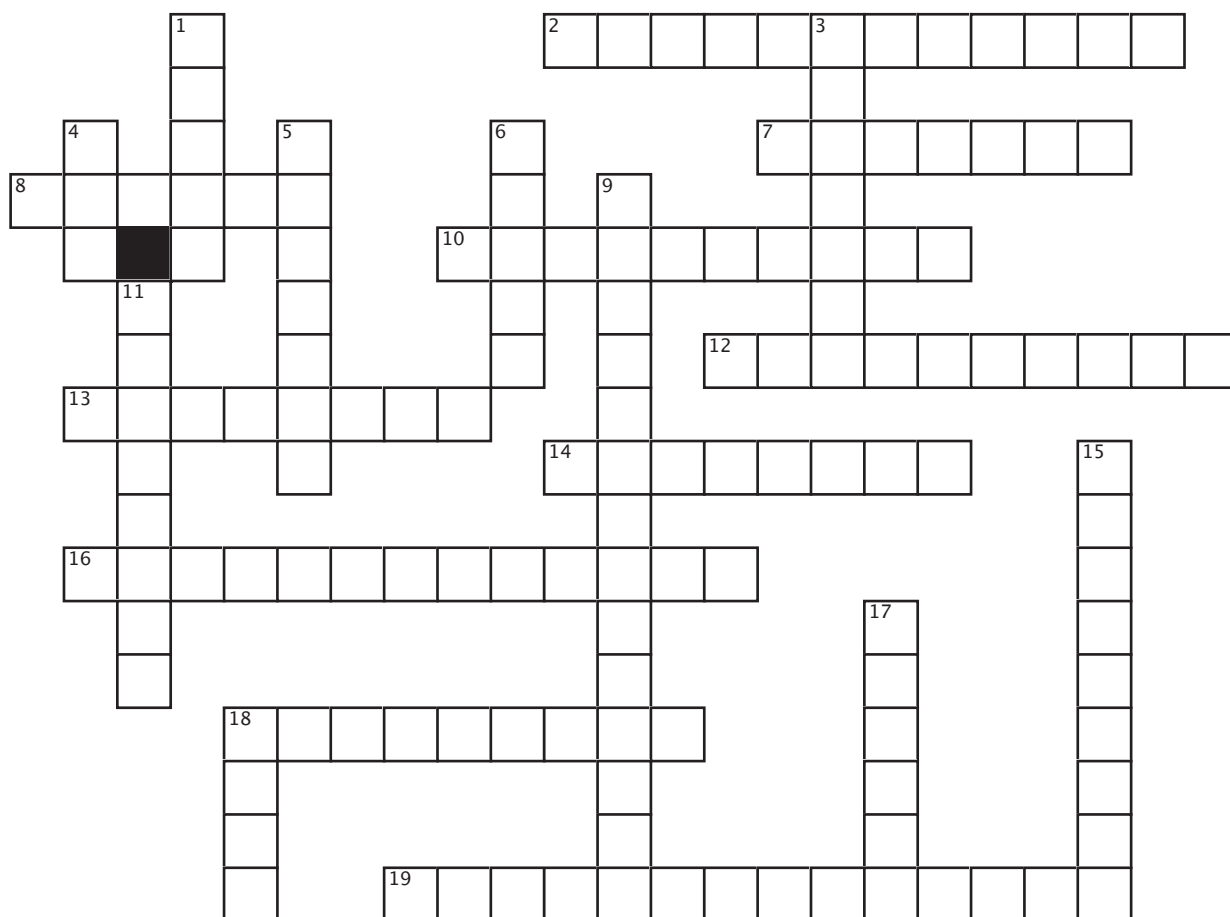
* electrical
* heat
* chemical
* mechanical

* proteins
* fat
* lipid
* vitamins

* amines
* amino acids
* Kilojoule
* respiration

* cellulose
* metabolism
* minerals
* nutrients

* lipid
* biuret
* iodine
* sugars



ACROSS

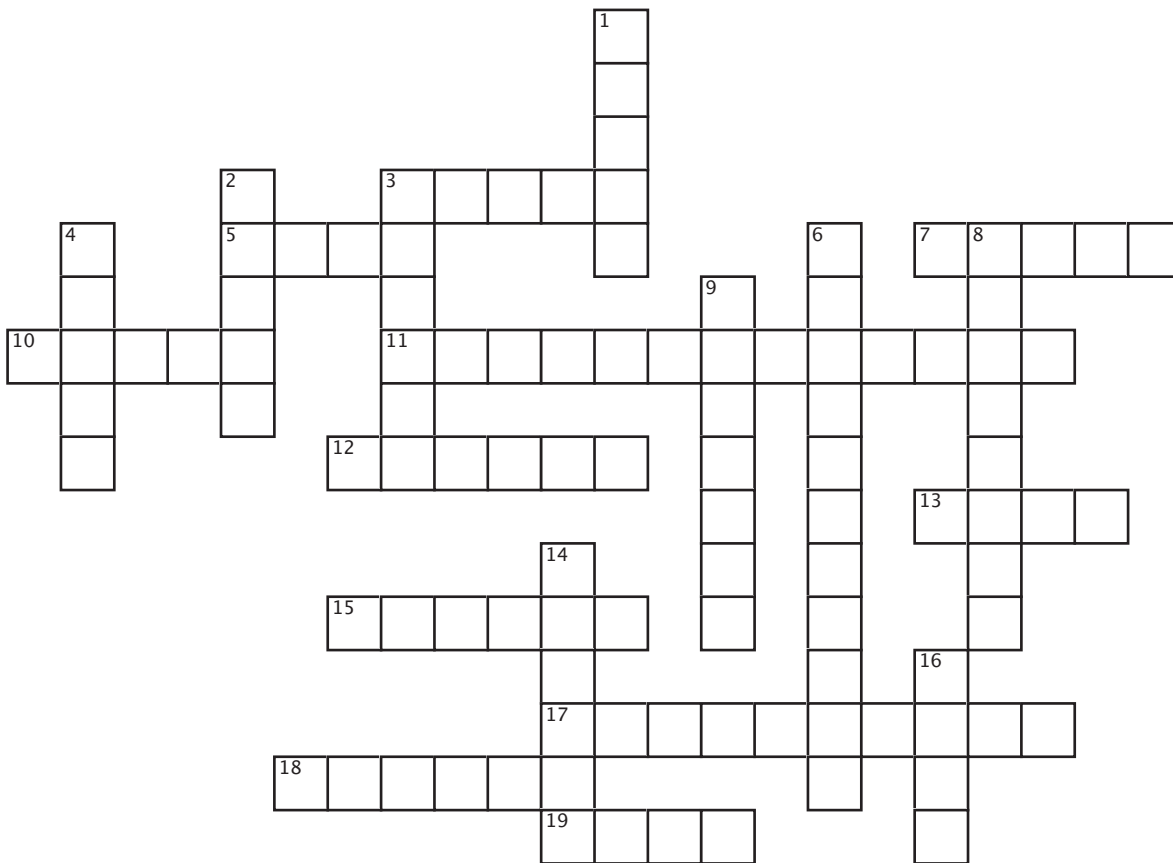
2. Acid found in stomach
7. Main component of meat and eggs
8. Liquid produced to predigest food in the mouth
10. Canal from mouth to anus
12. Food tube from mouth to stomach
13. Nutritional disease
14. Substance that can prevent scurvy
16. Plants make this for consumers
18. Solution used to test for glucose
19. Large diameter tube that extracts moisture from food residues which are later excreted as faeces

DOWN

1. Oranges, mangos, etc.
3. Name of food group containing wheat and rice
4. Needed for hormone production and cartilage manufacture
5. Fangs
6. Small finger-like projections in the intestines
9. Tube in body that secretes enzymes, and is the chief site of the absorption of digested nutrients
11. Essential nutrients from plants
15. Unit of energy
17. Used for grinding food in the mouth
18. Liquid produced by liver to break down fats

WORD LIST

- | | | | | |
|-----------------|-------------|----------------|-------------------|--------------|
| * alimentary | * Vitamin C | * fat | * small intestine | * canines |
| * protein | * benedicts | * anorexia | * large intestine | * saliva |
| * cereals | * molars | * minerals | * villi | * fruit |
| * carbohydrates | * bile | * hydrochloric | * Kilojoule | * oesophagus |



ACROSS

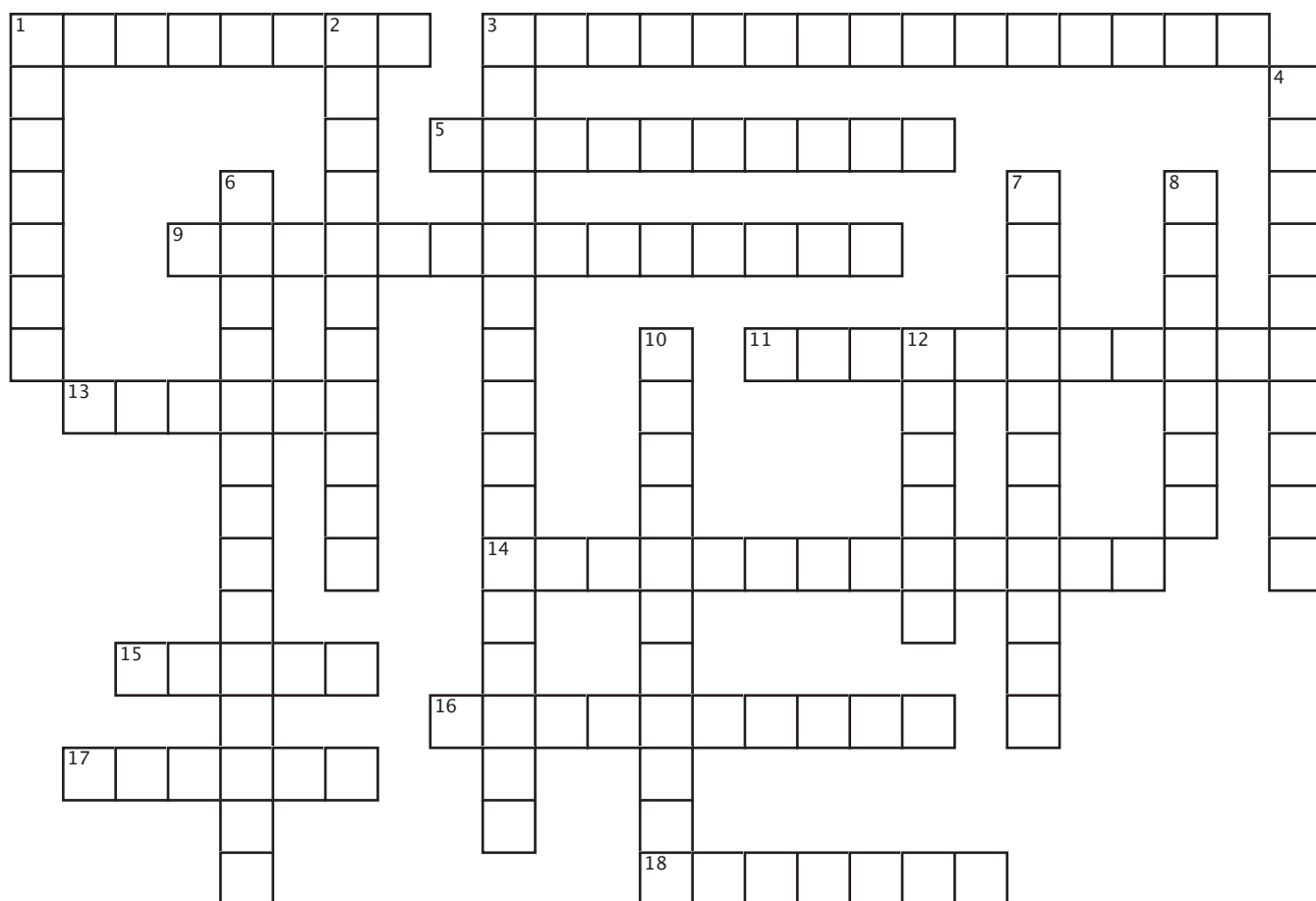
3. A small, usually single-celled reproductive body in certain bacteria, fungi, algae, and non-flowering plants
5. The underground portion that serves as the support of a plant
7. The main woody part of a tree
10. The kingdom that includes yeasts, moulds, smuts and mushrooms
11. Having only one seed leaf
12. The pollen-bearing part of the stamen
13. The tough, outer covering of a trunk
15. The food-conducting tissue of vascular plants
17. Botanical word for flowering plant
18. Ovule-bearing organ of a flower, including the stigma, style and ovary
19. Part of the ripe ovule containing an embryo

DOWN

1. The water conducting tissue in vascular plants
2. The ripened ovary, or ovaries, of a seed-bearing plant
3. The pollen-producing, reproductive organ consisting of a filament and an anther
4. A minute structure in seed plants that contains the embryo sac which develops into a seed after fertilisation
6. The chlorophyll-containing part found in algae and green plants
8. A thin, hair-like outgrowth of a plant root that absorbs water and minerals
9. A minute pore in a leaf or stem through which gases and water vapour pass
14. The brightly coloured parts of a flower surrounding the reproductive organs
16. The main organ of photosynthesis and transpiration of a plant

WORD LIST

* root hair	* trunk	* anther	* ovule	* angiosperm
* root	* bark	* stamen	* monocotyledon	* fungi
* stomate	* petals	* fruit	* chloroplast	* xylem
* leaf	* pistil	* seed	* spore	* phloem



ACROSS

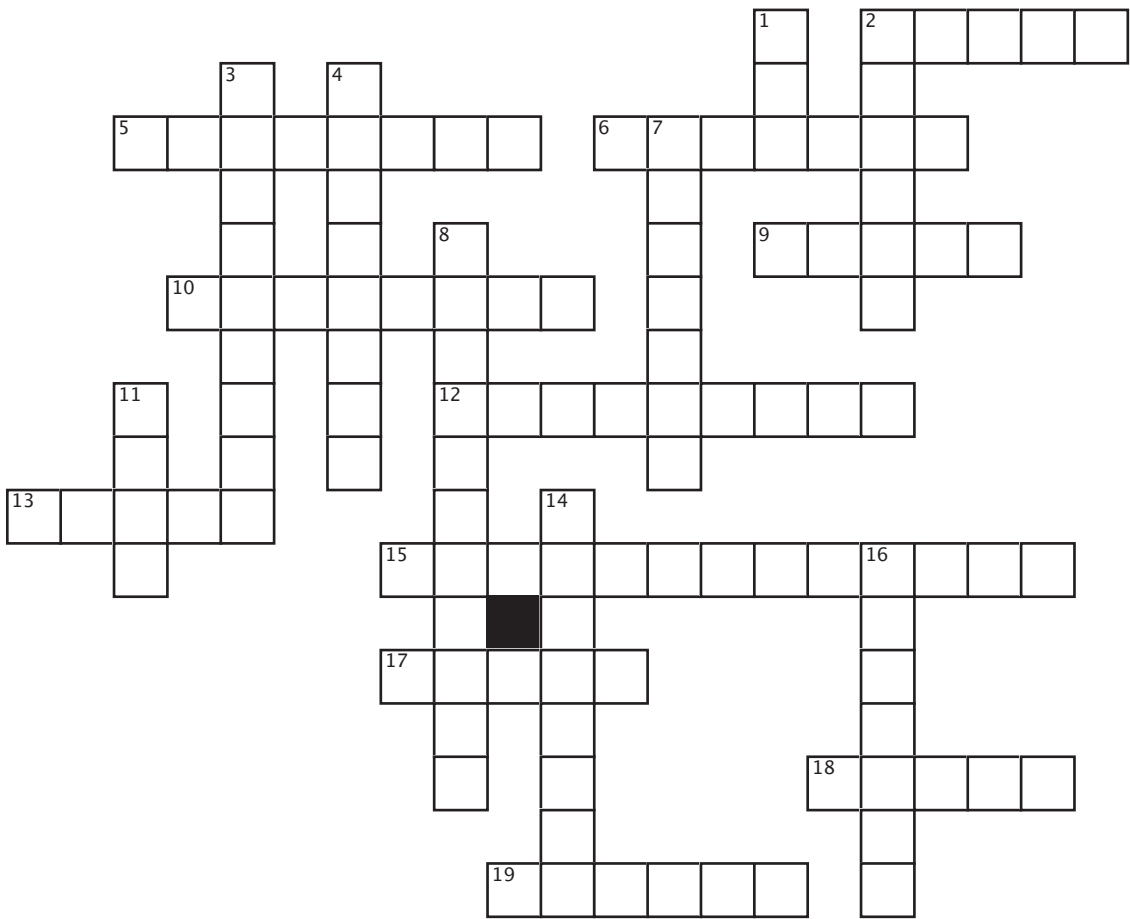
1. The baddies in Snugglepoot and Cuddlepoot
3. Floral emblem of South Australia
5. Australian tall trees with aromatic leaves that yield a medicinal oil and hardwood
9. Grass and palms belong to this classification
11. Victorian state flower
13. National flower; green and gold
14. National capital's flower emblem
15. Flowerless, spore-reproducing vascular plants that have rhizomes, stems and fronds
16. Living beech fossil from Gondwana
17. A fungus that grows symbiotically with algae
18. Pine that is a living fossil

DOWN

1. Tasmania's floral emblem
2. Botanical grouping for all flowering plants
3. Floral emblem of the Northern Territory
4. Mosses, liverworts and hornworts belong to this phylum
6. Queensland's state flower
7. Cycads and pines belong to this classification
8. A tall eastern Australian shrub with intense red flowers
10. Floral emblem of Western Australia
12. Green, usually small, non-vascular plants that live in moist conditions

WORD LIST

- | | | | | |
|---------------------|----------------------|-------------------|------------------|--------------|
| ✿ banksias | ✿ angiosperms | ✿ Cooktown orchid | ✿ kangaroo paw | ✿ ferns |
| ✿ blue gum | ✿ Sturts desert rose | ✿ gymnosperms | ✿ common heath | ✿ nothofagus |
| ✿ Sturts desert pea | ✿ bryophytes | ✿ waratah | ✿ mosses | ✿ lichen |
| | ✿ eucalyptus | ✿ monocotyledons | ✿ wattle | ✿ wollomi |
| | | | ✿ royal bluebell | |



ACROSS

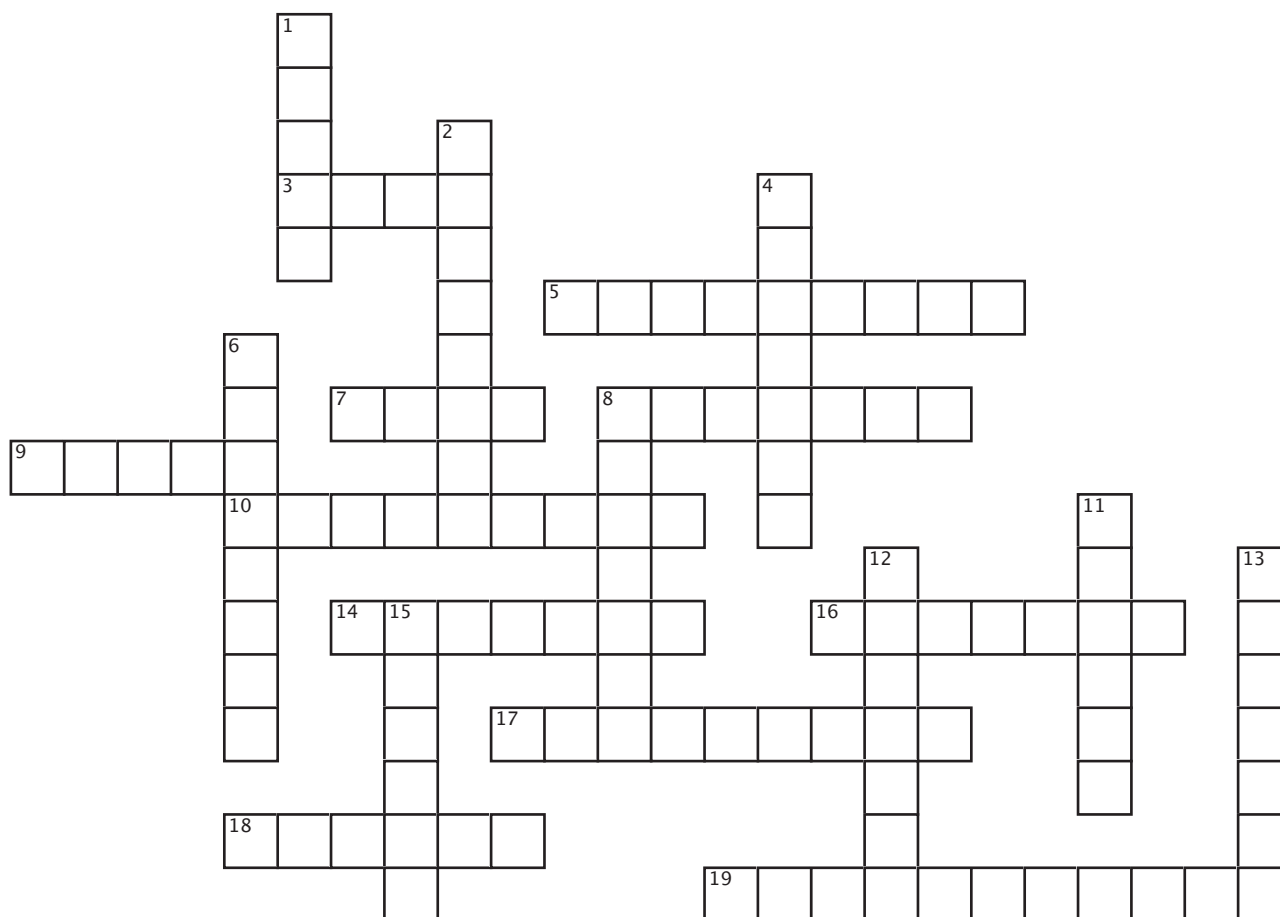
- 2. Neck joint
- 5. System for support, protection and movement
- 6. Another name for heart muscle
- 9. Long bone in leg
- 10. Blood vessel with muscular walls and no valves
- 12. Smallest blood vessel
- 13. Not a red cell
- 15. Bone joint for connecting arm to shoulder
- 17. Major artery coming from heart
- 18. Type of joint in finger
- 19. Receiving chamber of heart

DOWN

- 1. Blood cells that carry oxygen
- 2. Clear, yellowish fluid in which cells are suspended
- 3. Pumping chamber of heart
- 4. Bones that make up backbone
- 7. Grape-like structures in lungs used for diffusing gases
- 8. System made up of blood, blood vessels and heart
- 11. Blood vessel that carries deoxygenated blood
- 14. Disc-like body used for blood clotting
- 16. Important element used for bone strength

WORD LIST

- | | | | | |
|---------------|------------|------------|-------------------|------------|
| * circulatory | * white | * aorta | * atrium | * pivot |
| * red | * skeletal | * vertebra | * ventricle | * cardiac |
| * arteries | * hinge | * femur | * vein | * alveoli |
| * capillary | * plasma | * calcium | * ball and socket | * platelet |



ACROSS

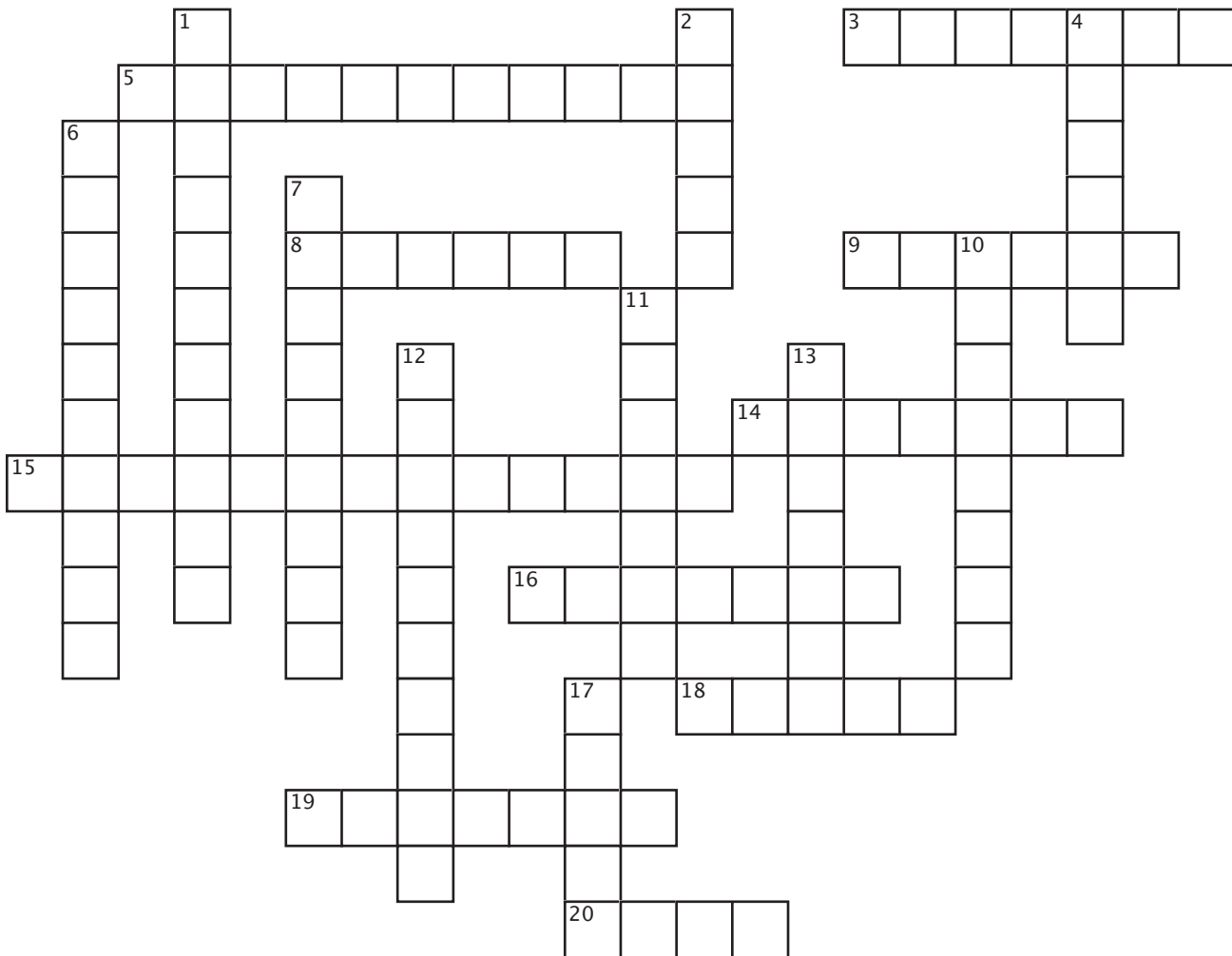
3. The bone that extends from the elbow to the wrist on the side opposite to the thumb
5. Cheek bone
7. Bones that go from sternum to backbone, forming a cage
8. Breastbone
9. Shin bone
10. Backbone bones
14. Correct name for skull
16. Wrist bones
17. Toe bones
18. The outer, narrow bone of the lower leg
19. Bones between fingers and wrist

DOWN

1. Thigh bone
2. Lower jaw bone
4. Long bone in arm
6. Collar bone
8. Shoulder blade
11. Girdle that supports the backbone and leg bones
12. Knee cap
13. Ankle bones
15. Bone that extends from the elbow to the wrist, and is thicker and slightly curved

WORD LIST

* cranium	* carpals	* fibula	* tarsals	* scapula
* sternum	* femur	* tibia	* patella	* ulna
* clavicle	* humerus	* phalanges	* mandible	* radius
* vertebrae	* pelvic	* metacarpals	* ribs	* zygomatic



ACROSS

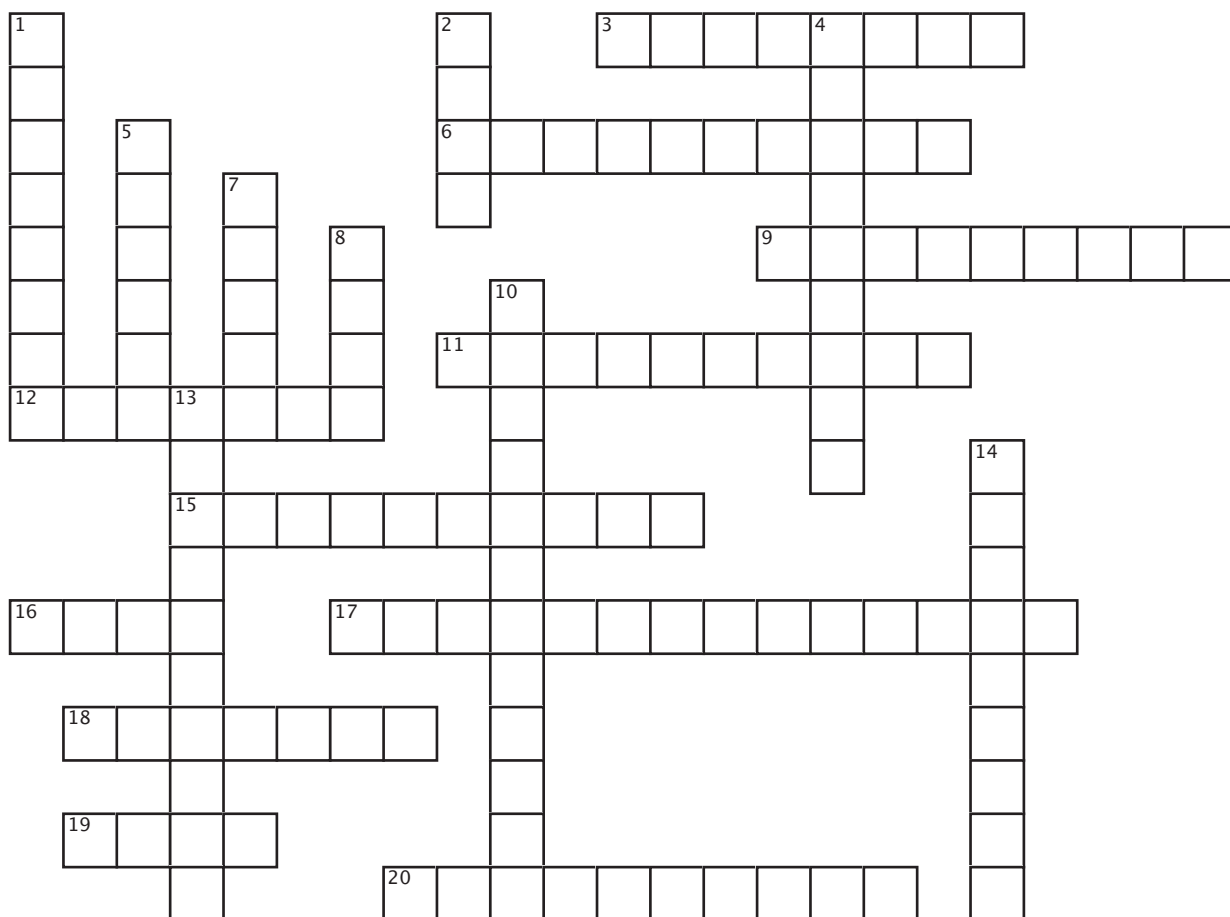
3. Cellular respiration requires oxygen and which other substance to occur?
5. A greenhouse gas exhaled from body
8. Drain from kidney to bladder
9. Organ used for processing and removing liquid waste from body
14. Inside part of the kidney
15. Main exhaled gas in animals
16. Tube from bladder for removing liquid waste
18. Organ used for exchanging gases in the body
19. Gas exchange units in lungs
20. Largest excretory organ of the human body

DOWN

1. Important oxygen-carrying protein
2. Name of liquid that comes from the kidney
4. Gas needed for cellular respiration
6. Loaded up with oxygen
7. Name of artery from heart to lungs
10. Medical method of cleaning blood
11. Liquid waste bag in body
12. Instrument used to measure lung capacity
13. Filtering unit of the kidney that removes waste from blood
17. A type of chemical substance excreted from the skin

WORD LIST

- | | | | | |
|------------------|----------------|--------------|-------------|---------------|
| * carbon dioxide | * water vapour | * ureter | * pulmonary | * medulla |
| * kidney | * nephron | * alveoli | * skin | * haemoglobin |
| * dialysis | * oxygen | * urethra | * salts | * urine |
| * lungs | * bladder | * spirometer | * glucose | * oxygenated |



ACROSS

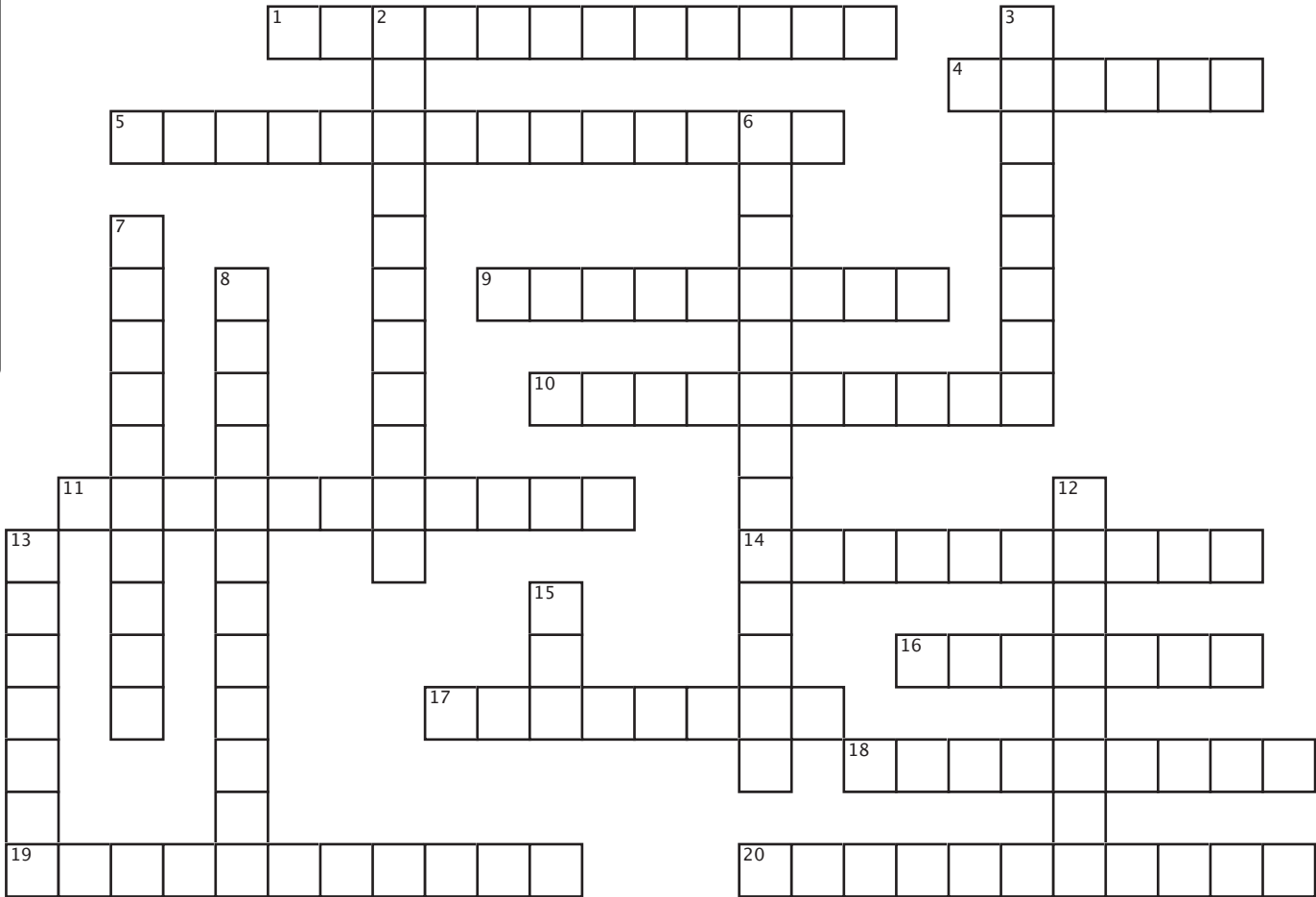
3. These organisms are harmless and grow on the skin
6. The endodermis contains fat for this purpose
9. Gland that secretes oil onto skin
11. Middle layer of skin
12. A protein that allows the skin to stretch
15. Innermost layer of skin
16. A solution that is in sweat
17. Human skin has approximately 700 000 of these
18. Pigment that gives skin its colouring
19. Common skin condition
20. Skin inflammation

DOWN

1. This contains the root of a hair
2. Grows out of the skin
4. Outermost layer of skin
5. A non-contagious skin inflammation that may weep and become encrusted and scaly
7. Skin used from one part of the body to cover a burnt area of skin
8. Largest body organ
10. Nerve ending that senses temperature
13. Perspiration comes from this
14. Skin is described as this type of organ

WORD LIST

- | | | | | |
|-----------------|-------------------|-------------|--------------|--------------|
| * endodermis | * touch receptors | * sebaceous | * bacteria | * eczema |
| * mesodermis | * follicle | * hair | * salt | * elastin |
| * epidermis | * skin | * melanin | * acne | * graft |
| * heat receptor | * sweat gland | * excretory | * dermatitis | * insulation |



ACROSS

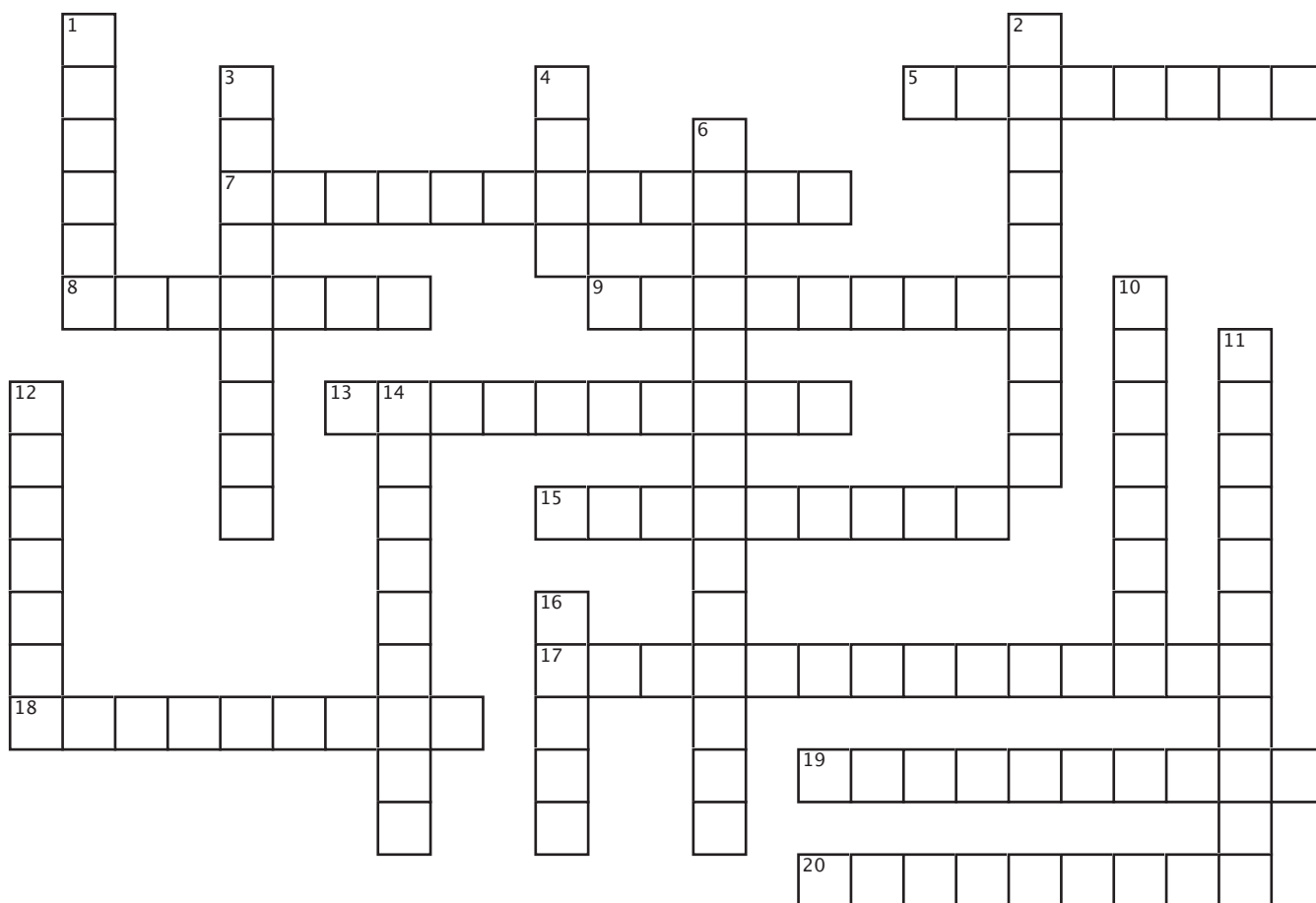
- 1. A human-designed ecosystem that reuses materials for growing plants and feeding animals
- 4. Australian droughts are caused by temperature and water circulation pattern
- 5. The process of converting carbon dioxide and water to oxygen, using light energy
- 9. Excessive burning of grasslands and woodlands causes the loss of ____
- 10. A feature that assists an organism to survive
- 11. Organisms that return nutrients to the soil
- 14. An ecosystem opposite to a desert ecosystem
- 16. A complex of interrelated food chains
- 17. An organism that preys on another
- 18. The flow of energy from one organism to another
- 19. This reacts with rain to assist germination in Australian plants
- 20. A representation that has producers at the base and end consumers at the top

DOWN

- 2. The conversion of chemical energy to other forms of energy
- 3. Microscopic marine organisms
- 6. Aquatic organisms with an exoskeleton
- 7. An adaptation of the bilby is its big ears. They are used to ____
- 8. A product of photosynthesis
- 12. An organism that makes its own food
- 13. The total mass of living matter within a given unit of environmental area
- 15. The source of all energy

WORD LIST

- | | | | | |
|--------------|------------------|-----------------|----------------|---------------|
| * adaptation | * consumer | * plankton | * smoke | * biomass |
| * food chain | * photosynthesis | * invertebrates | * food pyramid | * sun |
| * food web | * carbohydrate | * permaculture | * nutrients | * rain forest |
| * producer | * respiration | * El Nino | * decomposers | * remove heat |



ACROSS

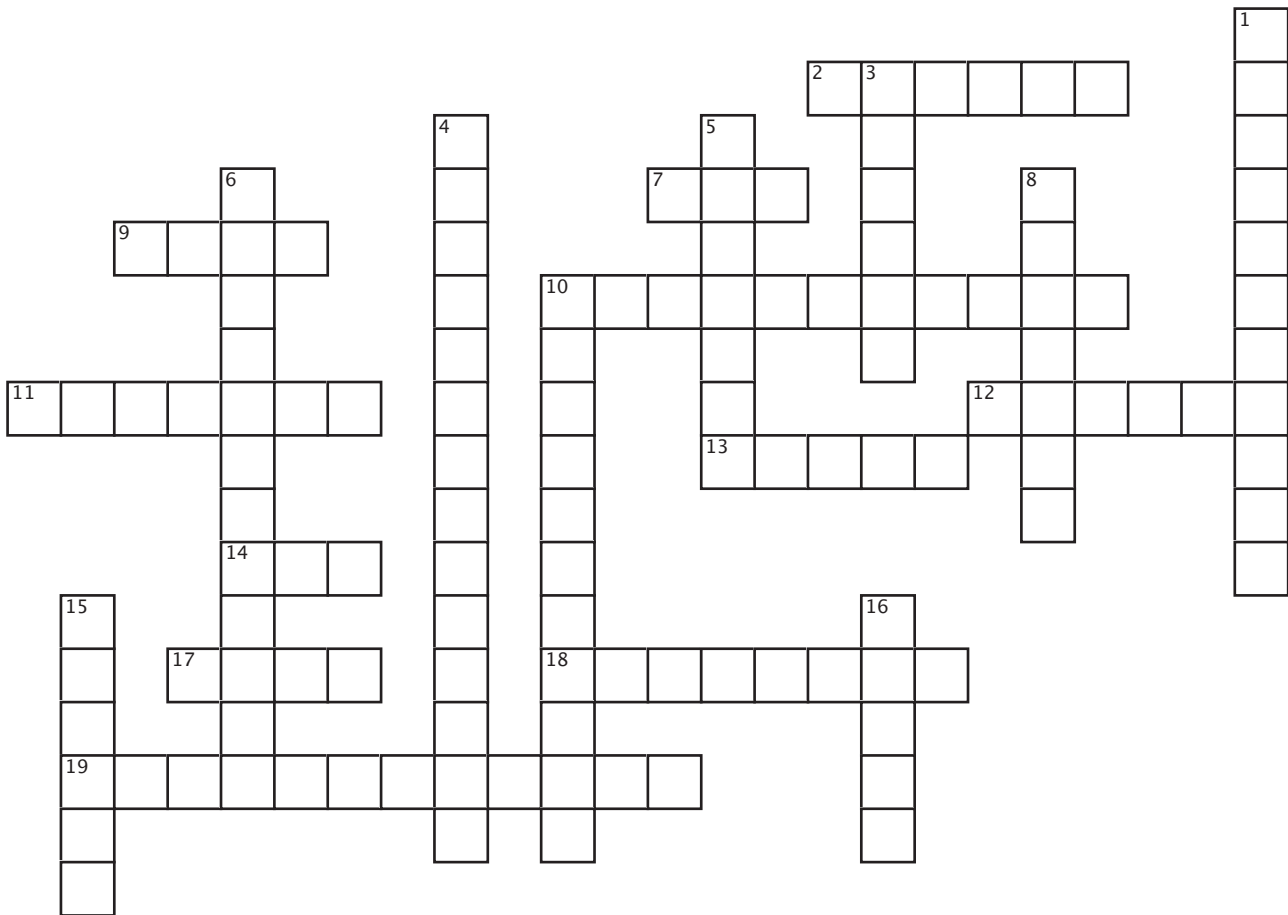
5. Another name for autotroph
7. Type of connection between organisms
8. Area or environment where an organism normally lives
9. Organism that makes its own food
13. Organism that breaks down dead organic matter
15. Organism that feeds mainly on flesh
17. Dependence on one another for survival
18. Two or more different organisms living and interacting with each other is called a _____
19. A group of organisms of the same species living in the same habitat
20. Animals which depend on other living things for food

DOWN

1. Edible product made by plants
2. Organisms linked together in series by feeding habit
3. An organism that feeds on plants
4. A requirement for survival
6. Process by which plants trap sunlight
10. Organism that feeds on both plants and flesh
11. Special characteristics that assist an organism to survive
12. Term used for non-living materials on which living things depend
14. Interactive system of living and non-living things
16. A particular area or position within a habitat where an organism lives

WORD LIST

- | | | | | |
|---------------|------------------|--------------|----------------|------------------|
| * habitat | * food | * decomposer | * omnivore | * photosynthesis |
| * niche | * interdependent | * autotroph | * producer | * starch |
| * population | * community | * herbivore | * consumers | * food chain |
| * adaptations | * abiotic | * carnivore | * relationship | * ecosystem |



ACROSS

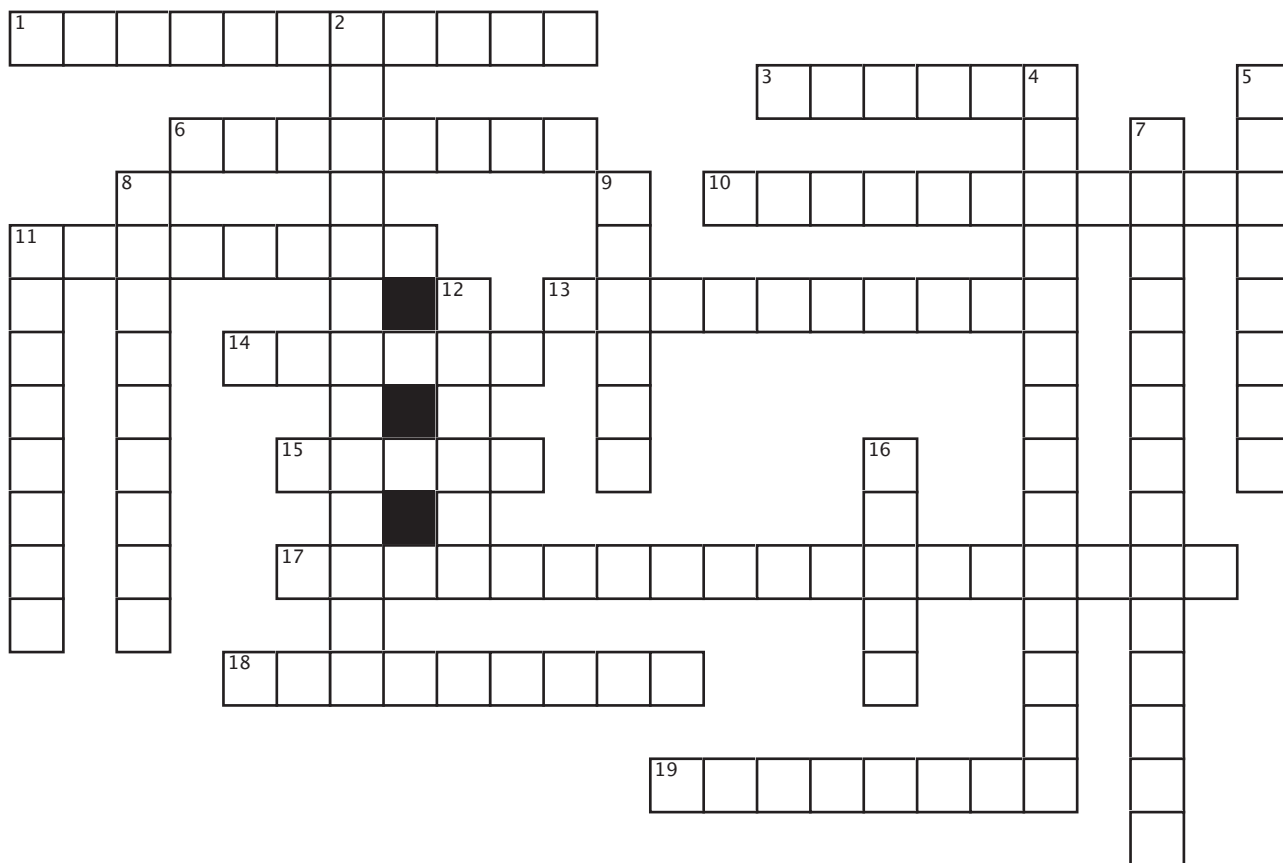
2. Australian term for wild horse
7. An animal with short legs, cloven hooves, bristly hair and a cartilaginous snout
9. A hollow-horned, bearded, ruminant mammal raised for meat and milk
10. Man's best friend gone wild
11. Finch-like bird with grey or brown plumage
12. Beast of burden
13. A freshwater game fish with a streamlined, speckled body and small scales
14. Red-brown dog with a tail like a brush
17. Large, goldfish-like fish
18. The perfect invasion machine
19. Flying insect with constricted abdomen, usually orange in colour

DOWN

1. A bird with a brown body and wing plumage, grey head and throat band, and bright yellow legs
3. Small, furry, hopping animal
4. Pale yellow ant that changes direction rapidly when disturbed
5. Omnivorous, ground-living insect that builds large nests and has a painful sting
6. Large cow-like animal that wallows in tropical wetlands
8. Mexican walking fish
10. A small, furry, carnivorous mammal that kills native animals and birds
15. A bird with a plump body, small head, short legs and usually grey in colour
16. A humped, long-necked animal

WORD LIST

- | | | | | |
|-----------------|----------|----------------|--------------------|------------|
| * camel | * brumby | * pig | * domestic cat | * pigeon |
| * cane toad | * rabbit | * goat | * Indian mynah | * sparrow |
| * European wasp | * carp | * donkey | * yellow crazy ant | * fire ant |
| * water buffalo | * fox | * domestic dog | * trout | * axolotl |



ACROSS

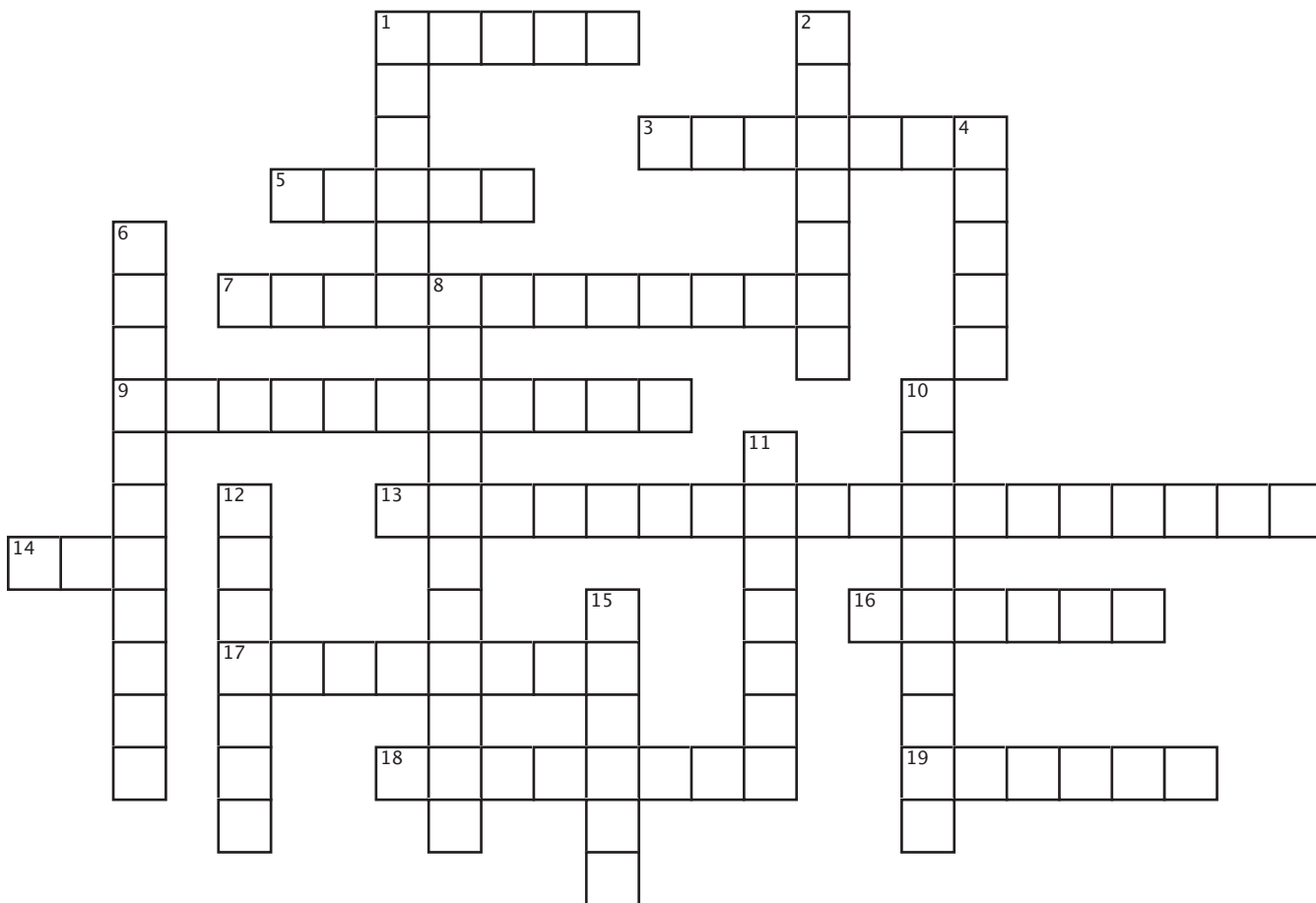
1. Small, nocturnal, grey, gliding animal with dark strip down its back which glides from tree to tree
3. Large, digging, nocturnal marsupial
6. Large, big-footed, hopping marsupial
10. World's deadliest snake
11. A brilliantly coloured parrot that feeds primarily on soft fruits or the nectar and pollen of flowers
13. Laughing Jackass
14. Nocturnal animal that eats leaves, fruit and flowers
15. Large-eared, desert-living bandicoot
17. Large estuarine and marine, carnivorous reptile
18. Largest Australian eagle
19. Largest Australian owl

DOWN

2. Reptile without limbs
4. Largest, carnivorous marsupial found in Tasmania
5. Tree-living amphibian, usually green
7. Small, hopping rodent with a pouch
8. This sun-loving lizard is the reptile emblem of Australia
9. A small, short-tailed wallaby
11. Bird with long tail feathers that are spread in a lyre-shaped display during courtship
12. Marsupial anteater with a bushy tail and striped back
16. Carnivorous cat-like marsupial

WORD LIST

- | | | | | |
|------------|-------------------|-----------------------|-------------------|--------------|
| * kangaroo | * wedge tail | * quokka | * lyrebird | * frill neck |
| * bilby | * lorikeet | * tree frog | * fierce snake | * numbat |
| * possum | * kookaburra | * legless lizard | * sugar glider | * powerful |
| * quoll | * marsupial mouse | * saltwater crocodile | * Tasmanian devil | * wombat |



ACROSS

1. When atoms vibrate extremely slowly, they are ____
3. When very hot particles move from liquid to gas state
5. All matter is made up of these
7. Cooling of particles from gas to liquid
9. To go from solid to gas without going through the liquid phase
13. Change in gas pressure is due to ____
14. When atoms vibrate extremely quickly, they are a ____
16. When atoms vibrate moderately quickly, they are ____
17. The particles and the dissolving substance combined is called a ____
18. When objects decrease in size
19. When objects increase in size

DOWN

1. A unit of time
2. The process of going from solid to liquid
4. Matter is commonly measured in these units
6. When gas atoms move extremely quickly, they are in a ____
8. The process of going from liquid to gas
10. Matter is made up of these
11. A substance that dissolves something is called a ____
12. Mass divided by volume, or how much matter there is in something
15. Particle motion is dependent on ____

WORD LIST

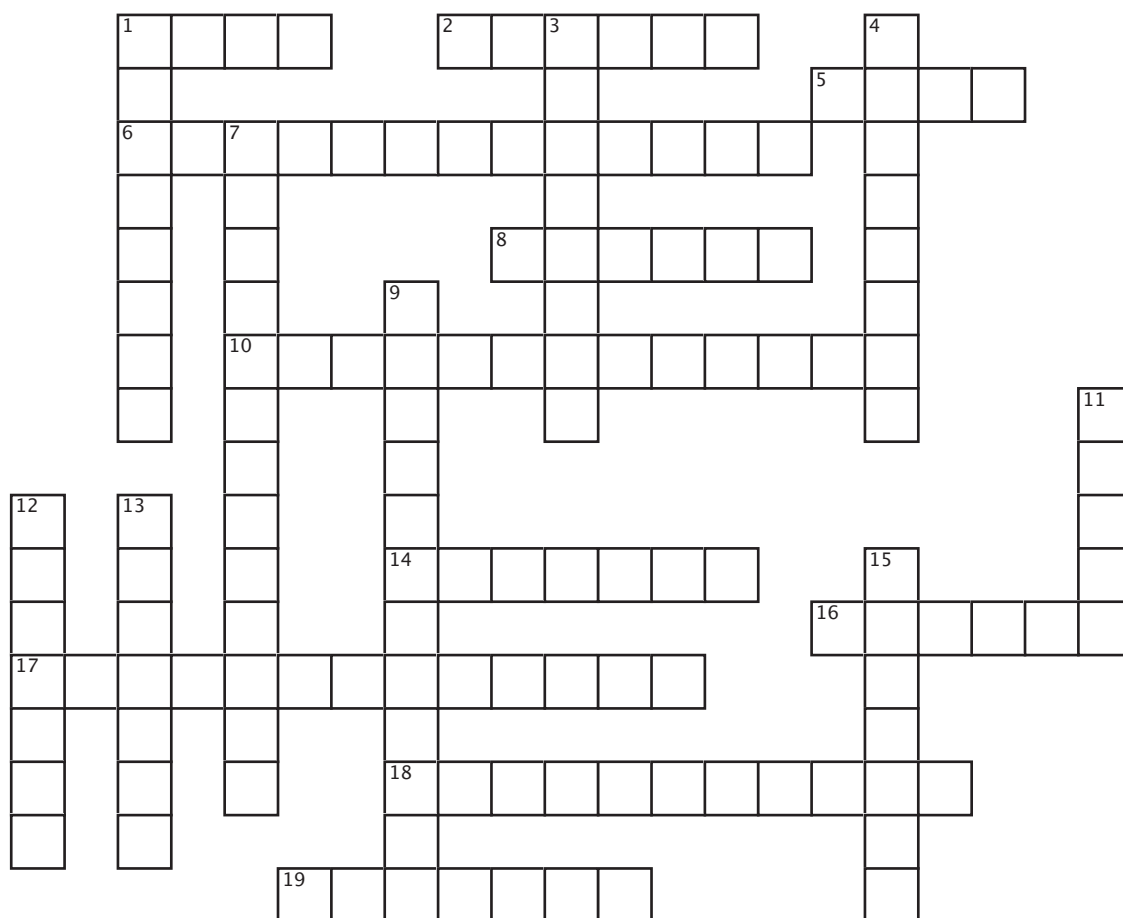
expand
 contract
 particles
 atoms

density
 solid
 liquid
 gas
 plasma state

energy
 melting
 evaporation
 boiling

condensation
 sublimation
 collision
 frequency

solvent
 solution
 grams
 second



ACROSS

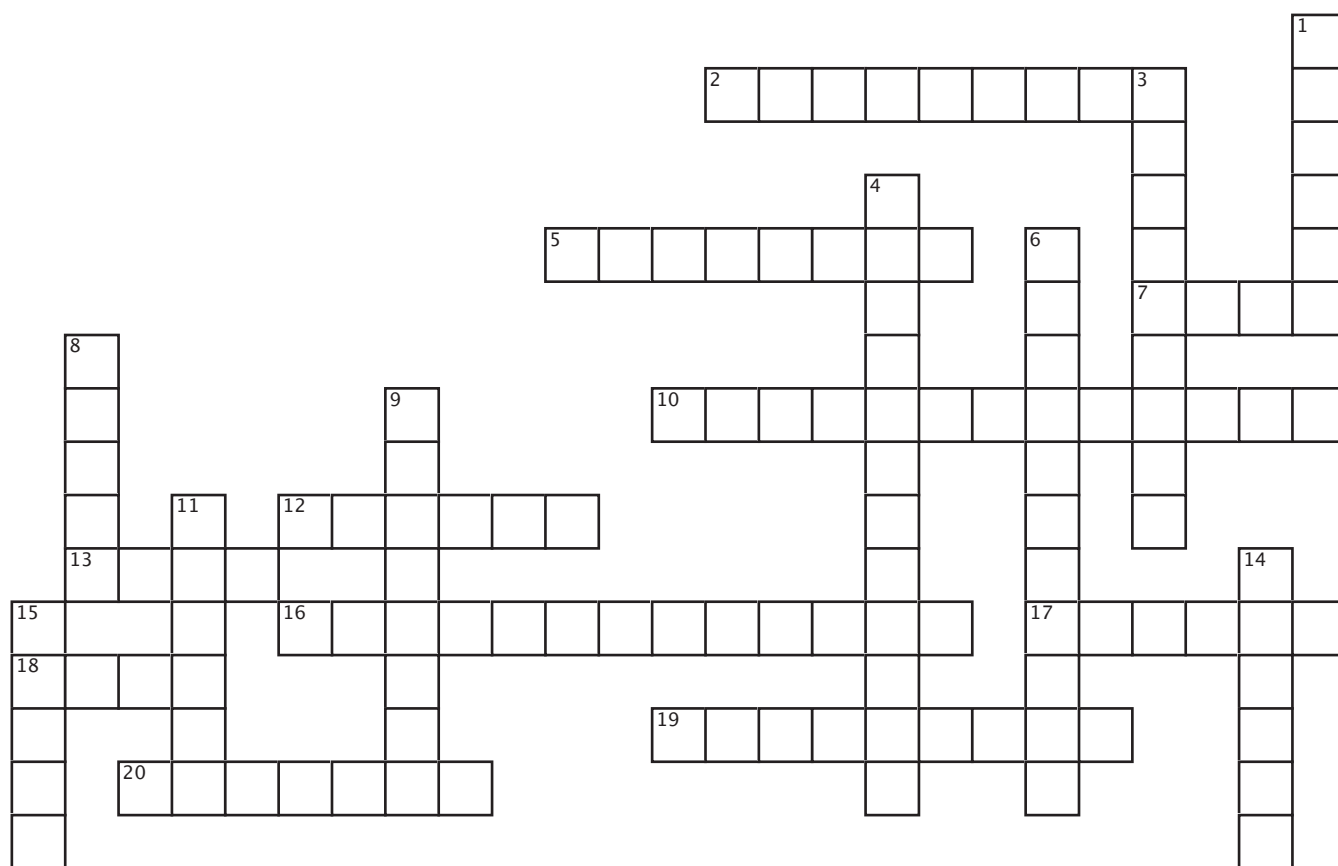
1. The amount of matter in an object
2. Very strong forces alter the shape of an object
5. Opposite to push
6. When an object falls, the force is ____
8. Opposite of weak
10. Force that is produced by a non-moving, electric charge
14. A black rod that produces a charge when rubbed
16. Unit of time
17. Used to measure force
18. The English scientist who discovered the law of gravity
19. Friction is this kind of force

DOWN

1. A field force associated with an electric current
3. A force that opposes motion
4. When a ship sinks it no longer has ____
7. Parachutes 'float' because of this force
9. Force equals mass times ____
11. Not a contact force
12. A stretched rubber band is an example of this force
13. Unlike forces do what to each other?
15. Force is measured in ____

WORD LIST

- | | | | | |
|---|--|--|--|--|
|  field |  friction |  electrostatic |  magnetic |  strong |
|  gravitational |  Newtons |  pull |  mass |  second |
|  buoyancy |  acceleration |  spring balance |  Isaac Newton |  deform |
|  tension |  contact |  attract |  ebonite |  air resistance |



ACROSS

2. Opposite of temporary
5. Shape of magnet in a magnetic resonance imaging chamber
7. Name of the end of a magnet
10. Direction in which a compass needle points
12. Name of Cretan shepherd who discovered magnetism
13. Used to demagnetise magnets
16. A temporary magnet made by coiling wire around an iron core
17. Metal used in magnets and named after goblins
18. Main metal used in making magnets
19. Name of a mineral that has weak, permanent, magnetic field
20. Opposite of repel

DOWN

1. Name given to a pair of equal and opposite magnetic poles, separated by a small distance
3. Name of a communication device that uses magnets
4. Instrument used for detecting magnetic properties; usually used in mineral exploration
6. Changing the magnetic field causes this to flow
8. Direction in which a south pole of a magnet points
9. Magnets only attract what kind of materials?
11. A coiled wire with electricity flowing through it, acts like one
14. Magnetic field lines form this type of loop
15. Magnetic forces are this kind of a force

WORD LIST

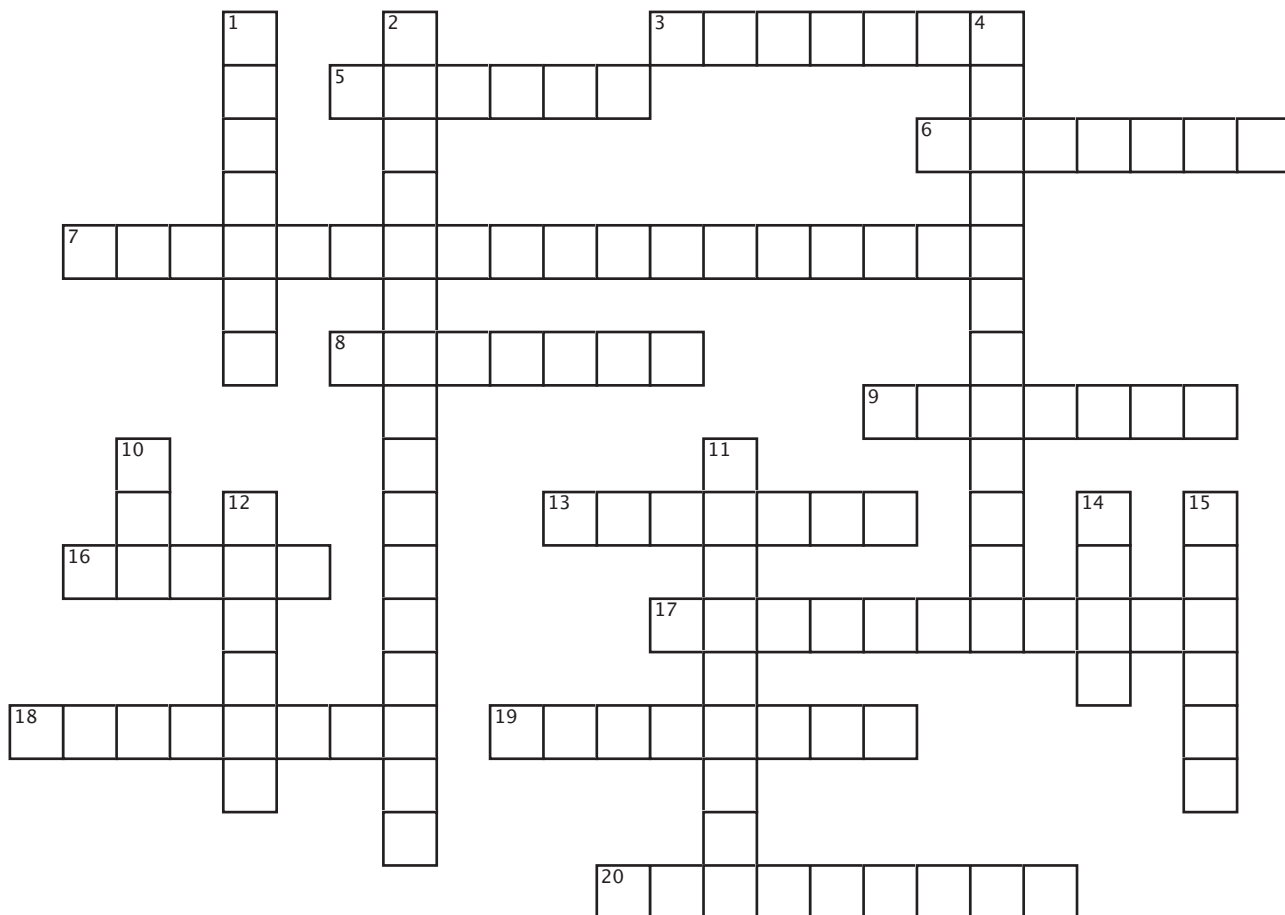
attract
 south
 field
 permanent

magnet
 magnetic
 Magnes
 magnetic north

circular
 heat
 cobalt
 magnetometer

electromagnet
 iron
 dipole
 pole

closed
 magnetite
 electricity
 telephone



ACROSS

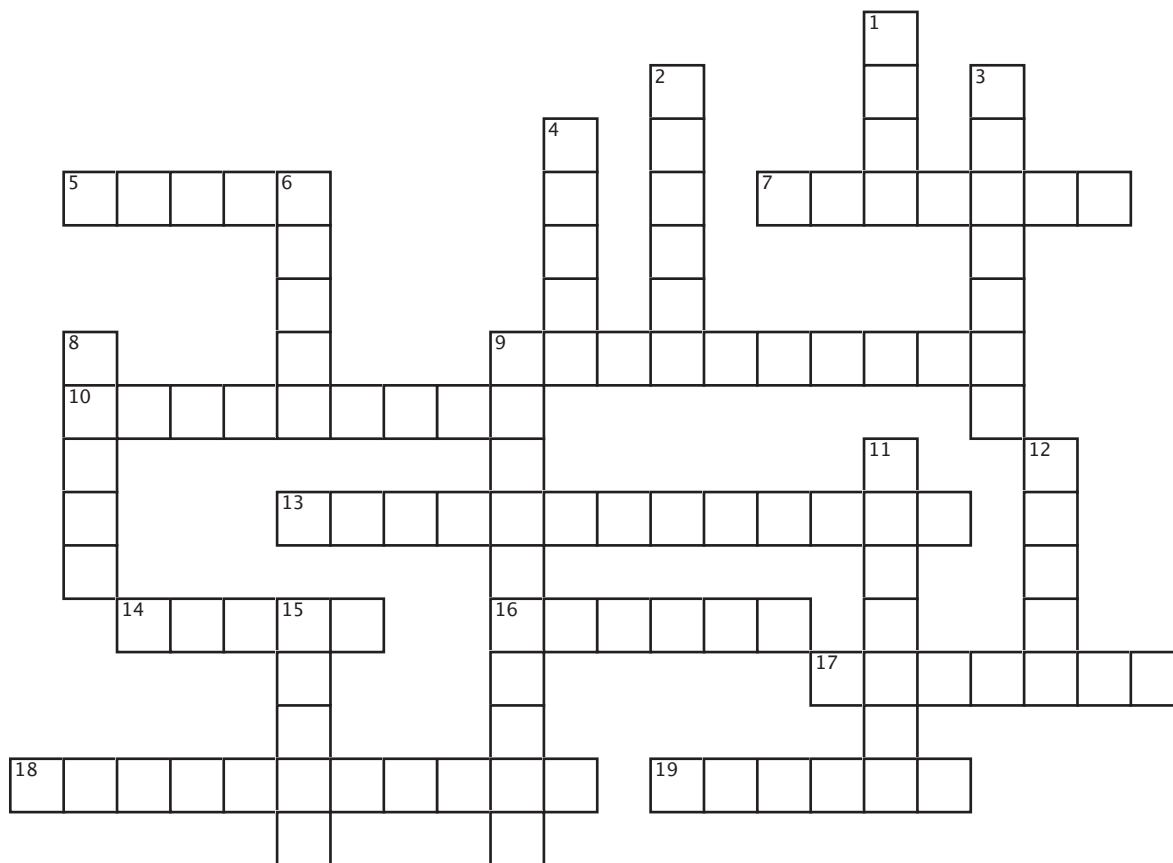
3. Current is measured in these units
5. Name of a circuit where components are sequential
6. The potential difference of electricity in a circuit is termed _____
7. An electric current that reverses direction in a circuit at regular intervals
8. Electricity requires this to flow
9. Name given to electrical flow
13. Instrument that measures electrical flow
16. Natural solidified resin that can create static electricity
17. Pioneer of electrical transmission
18. Name of a circuit where the current is split up to flow through components
19. Atomic particle that moves freely in metals
20. Term used for a substance that prevents electrical flow

DOWN

1. Device used for storing electrons
2. Name of the person who proved that lightning was electricity
4. Path taken by electricity current that flows to an area of low resistance and bypasses the rest of the circuit
10. Unit used for measuring resistance
11. Substances that hold onto electrons are termed _____
12. Substances that have free moving electrons
14. Device used to prevent electrical flow from causing a fire
15. The build up of an of electric charge on an insulated body

WORD LIST

- | | | | | |
|---------------------|-------------------|-----------|-----------|---------------|
| electron | metals | ammeter | ohm | short circuit |
| static | voltage | current | circuit | fuse |
| alternating current | Benjamin Franklin | amperes | series | battery |
| | Nikola Tesla | amber | parallel | |
| | | resistors | insulator | |



ACROSS

5. The point at which reflected light rays intersect
7. Name of light energy bundles
9. Name of angle at which light bounces off a surface
10. Name of angle at which light hits a surface
13. Reflection of sound in a room is called _____
14. Object that scatters light into a spectrum of colour
16. Part of eye responsible for refracting light
17. Part of ear that causes the middle ear bones to vibrate
18. Name given to the change in direction and intensity of waves
19. Opposite of concave

DOWN

1. Name given to sound reflection
2. The name of the angle when light hits a surface at right angles
3. Name of sound energy bundles
4. Name of a flat mirror
6. Travels at 330 m/s
8. Travels at 300 000 000 m/s
9. Name for bending light
11. Name of a mirror shaped like the inside of a ball
12. Short for light amplification by simulated emission of radiation
15. Short for sound navigation and ranging

WORD LIST

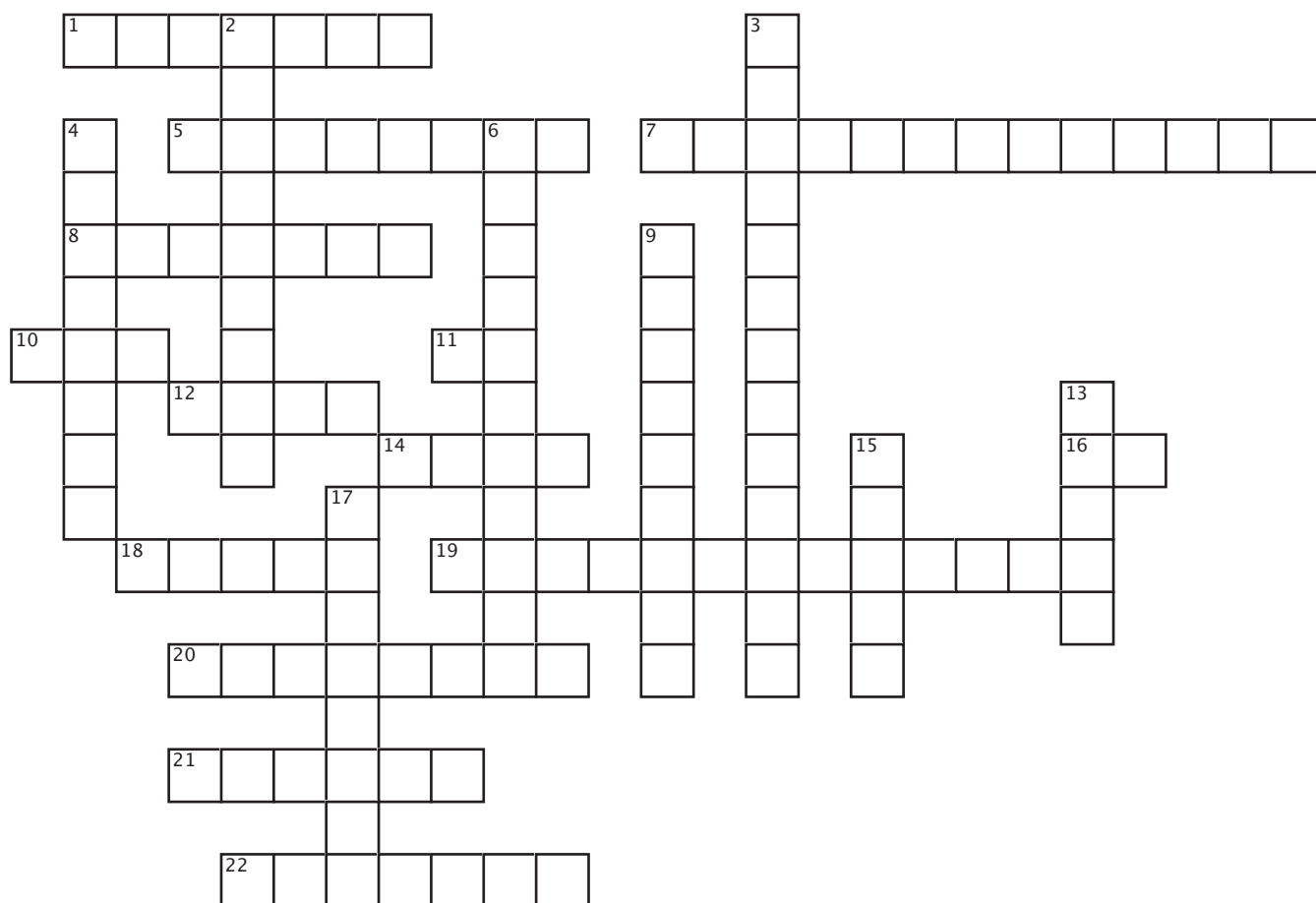
incidence
 reflection
 normal
 plane

focus
 concave
 convex
 echo

reverberation
 refraction
 diffraction
 light

sound
 prism
 laser
 cornea

eardrum
 phonons
 photons
 sonar



ACROSS

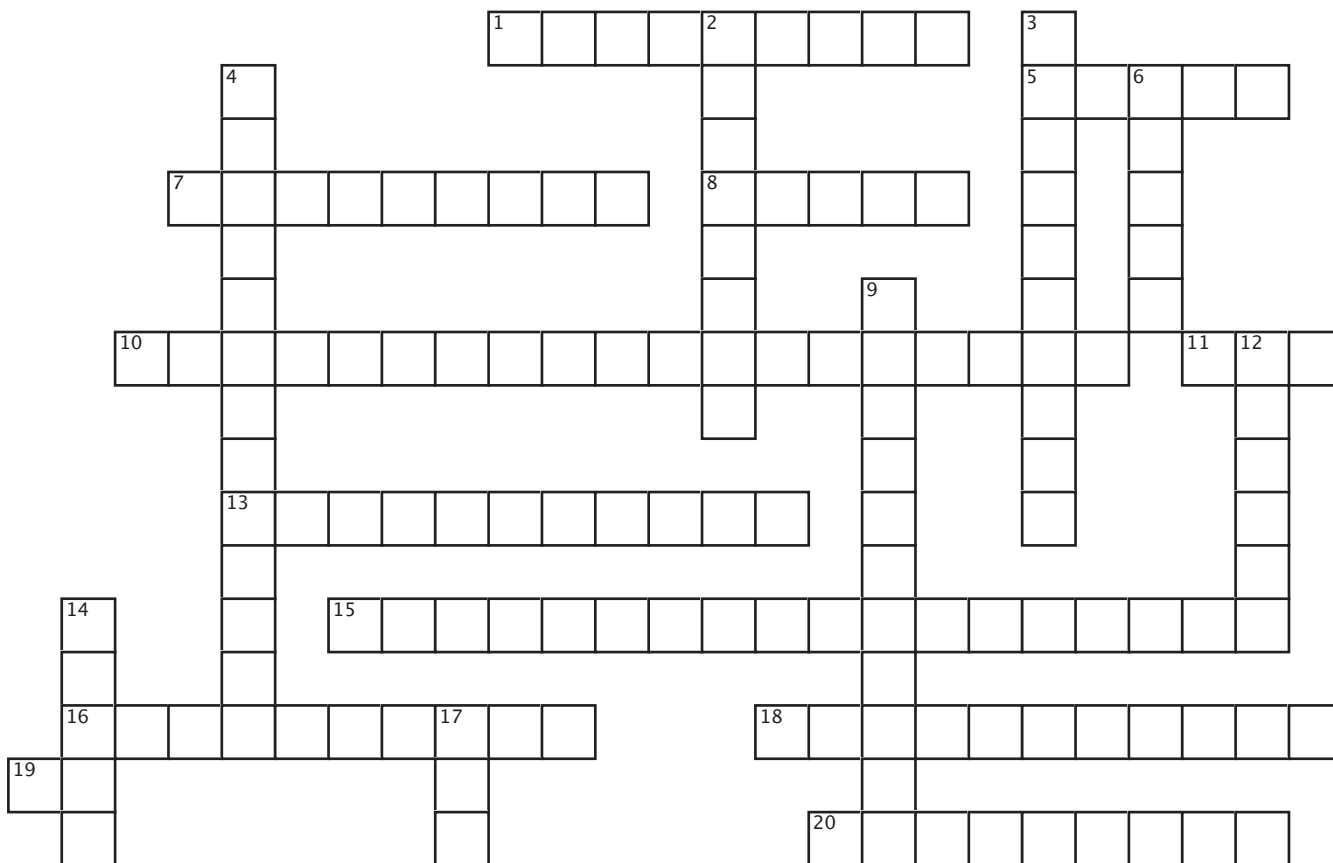
1. Liquid, non-metal element
5. Periodic Table is made up of these
7. Gas absorbed by plants
8. Liquid metal element
10. Hydrogen does this when near a flame
11. Symbol for iron
12. Second element in the Periodic Table
14. Name for Pb
16. Symbol for aluminium
18. Metals have this property when cut
19. Made from only one type of atom
20. Most common gas in atmosphere
21. A glowing splint identifies this gas
22. Non-metals cannot do this with heat or electricity

DOWN

2. Non-metal atoms or compounds
3. Invented by Mendeeleev as a system of element classification
4. Two or more elements joined together by chemical bonds
6. A chemical change is indicated by this
9. To make a compound from different elements
13. Last column of Periodic Table has only what kind of elements?
15. A solid mineral element, able to conduct heat and electricity
17. Explosive gas, first in Periodic Table

WORD LIST

☛ He	☛ compound	☛ pop	☛ elements	☛ nitrogen
☛ lead	☛ carbon dioxide	☛ temperature	☛ synthesis	☛ gases
☛ molecules	☛ hydrogen	☛ pure substance	☛ Al	☛ Mercury
☛ Periodic Table	☛ metal	☛ shiny	☛ Fe	☛ Bromine
	☛ oxygen	☛ conduct		



ACROSS

1. The height of an electromagnetic wave
5. Abbreviation for light amplification by the stimulated emission of radiation
7. Instrument used to detect expanding rocks or their layers
8. Short for radio detection and ranging
10. Used to smash atoms so that particles can be examined
11. Short for ultra high frequency waves
13. Wavelength used in remote sensing
15. A device used to see objects that cannot be seen using visible light
16. Mobiles and satellites use these for communication
18. A device used to graph the intensity of earthquakes
19. Frequency modulation abbreviation
20. Used to enlarge an object in the far distance

DOWN

2. Used in physiotherapy and deep space remote sensing
3. Instrument used to detect foetal heart beats
4. Instrument used to measure the intensity of radioactivity
6. A system used to detect and locate submerged objects or measure a distance to another object in water
9. Device used for examining broken limbs
12. Large space telescope
14. Rays used to sterilise medical equipment and kill bacteria
17. Used by aircraft to communicate (abbreviation)

WORD LIST

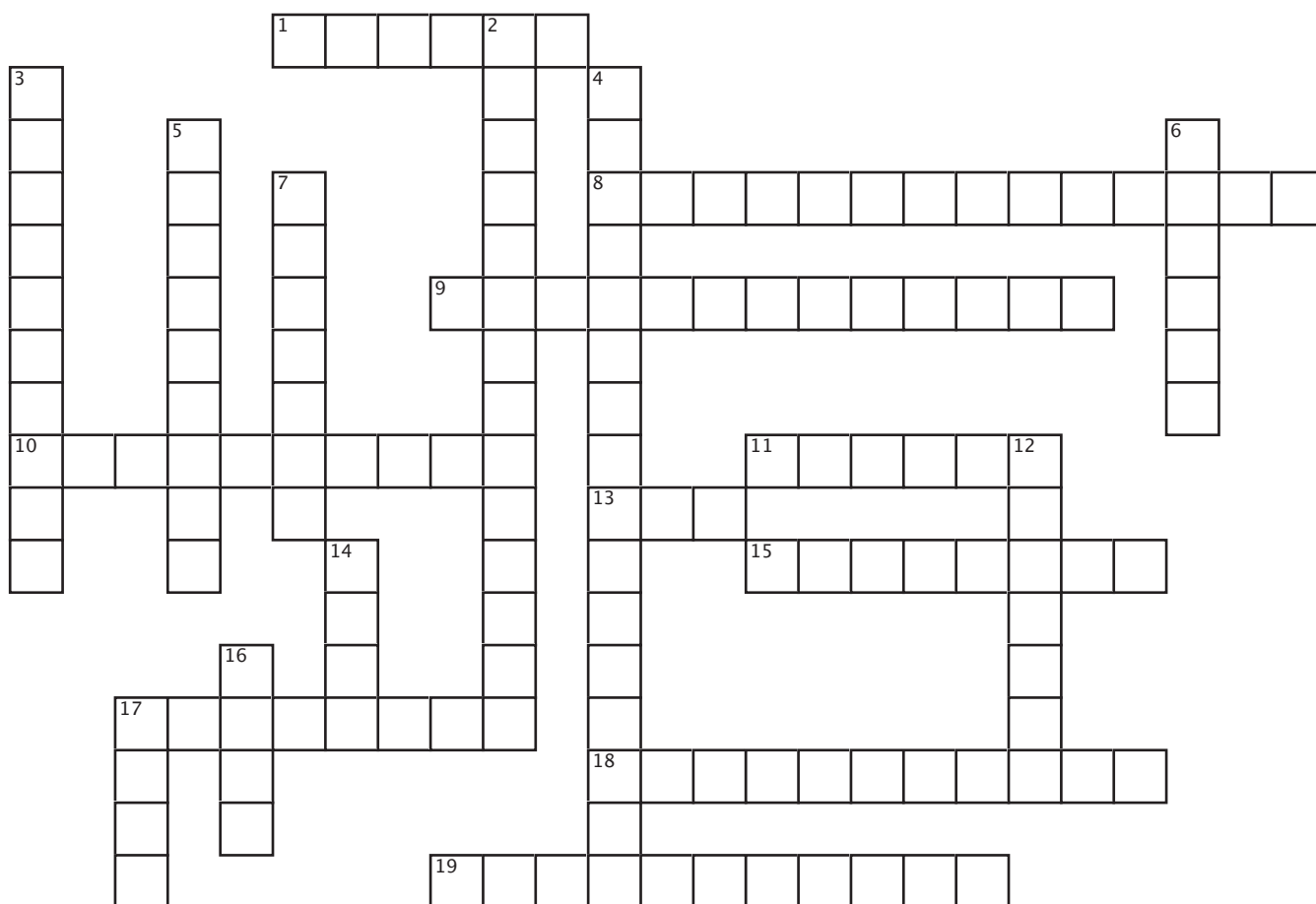
sonar
 radar
 seismograph
 ultra sound

geiger counter
 microwaves
 xray machine
 electron microscope

UHF
 amplitude
 FM
 VHF
 ultraviolet

infrared
 telescope
 particle accelerator

tiltmeter
 hubble
 gamma
 laser



ACROSS

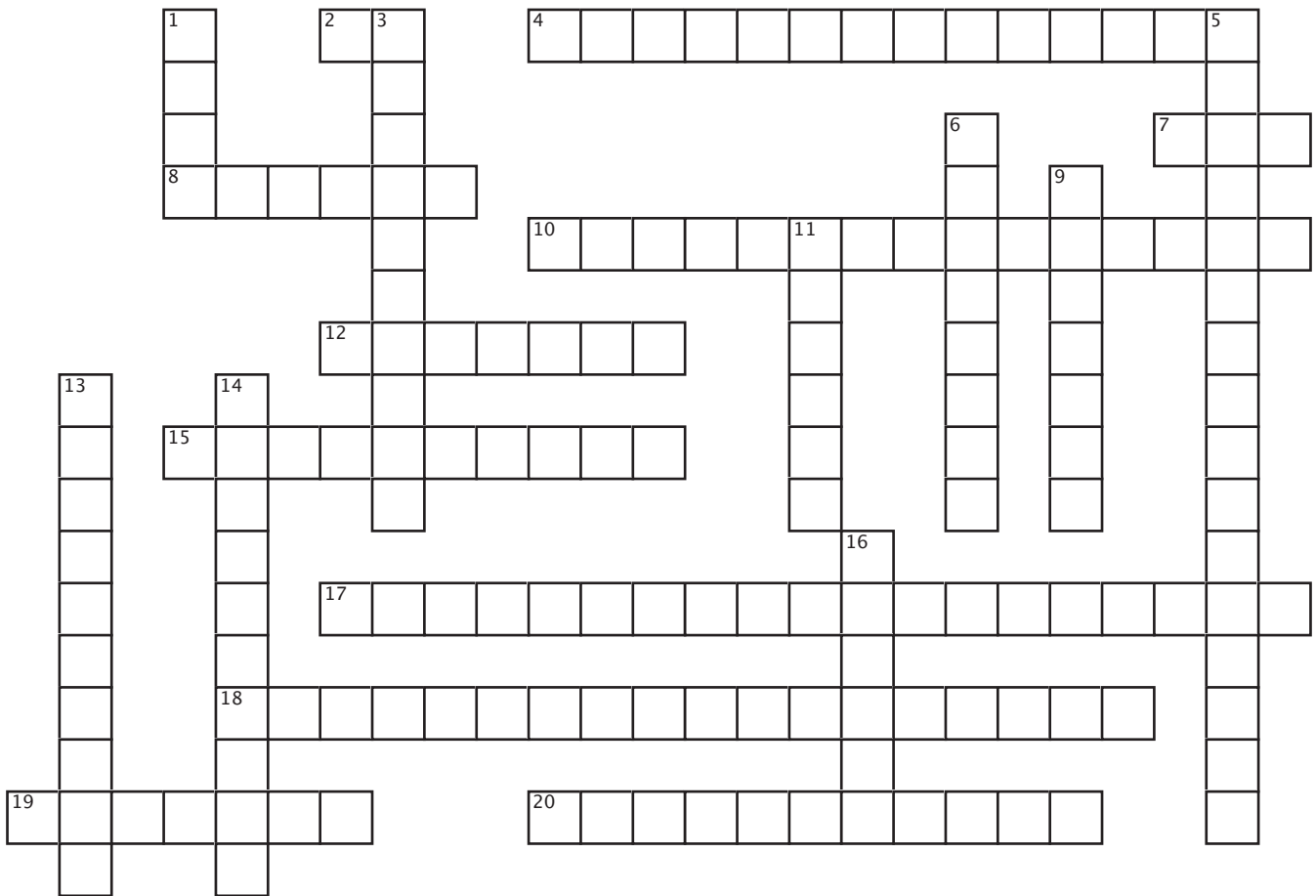
1. Person accountable for conducting an experiment
8. Management plan before experiment
9. Eye protection
10. Accidents need to be reported to _____
11. These are needed for handling concentrated acids
13. Never do this action in a laboratory
15. Used to remove noxious gases from a laboratory
17. Hot objects are placed on these
18. Work area in a laboratory should always be clean and _____
19. Made of fire resistant material and found in a red, square, covered jacket

DOWN

2. Liquid refreshment should not be taken from these
3. The drill to follow in event of a major fire
4. Essential to follow
5. The best treatment for burnt fingers
6. Safety flame colour
7. Course of action when breakages or spills occur
12. When dealing with liquids in bottles, this needs to be replaced firmly and properly
14. Always try to be _____ in an emergency
16. When using gas, ensure that these are off
17. If long, needs to be tied back

WORD LIST

- | | | | | |
|------------------|-------------------|--------------|---------------|-------------------|
| ☞ myself | ☞ gloves | ☞ run | ☞ stopper | ☞ laboratory taps |
| ☞ yellow | ☞ heat mats | ☞ clean up | ☞ uncluttered | ☞ calm |
| ☞ safety goggles | ☞ fire blanket | ☞ cold water | ☞ instructor | ☞ correct |
| ☞ fumehood | ☞ hair | ☞ evacuation | ☞ taps | ☞ procedure |
| | ☞ risk assessment | | | |



ACROSS

2. Unit used for measuring acidity
4. Unit of heat measurement
7. Statement of experimental purpose
8. Procedure followed in an experiment
10. Unit for measuring speed
12. Collection of observations and measurements as an outcome of the experiment, represented mathematically
15. Unit of volume
17. Unit of density
18. A way of writing decimal numbers between 1 and 10 multiplied by a power of 10
19. Unit of electrical charge equal to the quantity of charge transferred in one second by a steady current of one ampere
20. A variable that is not affected by any other variables

DOWN

1. Unit of mass
3. Possible answer to the question posed in an experiment
5. Process of conducting an experiment
6. Factor which affects the result of an experiment
9. Unit of sound intensity
11. Unit of time
13. Statement that either rejects or supports the experiment
14. Evaluation of experiment and result, including improvement of experimental design
16. Unit of force

WORD LIST

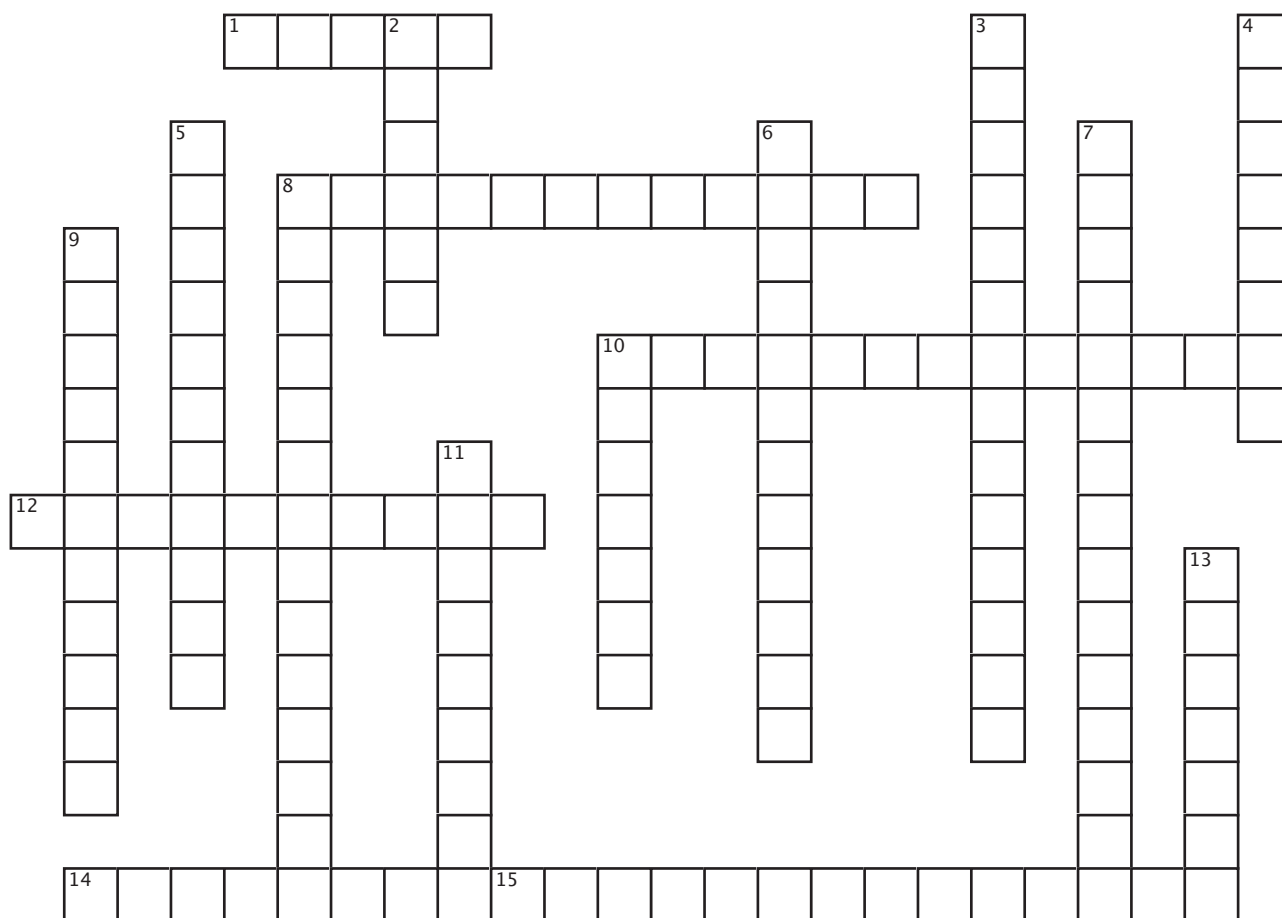
scientific
method
aim
hypothesis

method
results
discussion
conclusion
second

gram
millilitre
gram centimetre
cubed
degrees Celsius

variable
independent
metres per
second
decibel

Newton
pH
scientific
notation
coulomb



ACROSS

1. The universal solvent
8. The process of making something pure
10. The process of allowing solids to fall out of a liquid
12. The process of separating solids from liquids
14. A substance that cannot be separated into simpler substances by a chemical reaction
15. Changing a substance by reacting it with another

DOWN

2. All substances can transfer this
3. The separation of mixtures into their parts using a column of paper or another chemical such as silica
4. Two or more pure chemicals joined together
5. When something is measured, weighed or perceived, and then recorded
6. Separating liquids according to their boiling points
7. The process of removing the solvent and leaving behind a solid
8. Changing the phase of a substance
9. Tests or trials, in order to discover something unknown
10. Sorting from very large to very small
11. An aqueous, or water mixture
13. A hotch potch of substances

WORD LIST

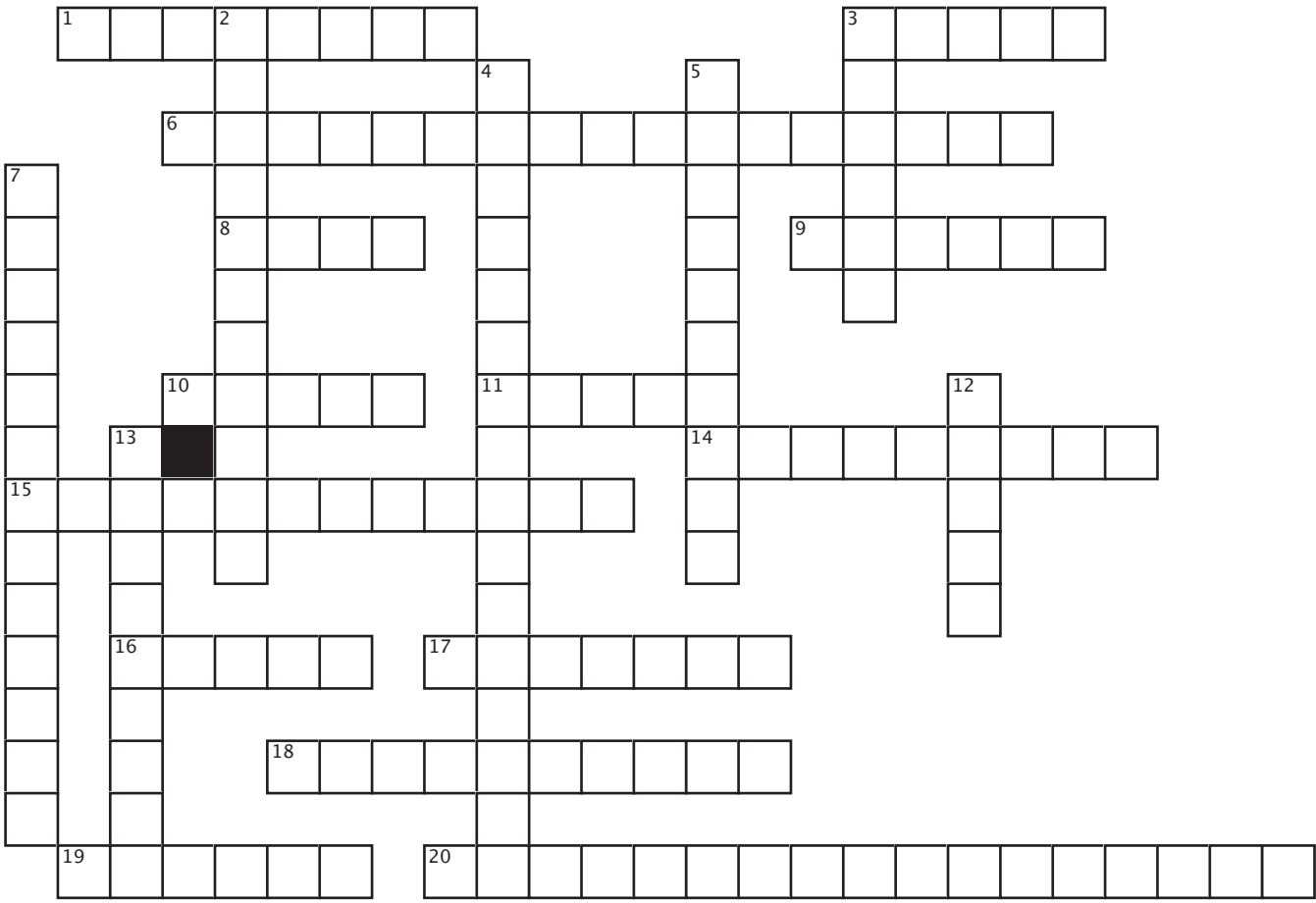
☞ crystallisation
☞ sieving
☞ sedimentation
☞ distillation

☞ purification
☞ physical change
☞ chemical change

☞ chromatography
☞ solution
☞ mixture
☞ compound

☞ element
☞ filtration
☞ water
☞ observation

☞ experiments
☞ energy



ACROSS

- 1. Small glass tube for testing substances
- 3. Glassware which can be conical or round in shape
- 6. Accurately measures volume
- 8. The amount of matter in something
- 9. Procedure
- 10. Device used for measuring something, such as volts
- 11. Made up of columns and rows
- 14. Used for measuring time
- 15. Named after Robert Bunsen
- 16. Used for measuring length
- 17. A two-dimensional drawing
- 18. Graphs are drawn on this
- 19. Used for holding a liquid
- 20. Used for measuring mass

DOWN

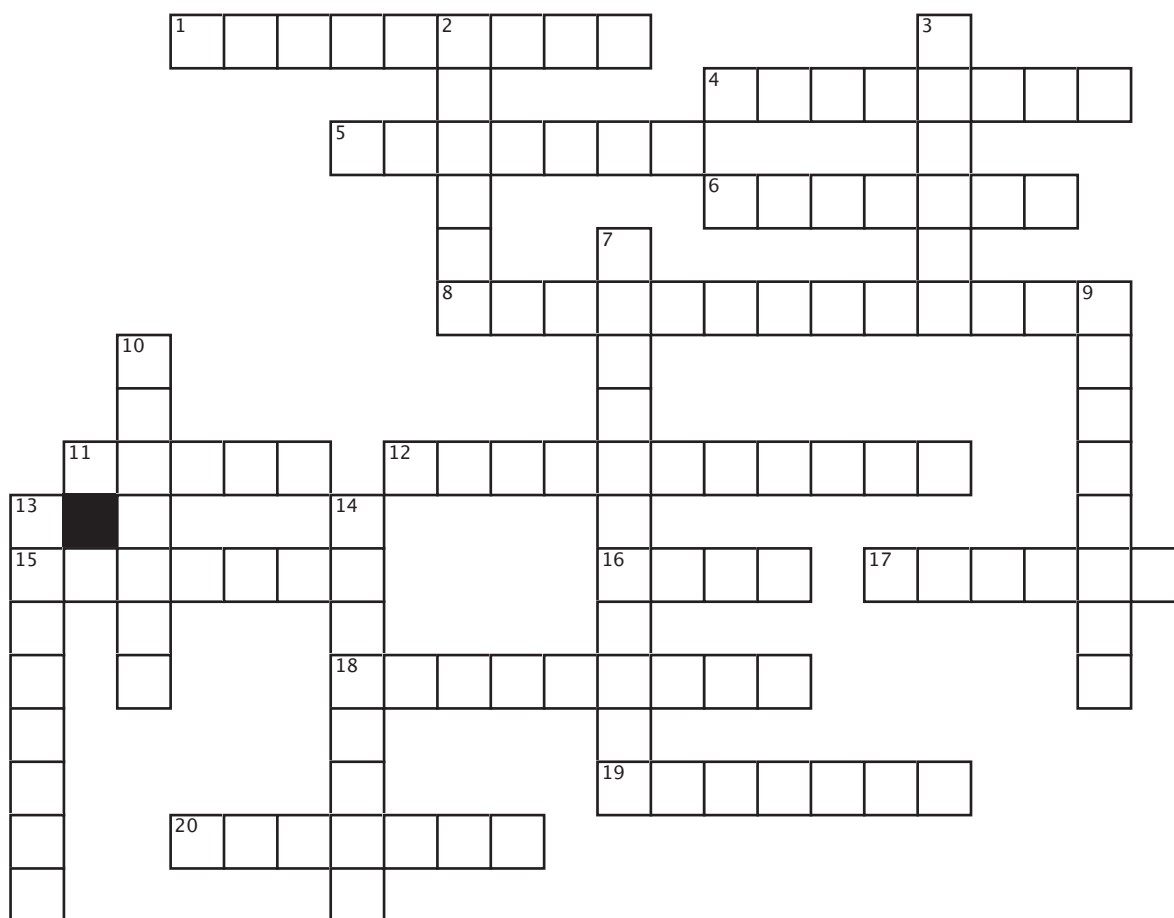
- 2. Instrument used for measuring temperature
- 3. Used for pouring a liquid into a container
- 4. Used for putting out fires
- 5. Best guess
- 7. Used for measuring weight
- 12. Wire mesh used with a tripod
- 13. Derive from known evidence

WORD LIST

- | | | | | |
|-----------------|-----------------------|----------------------|---------------------|------------------|
| ☞ graph paper | ☞ beaker | ☞ measuring cylinder | ☞ flask | ☞ inference |
| ☞ Bunsen burner | ☞ triple beam balance | ☞ table | ☞ fire extinguisher | ☞ mass |
| ☞ thermometer | ☞ stopwatch | ☞ gauze | ☞ hypothesis | ☞ meter |
| ☞ test tube | ☞ ruler | ☞ funnel | ☞ method | ☞ spring balance |



USEFUL CONCEPTS



ACROSS

1. The spreading out of particles in liquids and gases due to their random motion
4. A pure substance consisting of atoms of two or more different elements in fixed proportions which cannot be separated by physical means
5. A substance that dissolves another substance
6. A substance made up of two or more other substances which are not chemically joined
8. The emission of small bubbles of gas, usually in a liquid
11. A substance whose molecules move freely past each other and take the shape of the container
12. A solution that has a large amount of solute in it
15. Two or more atoms having the same atomic number but different mass numbers
16. The unit of matter; the smallest unit of an element
17. A substance that will change shape, but not volume
18. Something which cannot be dissolved
19. A substance that cannot be reduced to simpler substances by chemical means
20. Something that can be dissolved

DOWN

2. A substance dissolved in another substance
3. The amount of space a substance occupies
7. The measure of how hot something is
9. A suspension of small particles of one liquid in a second liquid in which the first will not mix
10. Liquid that contains particles that do not separate out
13. The liquid that passes through a filter
14. The curved surface of a liquid

WORD LIST

☞ meniscus
☞ soluble
☞ solvent
☞ insoluble

☞ solute
☞ diffusion
☞ effervescence
☞ colloid

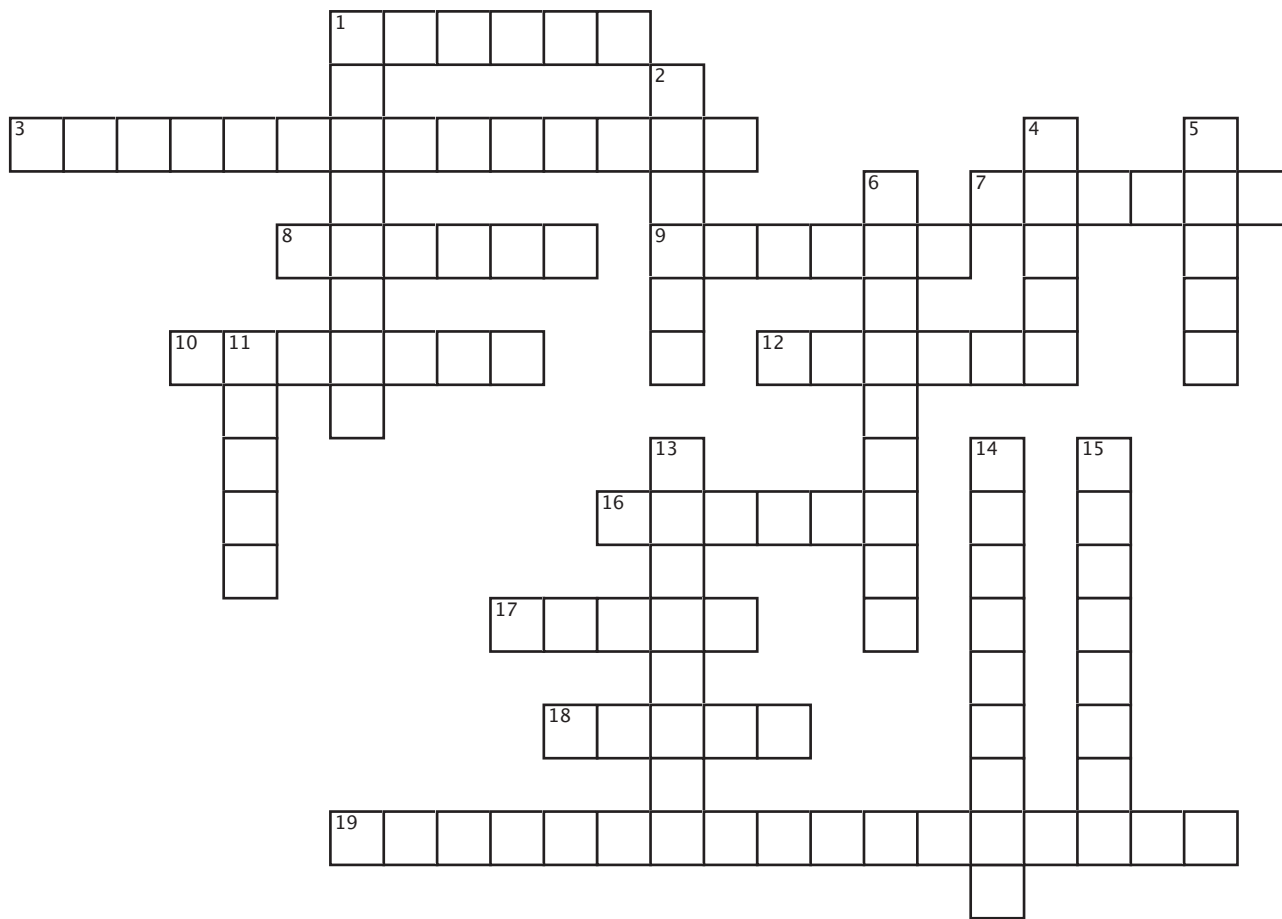
☞ emulsion
☞ concentrate
☞ filtrate
☞ liquid

☞ fluid
☞ mixture
☞ element
☞ compound

☞ atom
☞ isotope
☞ temperature
☞ volume



FAMOUS AUSTRALIAN SCIENTISTS



ACROSS

1. Developer of penicillin and Nobel prize winner
3. Marsupial biologist
7. Aerial surveyor and explorer of the Northern Territory
8. Nobel prize winner for physiology
9. Antarctic explorer and geologist
10. Nobel prize winner for medicine
12. Microbiologist responsible for the eradication of smallpox
16. Famous Australian meteorologist
17. Australian zoologist famous for her work on dugongs
18. Nobel prize winner for physics
19. Won a Nobel prize for immunology

DOWN

1. Curator of Palaeontology at the Australian Museum
2. Physiologist famous for her work in neurotransmitters
4. Principal CSIRO researcher in Australian geology
5. Geophysicist who made major contributions to understanding the Earth's interior
6. Nobel prize winner for chemistry
11. Famous Australian marine scientist
13. Winner of a Nobel prize for determining that a bacterium causes stomach ulcers
14. Pioneer of polio treatment and research
15. Famous Australian neurobiologist

WORD LIST

Doherty

Florey

MacFarlane-Burnett

MacKay

Hales

Olley

Fletcher

Marshall

Taylor

Bragg

Marsh

Cornforth

Fenner

Mawson

Baker

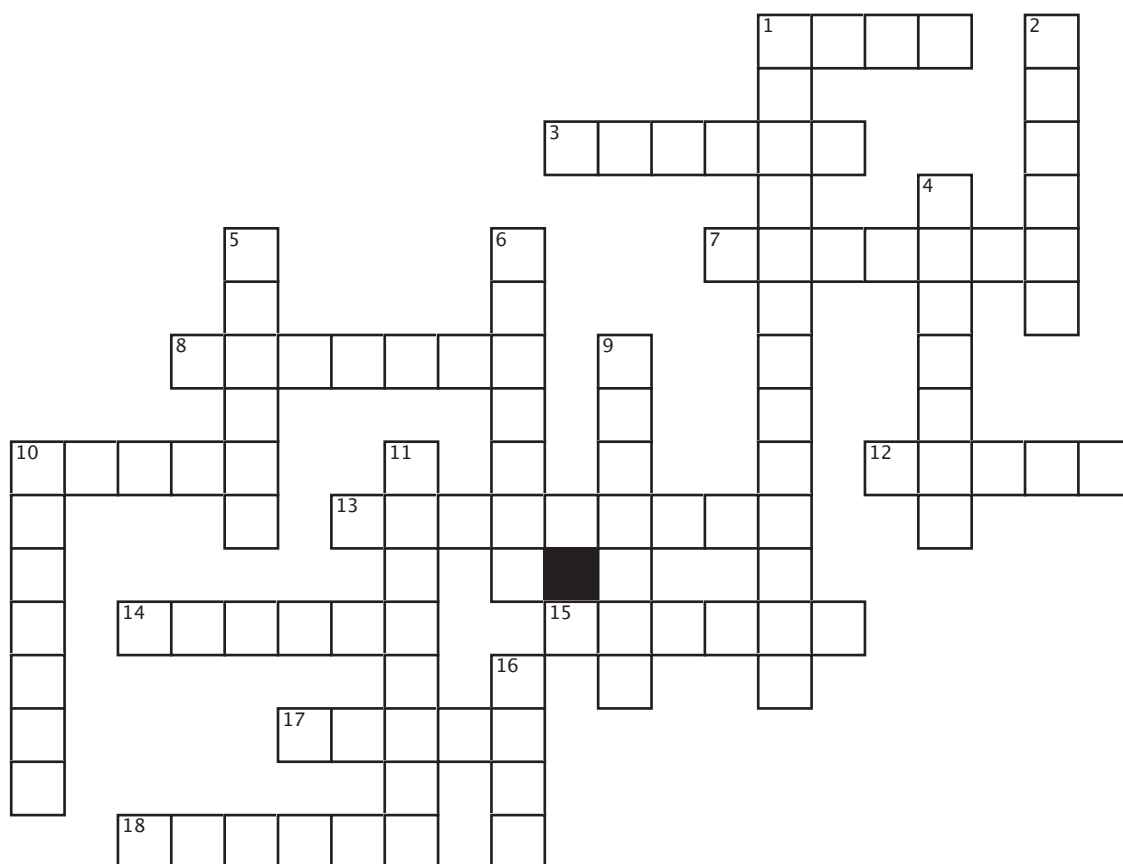
Holman

Horridge

Eccles

Tyndale-Biscoe

Macnamara



ACROSS

1. Discovered the method of determining which pathogen causes disease
3. Has a space telescope named after him
7. Discovered a method of destroying bacteria without harming food
8. Renowned primatologist and conservationist
10. Determined elasticity
12. Discovered radioactivity
13. Discovered soda water, laughing gas and oxygen
14. Formulated the laws of gravity
15. Discovered the process of vaccination using cowpox
17. Discovered that electricity had pressure
18. Author of *The Origin of Species*

DOWN

1. Inventors of the silicon chip
2. Discoverer of electrical current
4. Worked out the scale on thermometers used in science
5. Maker of the first antibiotic
6. Discovered that different masses hit the ground simultaneously
9. Determined the processes of nuclear fission
10. Determined the structure of penicillin and took the first images of vitamin B12
11. Used X-ray crystallography to obtain an image of DNA
16. Inventor of the piston steam engine

WORD LIST

☞ Curie
☞ Franklin
☞ Hodgkin
☞ Meitner

☞ Kilby and Noyce
☞ Florey
☞ Watt
☞ Volta

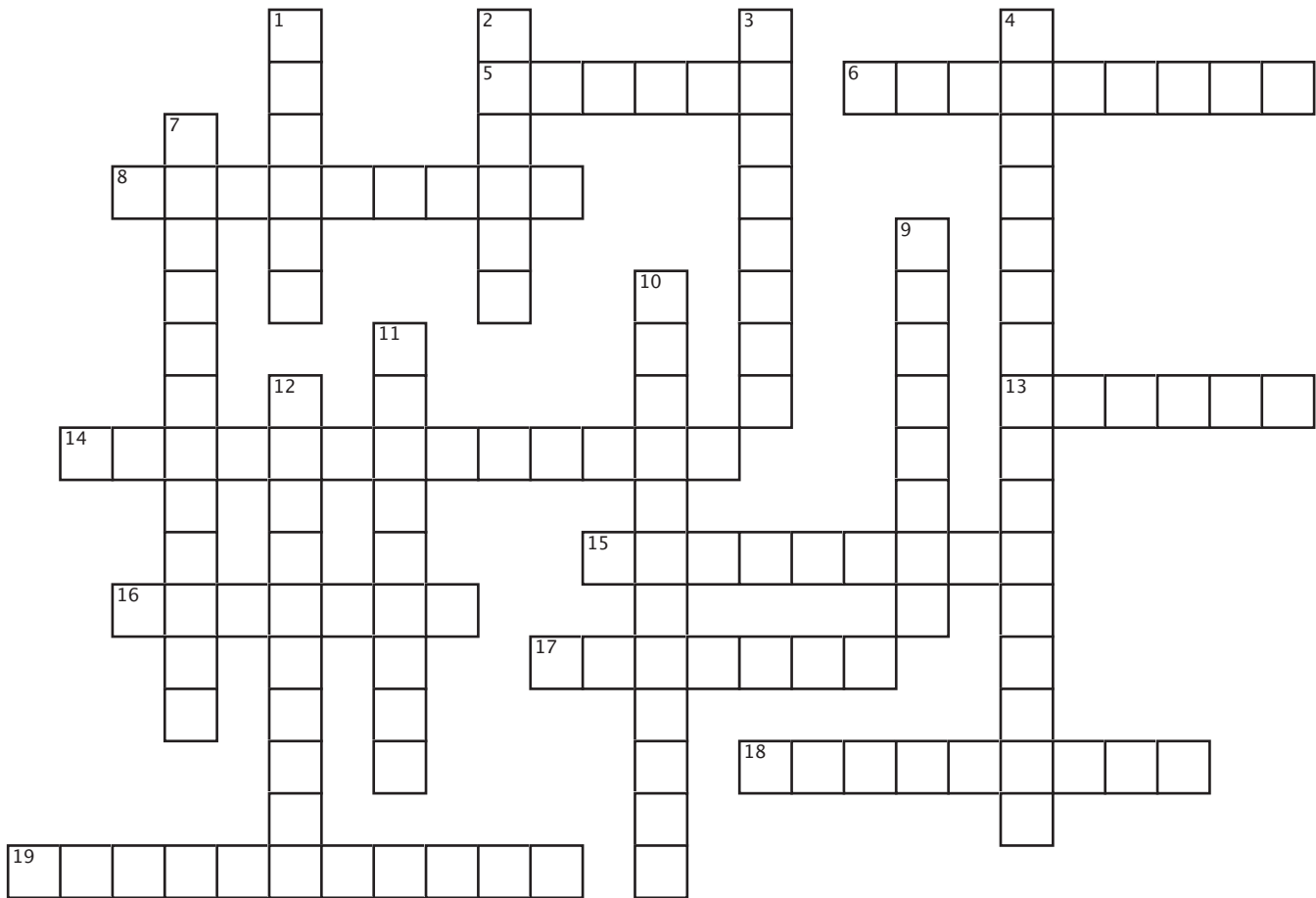
☞ Goodall
☞ Hooke
☞ Pasteur
☞ Koch

☞ Celsius
☞ Jenner
☞ Hubble
☞ Priestley

☞ Galileo
☞ Newton
☞ Darwin
☞ Ampere



SCIENTISTS AND THEIR 'OLOGIES'



ACROSS

- 5. A telescope in space is named after him
- 6. Australian Chemist who was awarded the Nobel prize in chemistry in 1975
- 8. Science of the stars
- 13. Geologist and Antarctic explorer
- 14. Volcano scientist
- 15. Scientist who studies animal structure and function
- 16. Science of living things
- 17. Science of living systems
- 18. Science of the composition of substances
- 19. Science of climate

DOWN

- 1. Discoverer of penicillin
- 2. Australian astronaut
- 3. Science of healing
- 4. Scientist who studies the processes of all living and non-living things
- 7. Scientist who studies human behaviour
- 9. He wrote the theory of relativity
- 10. Scientist who studies bodily processes
- 11. Scientist who studies the Earth's history, structure and composition
- 12. Scientist who studies microscopic matter

WORD LIST

☞ Einstein
☞ Hubble
☞ Mawson
☞ Florey

☞ Cornforth
☞ Thomas
☞ biology
☞ vulcanologist
☞ zoologist

☞ climatology
☞ medicine
☞ ecology
☞ geologist

☞ astronomy
☞ nanologist
☞ chemistry
☞ environmentalist

☞ physiologist
☞ psychologist

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